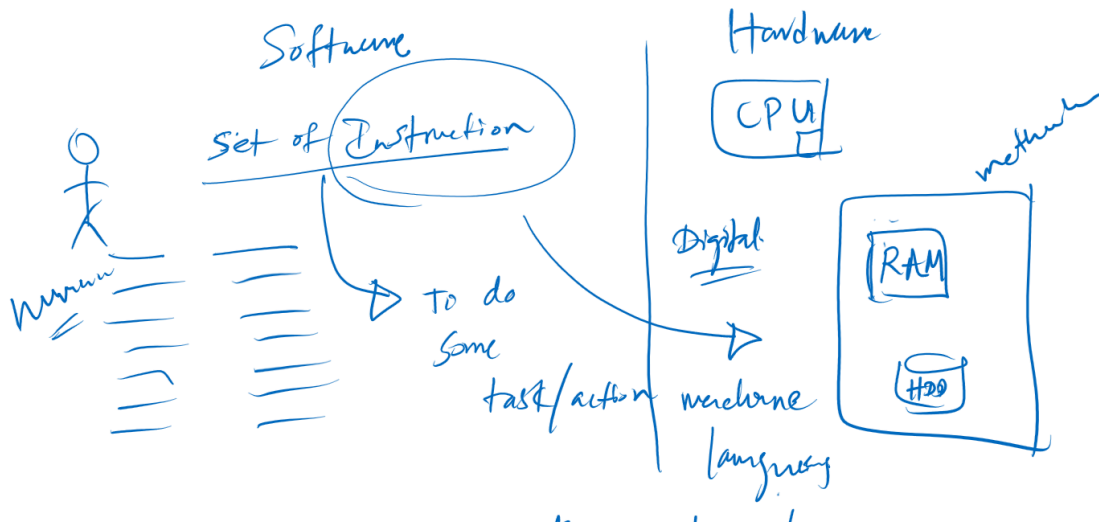


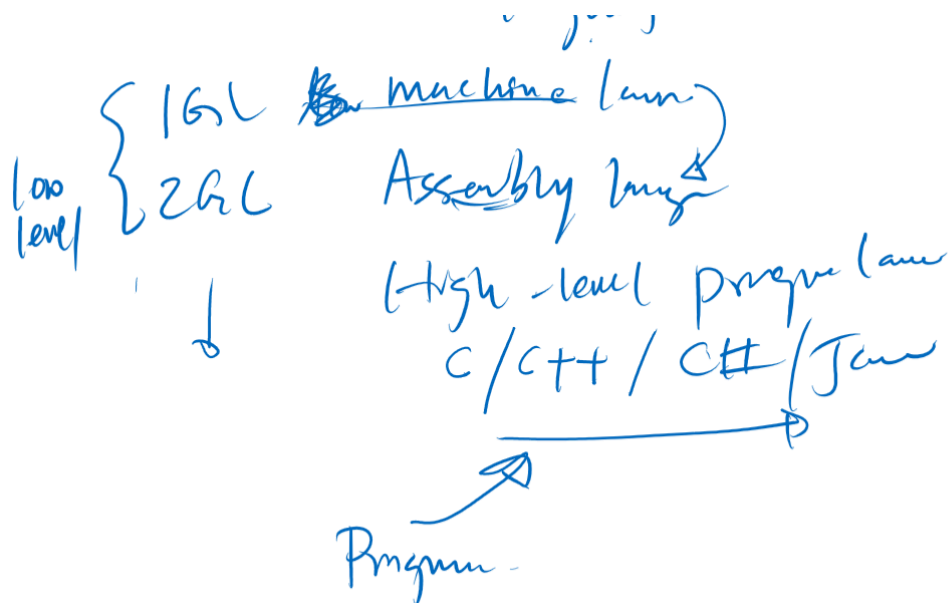
Programming Fundamentals

=====

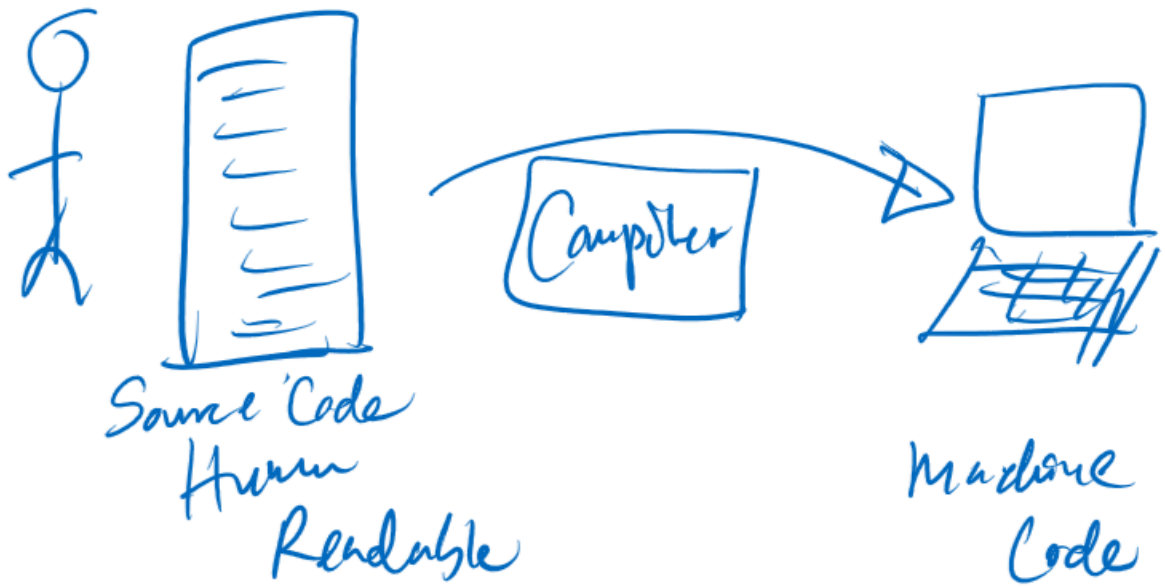
Q1



Q2

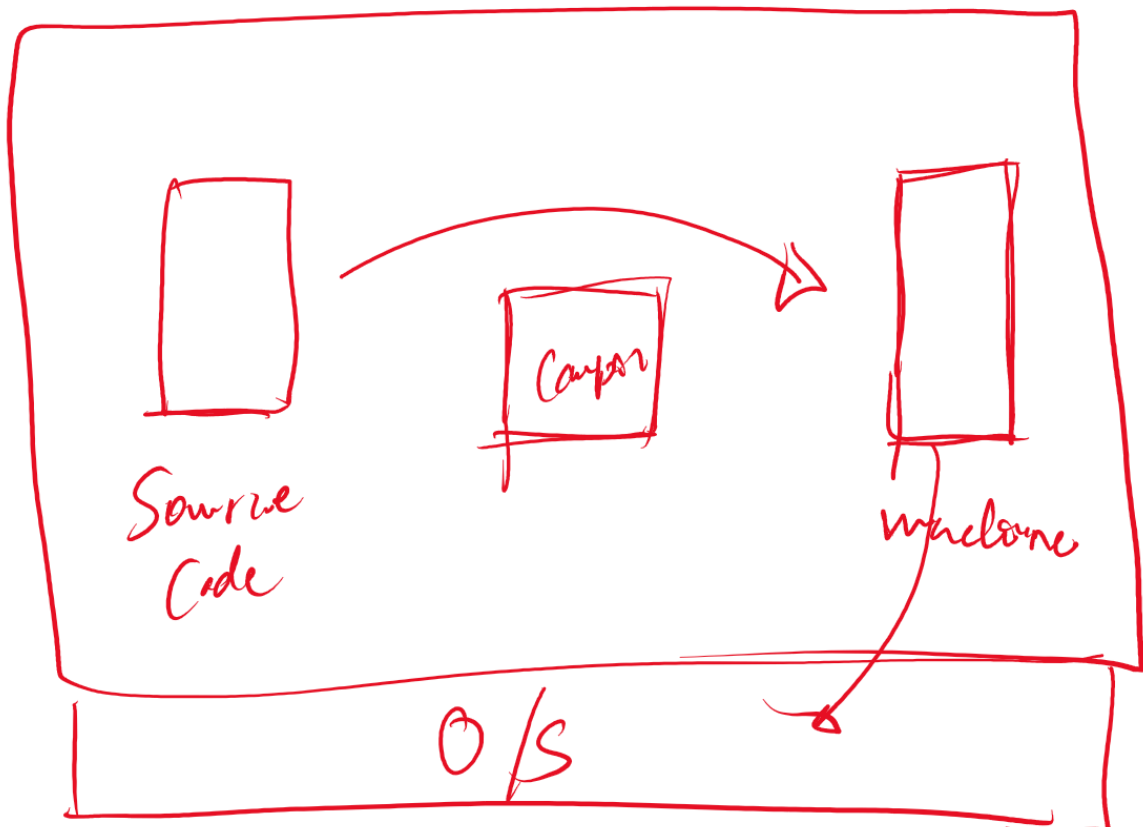


Q3



////////////////////////////////////

Q4



////////////////////////////////////

Q5

(992-3-4)

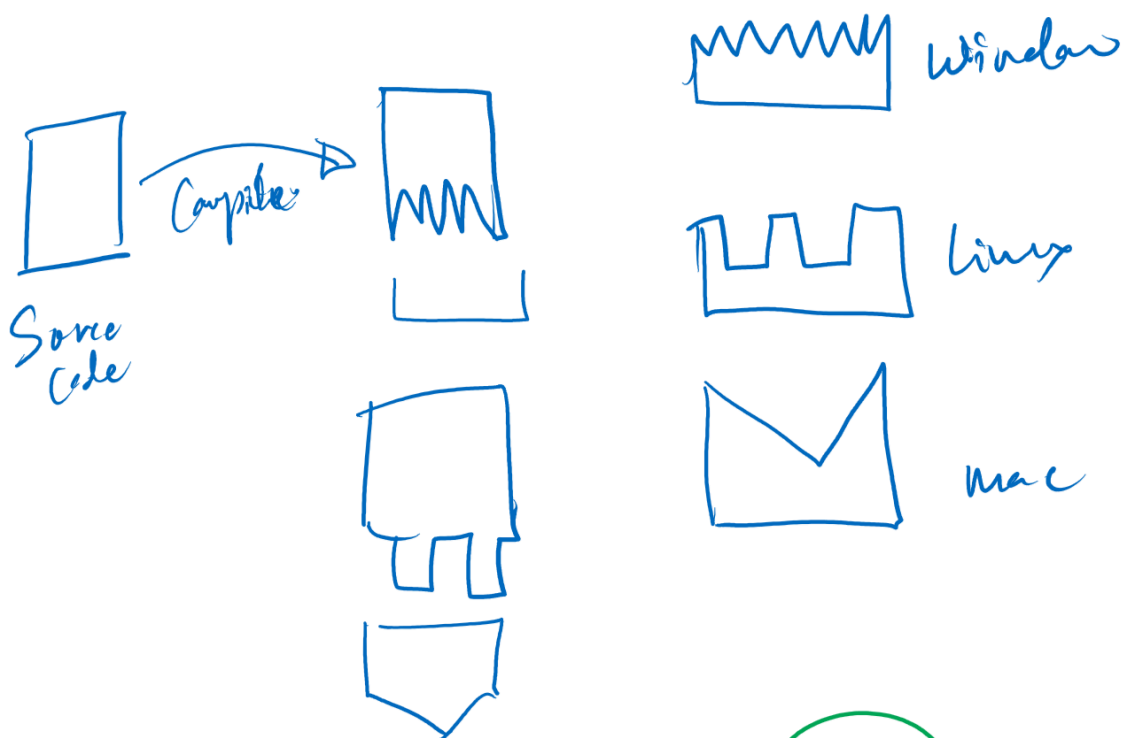
→ Java

Write Once, Run Anywhere

Platform Independence

////////////////////////////////////

Q6



////////////////////////////////////

Q7


```
class Example{
    public static void main(String args[]){
        system.out.println("Hello World!");
    }
}
```

```
E:\ICM120_ICD121>javac Example.java
Example.java:3: error: package system does not exist
        system.out.println("Hello World!");
        ^
1 error
E:\ICM120_ICD121>
```

//

Q12

```
E:\ICM120_ICD121>java Example
Hello World!

E:\ICM120_ICD121>
```

//

Q13

Once we compile and create the bytecode, we no longer needed the source code to execute the program

//

Q14

```
class Example{
    public static void main(String args[]){
```

Q16 main method - Java Entry point

```
File Edit View

class Example{
    public static void mai(String args[]){
        System.out.println("A");
        System.out.println("B");
        System.out.println("C");
        System.out.println("D");
        System.out.println("E");
    }
}

E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>javac Example.java

E:\ICM120_ICD121>java Example
Error: Main method not found in class Ex
ample, please define the main method as:
    public static void main(String[] args
)
or a JavaFX application class must exten
d javafx.application.Application

E:\ICM120_ICD121>
E:\ICM120_ICD121>
```

```
class Example{
    public static void mai(String args[]){
        System.out.println("A");
        System.out.println("B");
        System.out.println("C");
        System.out.println("D");
        System.out.println("E");
    }
}
```

////////////////////////////////////

class name

```
class Demo{
    public static void main(String[] args){
        System.out.println("Hello!");
    }
}
```

Name	Date modified	Type
Demo.class	3/1/2026 11:35 AM	CLASS File
Example.java	3/1/2026 11:32 AM	JAVA File

```
E:\ICM120_ICD121>
E:\ICM120_ICD121>javac Example.java

E:\ICM120_ICD121>java Demo
A
B
C
D
E
F

E:\ICM120_ICD121>
```

//

Q17 Legal Main Method

```
public static void main(String []args){
```

//

Q18

```
class Example{
    public static void main(String[] args){

    }
}
```

//

Q19

```
class Example{
    public static void main(String[] args){
        System.out.println("ABC");
        System.out.println();
        System.out.println("XYZ");
    }
}
```

//

Q20 - System.out.println();

```
class Example{
    public static void main(String[] args){
        System.out.println("ABC");
        System.out.println();
        System.out.println("XYZ");
    }
}
```

//

Q21 - System.out.println(); vs System.out.print();

```
class Example{
    public static void main(String[] args){
        System.out.println("A");
        System.out.print("B");
        System.out.println("C");
        System.out.print("D");
        System.out.println("E");
    }
}
```

```

class Example{
    public static void main(String[] args){
        System.out.println("A");
        System.out.print("B");
        System.out.println("C");
        System.out.print("D");
        System.out.println("E");
    }
}

```

Entry Point



```

E:\ICM120_ICD121>java Example
A
BC
DE

```

System.out.println (" ^{data} ");

System.out.print (" ");

== [] | _

mm
[]

////////////////////////////////////

Q22

```

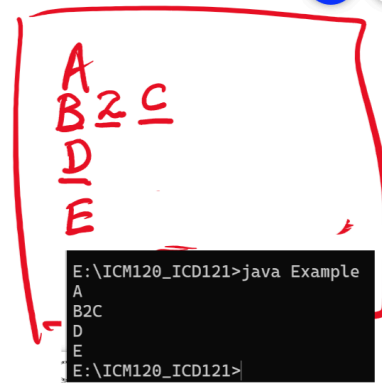
class Example{
    public static void main(String[] args){
        System.out.println("A");
        System.out.print("B");
        System.out.print("2");
        System.out.println("C");
        System.out.println("D");
        System.out.print("E");
    }
}

```

```

class Example{
    public static void main(String[] args){
        System.out.println("A");
        System.out.print("B");
        System.out.print("2");
        System.out.println("C");
        System.out.println("D");
        System.out.print("E");
    }
}

```



////////////////////////////////////

Q23

```

class Example{
    public static void main(String[] args){
        System.out.print("A");
        System.out.print("B");
        System.out.print("");
        System.out.println("C");
        System.out.print();
        System.out.print("D");
        System.out.println();
        System.out.print("E");
        System.out.print("E");
    }
}

```

}


```

        public static void main(String[] args){
            System.out.println("A");
            System.out.println("B"); /*laksjdffhskdfh*/
System.out.println("C");
            System.out.println("C");
            System.out.println("D");
            System.out.println("E");
            System.out.println("F");
        }
    }

```

//

Java Literals

//

Q29

```

class Example{
    public static void main(String[] args){
        System.out.println("Hello!");
        System.out.println(10);
        System.out.println(22.34);
    }
}

```

//

Q30



- (1) Integer Literals
- (2) Floating Point Literals
- (3) Character Literals
- (4) String Literals
- (5) Boolean Literals
(true, false)
- (6) null Literals

literals

directly $\rightarrow$ value

Q31

}

Q32

```
class Example{
    public static void main(String[] args){
```

```
System.out.println("Hello!"); // String Literals
```

```
System.out.println(10); // Integer Literals
```

```
System.out.println(22.345465); // Floating Point  
Literals
```

```
System.out.println(-22.345465); // Floating  
Point Literals
```

```
System.out.println('A'); // Character Literals
```

```
System.out.println(true); // Boolean Literals  
System.out.println(false);
```

```
String n = null; // null Literal  
}  
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q33 Integer Literals

```
=====
```

```
class Example{  
    public static void main(String[] args){  
        System.out.println(100); // Decimal Literal  
        System.out.println(0144); // Octal Literal  
    }  
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q34

```

file Edit View
class Example{
    public static void main(String[] args){
        System.out.println(100); // Dec
        System.out.println(0148); // Octal Literal
    }
}

E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>javac Example.java
Example.java:4: error: integer number too large
        System.out.println(0148);
                           ^
// Ocatal Literal

1 error

E:\ICM120_ICD121>
```

Q35

Literal (0-1)

Literal (0-15)

Literal (0-9, A-F)

}

Q36

Literal (0-1)

Literal (0-15)

Literal (0-9, A-F)

```

System.out.println("0"+Integer.toOctalString(100));
    }
}

```

//

Q37

```

class Example{
    public static void main(String[] args){
        System.out.println(100); // Decimal Literal (0-9)
        System.out.println(0144); // Octal Literal (0-7)
        System.out.println(0b1100100); // Binary Literal
(0-1)
        System.out.println(0x64); // Hexa Decimal
Literal (0-15)
        System.out.println(0x1A5F); // Hexa Decimal
Literal (0-9, A-F)

        System.out.println(Integer.toBinaryString(100));
    }
}

```

//

Q38 Floating Point Literals

```

class Example{
    public static void main(String[] args){
        System.out.println(100.0154);
        System.out.println(1.000154E2);
        System.out.println(10.00154E1);
    }
}

```

$$5.243 \times 10^{-3}$$

```
class Example{
    public static void main(String[] args){
        System.out.println(100.0154);
        System.out.println(1.000154E2);
        System.out.println(10.00154E1);
        System.out.println(0.005243); // Prints ->
0.005243
        System.out.println(); // Prints -> 0.005243
    }
}
```

Q40 From Q39

////////////////////////////////////

Q41

////////////////////////////////////

Q42 Character Literals

////////////////////////////////////

```
class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.println('B');
        System.out.println('\b'); // Backspace
        System.out.println('C');
        System.out.println('D');
    }
}
```

```
E:\ICM120_ICD121>javac Example.java

E:\ICM120_ICD121>java Example
A
B

C
D

E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
```

Q44

```
class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.print('B');
        System.out.println('\b'); // Backspace
        System.out.println('C');
        System.out.println('D');
    }
}
```

```

class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.print('B');
        System.out.println('\b'); // Backspace
        System.out.println('C');
        System.out.println('D');
    }
}

```

```

E:\ICM120_ICD121>javac Example.java
E:\ICM120_ICD121>java Example
A
B
C
D
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>

```

//

Q45

```

class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.print('B');
        System.out.print('\b'); // Backspace
        System.out.println('C');
        System.out.println('D');
    }
}

```

```

class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.print('B');
        System.out.print('\b'); // Backspace
        System.out.println('C');
        System.out.println('D');
    }
}

```

```

E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>javac Example.java
E:\ICM120_ICD121>java Example
A
C
D
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>

```

//

Q46

```
class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.println('B');
        System.out.println('\b'); // Backspace
        System.out.println('C');
        System.out.println('D');
    }
}
```

A
B
C
D

A
B
C
D

```
class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.print('B');
        System.out.println('\b'); // Backspace
        System.out.println('C');
        System.out.println('D');
    }
}
```

A
B
C
D

A
B
C
D

```
class Example{
    public static void main(String[] args){
        System.out.println('A');
        System.out.print('B');
        System.out.print('\b'); // Backspace
        System.out.println('C');
        System.out.println('D');
    }
}
```

A
C
D

////////////////////////////////////

Q47 = Escape Characters

=====

\b

\t

\n

\r

```

class Example{
    public static void main(String[] args){
        System.out.println("ABC\bEFG");
        System.out.println("ABC\tEFG");
        System.out.println("ABC\nEFG");
    }
}

```

```

E:\ICM120_ICD121>javac Example
E:\ICM120_ICD121>java Example
ABEFG
ABC      EFG
ABC
EFG
E:\ICM120_ICD121>
E:\ICM120_ICD121>

```

//

Q48

```

class Example{
    public static void main(String[] args){
        System.out.println("\\");
    }
}

```

```

E:\ICM120_ICD121>javac Example
E:\ICM120_ICD121>java Example
\
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>

```

//

Q49

```

class Example{
    public static void main(String[] args){
        System.out.println("\\");    // print \
        System.out.println("\\\\");  // print \\
        System.out.println("\\\\\\""); // print \\\
    }
}

```

```

class Example{
    public static void main(String[] args){
        System.out.println("\\");
        System.out.println("\\\\");
        System.out.println("\\\\\\"");
    }
}

```

```

E:\ICM120_ICD121>
E:\ICM120_ICD121>javac Example
E:\ICM120_ICD121>java Example
\
\\
\\\
E:\ICM120_ICD121>
E:\ICM120_ICD121>

```

//

Q50

```

class Example{
    public static void main(String[] args){

```

```

        System.out.println("\\");
        System.out.println("");
    }
}

```

<pre> class Example{ public static void main(String[] args){ System.out.println("\\"); System.out.println(''); } } </pre>	<pre> E:\ICM120_ICD121>javac Example E:\ICM120_ICD121>java Example " " E:\ICM120_ICD121> </pre>
---	--

//

Q51

```

class Example{
    public static void main(String[] args){
        System.out.println(""); // Legal
        System.out.println(""); // Illegal
    }
}

```

//

Q52

```

class Example{
    public static void main(String[] args){
        System.out.println(""); // Legal
        System.out.println("\");
    }
}

```

<pre> class Example{ public static void main(String[] args){ System.out.println(""); // Legal System.out.println('\'); } } </pre>	<pre> E:\ICM120_ICD121>javac Example E:\ICM120_ICD121>java Example ' ' E:\ICM120_ICD121> </pre>
---	--

Q53 String Literals

```
class Example{
    public static void main(String[] args){
        System.out.println("Dinuka
Kodithuwakku".length());
        System.out.println(162.length()); // illegal
    }
}
```

```
class Example{
    public static void main(String[] args){
        System.out.println("Dinuka".length());
        System.out.println(162.length()); // illegal
    }
}
```

Q54

```
class Example{
    public static void main(String[] args){
        System.out.println("Dinuka
Kodithuwakku".substring(7));
        //System.out.println(162.length()); // illegal
    }
}
```

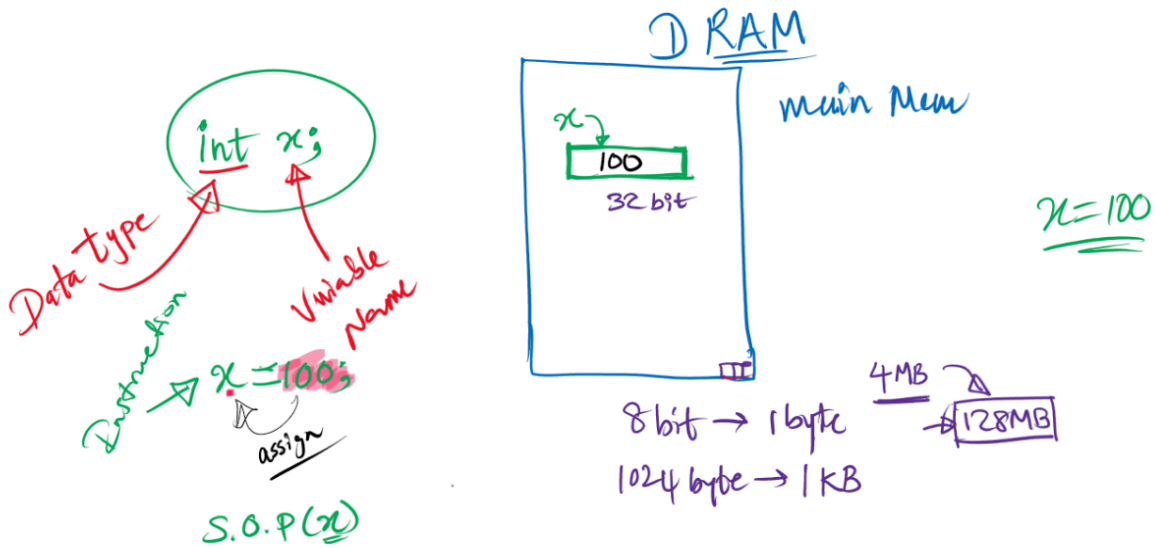
Variables

Q55

```
class Example{
    public static void main(String[] args){
        int x;
        x = 100;
        System.out.println(x);
    }
}
```

```
class Example{
    public static void main(String[] args){
        int x;
        x = 100;
        System.out.println(x);
    }
}
```

```
E:\ICM120_ICD121>javac Example.java
E:\ICM120_ICD121>java Example
100
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
```



////////////////////////////////////

Q56

```
class Example{
    public static void main(String[] args){
        int x;
        x = 100;
        x = 200;
        System.out.println(x);
    }
}
```

```
}
```

//

Q57 Compile Err

```
class Example{
    public static void main(String[] args){
        int x;

        System.out.println(x);
    }
}
```

<pre>class Example{ public static void main(String[] args){ int x; System.out.println(x); } }</pre>	<pre>200 E:\ICM120_ICD121>javac Example.java Example.java:5: error: variable x might not have been initialized System.out.println(x); ^ 1 error E:\ICM120_ICD121></pre>
--	---

//

Q58 Compile Err

```
class Example{
    public static void main(String[] args){
        int x;

        System.out.println(x);
        x = 100;
    }
}
```

//

Q59

```
class Example{
```

////////////////////////////////////

////////////////////////////////////

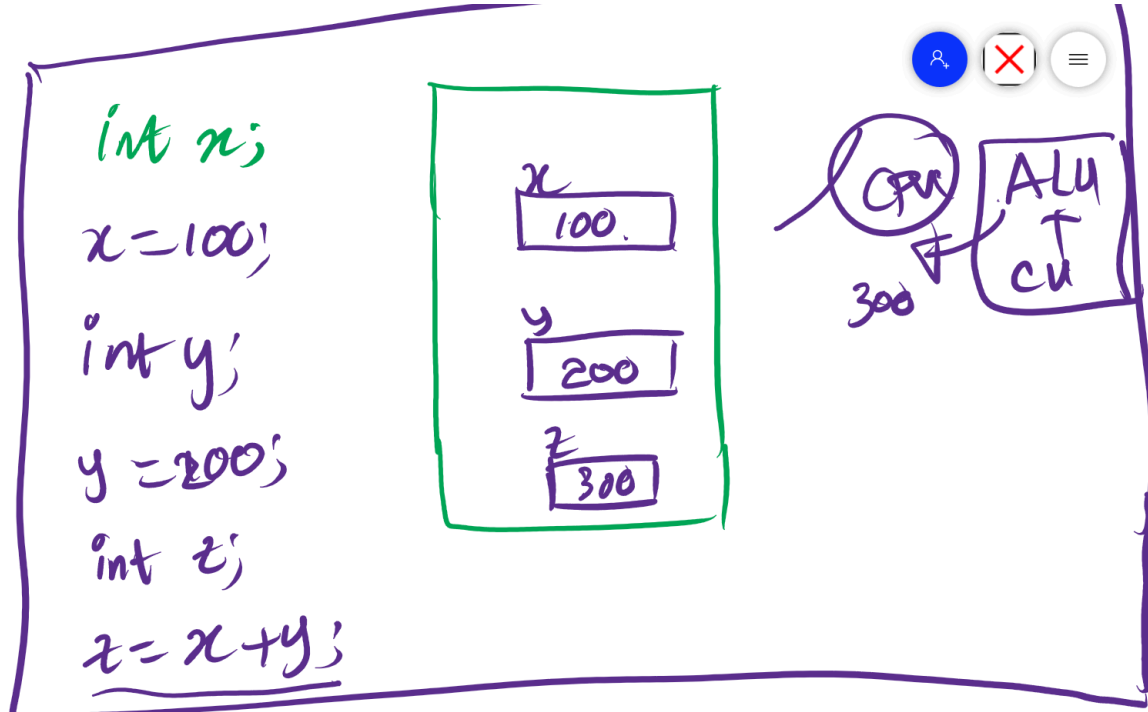
```
class Example{
    public static void main(String[] args){
        int x;
        x = 100;
        int y;
        y = 200;

        System.out.println(x);
        System.out.println(y);
    }
}
```

////////////////////////////////////

```
class Example{
    public static void main(String[] args){
        int x;
        x = 100;
        int y;
        y = 200;

        int z;
        z = x + y;
        System.out.println(x);
        System.out.println(y);
        System.out.println(z);
    }
}
```



```

class Example{
    public static void main(String[] args){
        int x;
        x = 100;
        int y;
        y = 200;
        x = 500;
        int z;
        z = x + y;
        System.out.println(x);
        System.out.println(y);
        System.out.println(z);
    }
}

```

Handwritten annotations on the code block:

- Variable Declaration:** Points to `int x;` and `int y;`.
- Variable Initialization:** Points to `x = 100;` and `y = 200;`.
- Assignment:** Points to `x = 500;` and `z = x + y;`.

////////////////////////////////////

Q63

```

class Example{
    public static void main(String[] args){
        int x = 100;

        int y, z;
        y = 200;

        z = x + y;
        System.out.println(x);
        System.out.println(y);
    }
}

```

////////////////////////////////////

////////////////////////////////////

```
class Example{
    public static void main(String[] args){
        int x = 100, y, z;

        y = 200;

        int y; // Compile Error
    }
}
```

```

        z = x + y;
        System.out.println(x);
        System.out.println(y);
        System.out.println(z);
    }
}

```

//

Q66

```

class Example{
    public static void main(String[] args){
        int x, y, z;

        x = y = z = 100;

        System.out.println(x);
        System.out.println(y);
        System.out.println(z);
    }
}

```

//

String Concatenation And Arithmetic Addition

//

Q67

```

class Example{
    public static void main(String[] args){
        System.out.println(10 + 20); // 30
        System.out.println("10" + "20"); // 1020
    }
}

```

//

Q68

```
class Example{
    public static void main(String[] args){
        System.out.println(10 + 20); // 30
        System.out.println("10" + "20"); // 1020
        System.out.println(10 + "20"); // 1020
        System.out.println("10" + 20); // 1020
    }
}
```

<pre>class Example{ public static void main(String[] args){ System.out.println(10 + 20); // System.out.println("10" + "20"); System.out.println(10 + "20"); / System.out.println("10" + 20); / } }</pre>	<pre>E:\ICM120_ICD121>javac Exampl E:\ICM120_ICD121>java Example 30 1020 1020 1020 E:\ICM120_ICD121> E:\ICM120_ICD121></pre>
--	--

//

Q69

```
class Example{
    public static void main(String[] args){
        int x = 100, y = 200, z;

        z = x + y;
        System.out.println(); // 100 + 200 = 300;
    }
}
```

//

Q70

```
class Example{
    public static void main(String[] args){
        int x = 100, y = 200, z;

        z = x + y;
        System.out.println(""); // 100 + 200 = 300

        System.out.println(x+"ABC"); // 100ABC
        System.out.println(10 + "ABC"); // 10ABC
    }
}
```

E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>javac Example.java
E:\ICM120_ICD121>java Example
100ABC
10ABC
E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>

//

Q71

```
class Example{
    public static void main(String[] args){
        int x = 101, y = 201, z;

        z = x + y;
        System.out.println(x + " + " + y + " = " + z); //
100 + 200 = 300
    }
}
```

Q72

```
import java.util.*;
class Example{
    public static void main(String[] args){
        Scanner input = new Scanner(System.in);

        int x, y, z;

        x = input.nextInt();
        y = input.nextInt();

        z = x + y;
        System.out.println(x + " + " + y + " = " + z); //
100 + 200 = 300

    }
}
```

```
import java.util.*;
class Example{
    public static void main(String[]
        Scanner input = new Scann

        int x, y, z;

        x = input.nextInt();
        y = input.nextInt();

        z = x + y;
        System.out.println(x + "

    }
}
```

```
symbol:   variable input
location: class Example
Example.java:7: error: cannot find symbol
        y = input.nextInt();
              ^
symbol:   variable input
location: class Example
2 errors

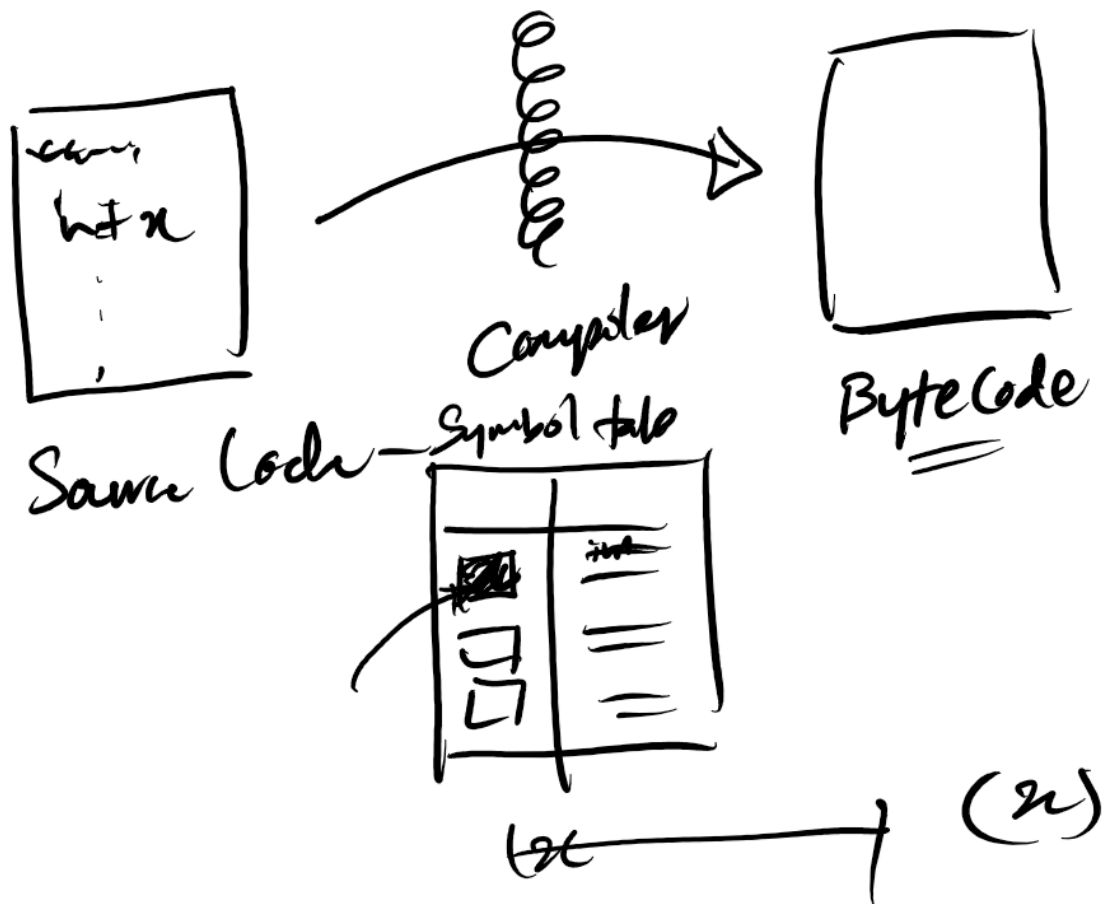
E:\ICM120_ICD121>javac Example.java

E:\ICM120_ICD121>java Example
10
20
10 + 20 = 30

E:\ICM120_ICD121>java Example
152
654
152 + 654 = 806

E:\ICM120_ICD121>java Example
85
65
85 + 65 = 150

E:\ICM120_ICD121>
E:\ICM120_ICD121>
E:\ICM120_ICD121>
```

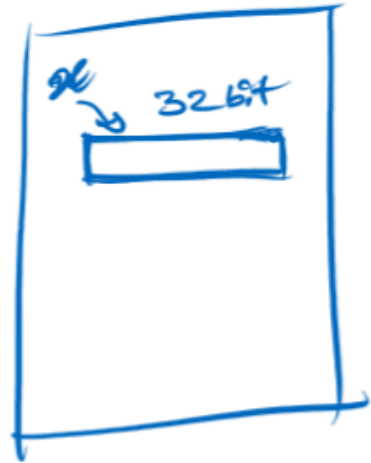


////////////////////////////////////

Primitive Data Types

////////////////////////////////////

int x;
 ↑ ↑
 data type variable
 name
 (identified)



8 primitive data types

byte }
 short } Integer
 → int } values
 long }

float } floating
 double } Point values

char — characters

boolean — (logic)

byte	8 bits
short	16 bits
int	32 bits
long	64 bits
float	32 bits
double	64 bits
char	16 bits
boolean	1 bit

//

Q75

```
import java.util.*;
class Example{
    public static void main(String[] args){
        byte x;
        x = 100;
        System.out.println(x);
    }
}
```

//

Q76

```

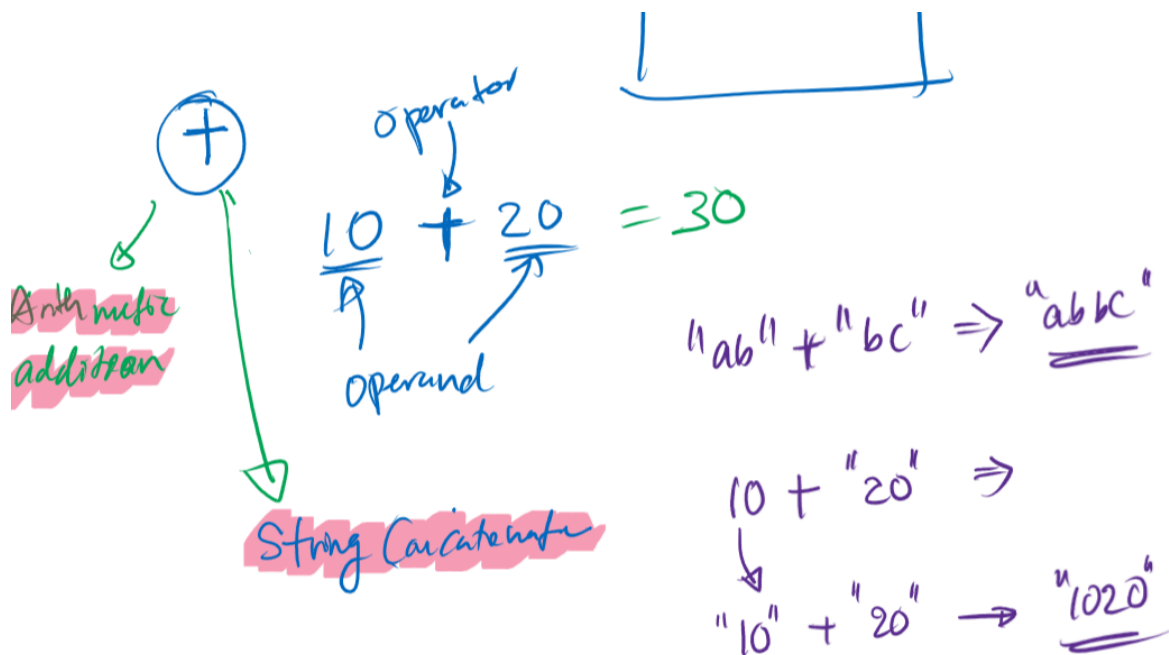
import java.util.*;
class Example{
    public static void main(String[] args){
        long y = 2154525852; // Compile Err.
        long x;
        x = 2154525852L; // Tells compiler to take this
as long Literal
        System.out.println(x);

        float f = 1.5f; // Tells compiler to take this as
float Literal
    }
}

```

////////////////////////////////////

Q77



////////////////////////////////////

Q78

```

import java.util.*;
class Example{

```



```

    int y = x;
    System.out.println("x : " + x);
    System.out.println("y : " + y);
}
}

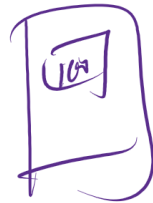
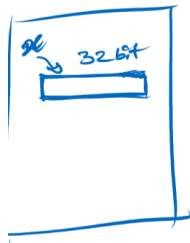
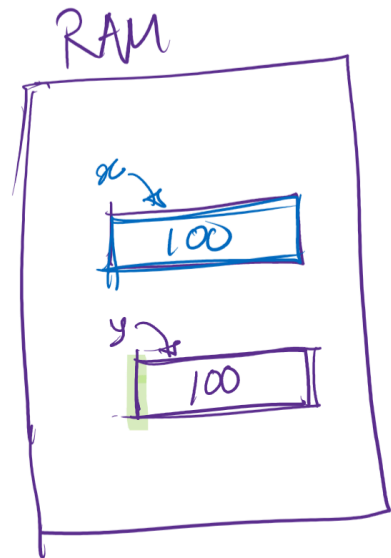
```

```

import java.util.*;
class Example{
    public static void main(String []args){
        int x = 100;
        int y = x;
        System.out.println("x : " + x);
        System.out.println("y : " + y);
    }
}

```

~~int x;~~



~~int x;~~
32
~~int y;~~

////////////////////////////////////

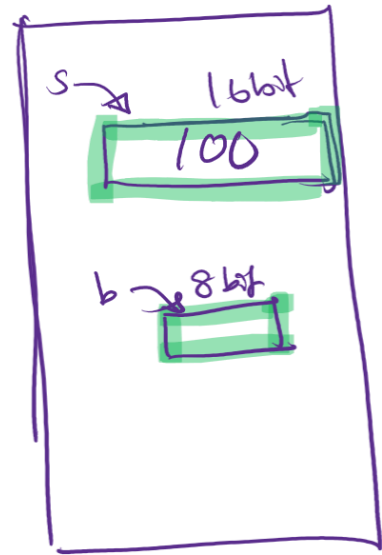
Q80

```

import java.util.*;
class Example{
    public static void main(String []args){
        short s = 100;
        byte b = s; // Compile Error
    }
}

```

```
import java.util.*;
class Example{
    public static void main(String []args){
        short s = 100;
        byte b = s;
    }
}
```



////////////////////////////////////

Q81

```
import java.util.*;
class Example{
    public static void main(String []args){
        byte b = 100;
        short s = b;
    }
}
```

////////////////////////////////////

Key Board Input II - User Interaction

////////////////////////////////////

Q82 Exercise

=====

////////////////////////////////////

=====

////////////////////////////////////

=====

```
import java.util.*;
class Example{
    public static void main(String []args){
        Scanner input = new Scanner(System.in);
        double x;
        System.out.print("Input number: ");
        x = input.nextDouble();

        System.out.println("x : " + x);
    }
}
```

//

Q85 - I String variable

=====

```
class Example{
    public static void main(String []args){
        String name;
        name = "dinuka";

        System.out.println("My Name is: " + name);
    }
}
```

//

Q86 String Input

=====

```
import java.util.*;
class Example{
    public static void main(String []args){
```

```
E:\ICM120_ICD121>java Example
Enter your name: Dinuka
Your Name is: Dinuka

E:\ICM120_ICD121>java Example
Enter your name: Dinuka Kodithuwakku
Your Name is: Dinuka
```

```

////////////////////////////////////////////////////////////////////////////////////////////////
Q87 input.nextLine();
=====
import java.util.*;
class Example{
    public static void main(String []args){
        Scanner input = new Scanner(System.in);
        String name;
        System.out.print("Enter your name: ");
        name = input.nextLine();
    }
}

```

```
        System.out.println("Your Name is: " + name);
    }
}
```

```
E:\ICM120_ICD121>java Example
Enter your name: Dinuka Kodithuwakku
Your Name is: Dinuka Kodithuwakku
```

//

Q88

```
import java.util.*;
class Example{
    public static void main(String []args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your NIC: ");
        String nic = input.next();

        String name;
        System.out.print("Enter your name: ");
        name = input.nextLine();

        System.out.println("Your Name is: " + name);
    }
}
```

```
E:\ICM120_ICD121>java Example
Enter your NIC: 564651656V
Enter your name: Your Name is:
```

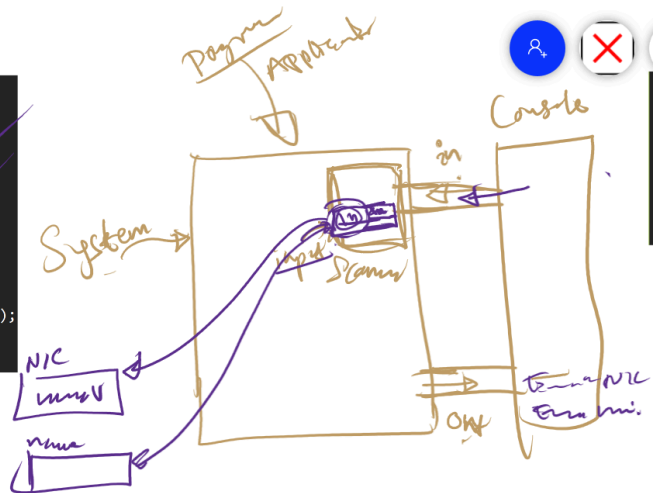
Q89

```
import java.util.*;
class Example{
    public static void main(String []args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your NIC: ");
        String nic = input.next();

        String name;
        System.out.print("Enter your name: ");
        name = input.nextLine();

        System.out.println("Your Name is: " + name);
    }
}
```



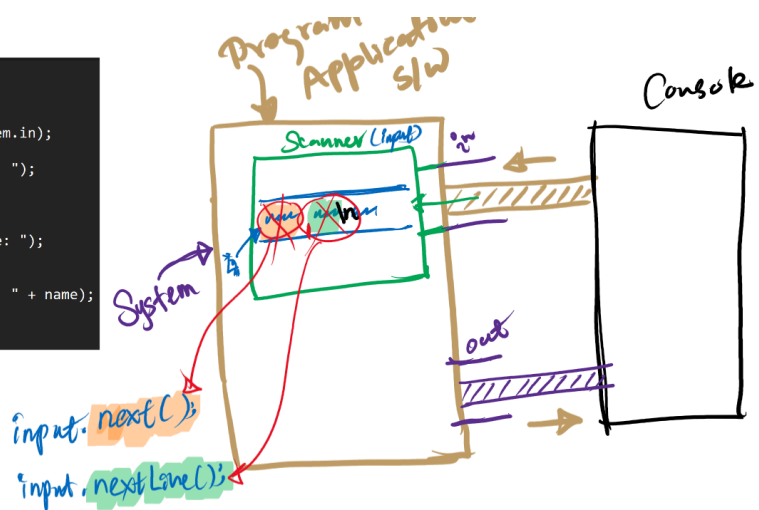
input.nextLine();

```
import java.util.*;
class Example{
    public static void main(String []args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your NIC: ");
        String nic = input.next();

        String name;
        System.out.print("Enter your name: ");
        name = input.nextLine();

        System.out.println("Your Name is: " + name);
    }
}
```

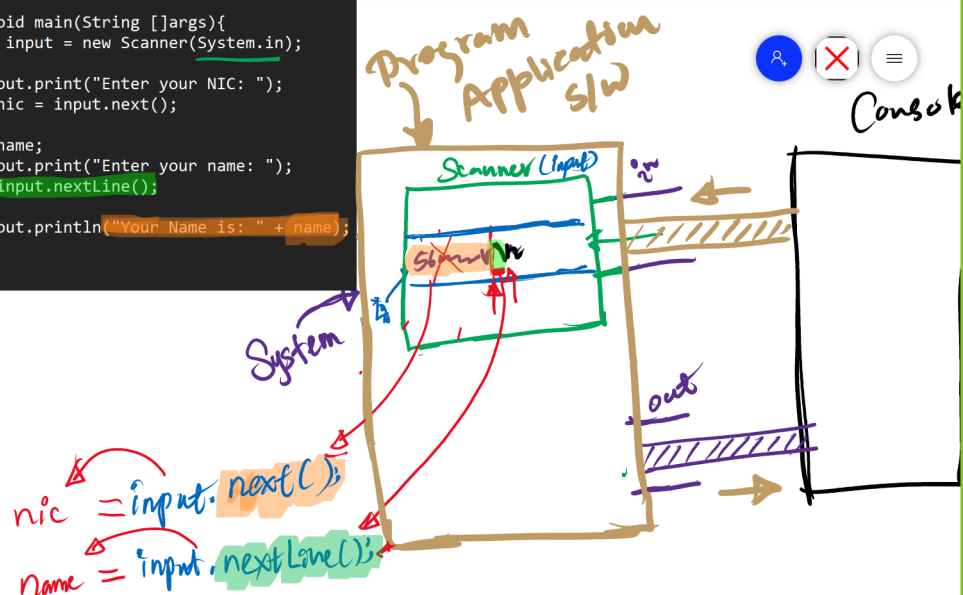


```
class Example{
    public static void main(String []args){
        Scanner input = new Scanner(System.in);

        System.out.print("Enter your NIC: ");
        String nic = input.next();

        String name;
        System.out.print("Enter your name: ");
        name = input.nextLine();

        System.out.println("Your Name is: " + name);
    }
}
```



Q90 From 88

```
import java.util.*;
```

```
class Example{
```

```
public static void main(String []args){
```

```
Scanner input = new Scanner(System.in);
```

```
System.out.print("Enter your NIC: ");
```

```
String nic = input.next();
```

```
input.nextLine(); // read unnecessary new line
```

String name;

```
System.out.print("Enter your name: ");
```

```
name = input.nextLine();
```

```
System.out.println("Your Name is: " + name);
```

}

}

```
E:\ICM120_ICD121>java Example
```

Enter your NIC: 892379472v

```
Enter your name: Dinuka Kodithuwakku
```

Your Name is: Dinuka Kodithuwakku

Variable Naming Rules & Conventions

Q91

```
import java.util.*;
class Example{
    public static void main(String []args){
        int x;
        int name;
        int x1;
        //int 1x;
        int ab_;
        int __;
        int _x;
        // int _; // Java 9 - 21 versions Error, 8- OK |
```

22+ Ok.

```
        int ab$;
        int x#;
    }
}
```

Q92



- special characters
- (1) a-z A-Z 0-9 \$ _ only
- (2) first letter → character (a-z A-Z)
- (3) Can be any length
- (4) Case Sensitive
- (5) ~~Cannot~~ use keywords or ~~reserved~~ words
- (6) Variable name first letter → lower case
- (7) method " " " "
- (8) Class name first letter → upper case
- (9) Camel Case / Snake Case
- Rules
- int a; ✓
int A; ✓
- converting

Q93

```
import java.util.*;
class Example{
    public static void main(String []args){
        int x;
        int name;
        int x1;
        //int 1x;
        int ab_;
        int __;
        int _x;
        // int _; // Java 9 - 21 versions Error, 8- OK |
```

22+ Ok.

```
int ab$;
//int x#;
int $ab;
```

```
int name;  
int Name;  
int my name; // invalid  
int myName;  
int my_name;  
int int; // Error  
int class; // Error  
int main;  
int true; // Error  
}  
}
```

////////////////////////////////////
Q94

Contents of Source Code

- (1) Keywords & reserve words
- (2) identifiers , literals
- (3) Java Comments
- (4) Whitespace
- (5) Operators & Separators
+ , - , / , * , % , ? ; : ,

100Dinuka

```

    }
}

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
Q97
import java.util.*;
class Example{
    public static void main(String []args){
        System.out.printf("Hello World!\n");

        int x = 100;
        System.out.println(x); // 100

        //System.out.printf(x); // Compile Error

        String name = "Dinuka";

        System.out.printf("%d : %s", x, name); // 100 :
Dinuka
    }
}

```

```

E:\ICM120_ICD121>java Example
Hello World!
100
100Dinuka
E:\ICM120_ICD121>javac Example.java

E:\ICM120_ICD121>java Example
Hello World!
100
100 : Dinuka
E:\ICM120_ICD121>

```

Q99

Data Representation in Computer Memory

Q100 Positive Integer Numbers

Data Representation in Computer Memory

8	byte	8 bits
	Short	16 bits
	int	32 bits
	long	64 bits
	float	32 bits
	double	64 bits
	char	16 bits
	boolean	1 bit

Diagram illustrating memory access:

DRAM - Random Access Memory

main memory

working memory

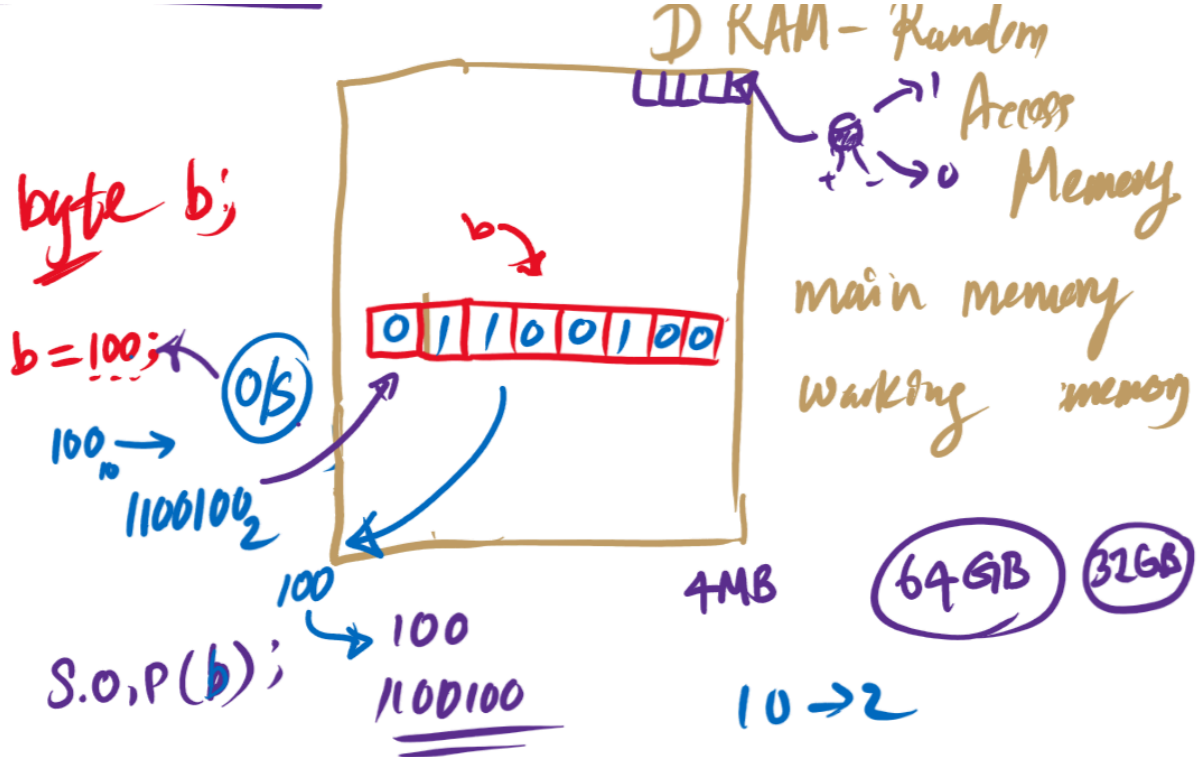
4MB

64GB

32GB

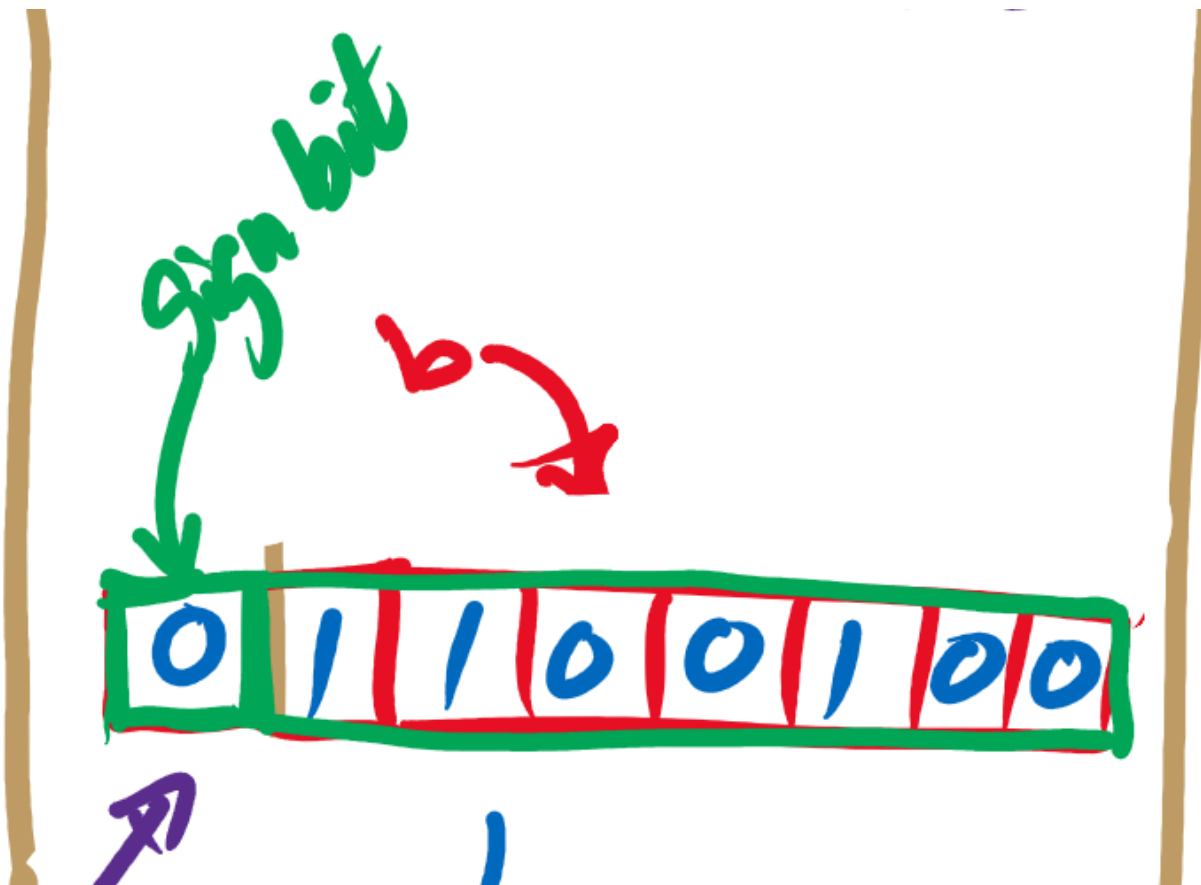
Q101

```
import java.util.*;
class Example{
    public static void main(String []args){
        byte b;
        b = 100;
        System.out.println(b);
    }
}
```



////////////////////////////////////

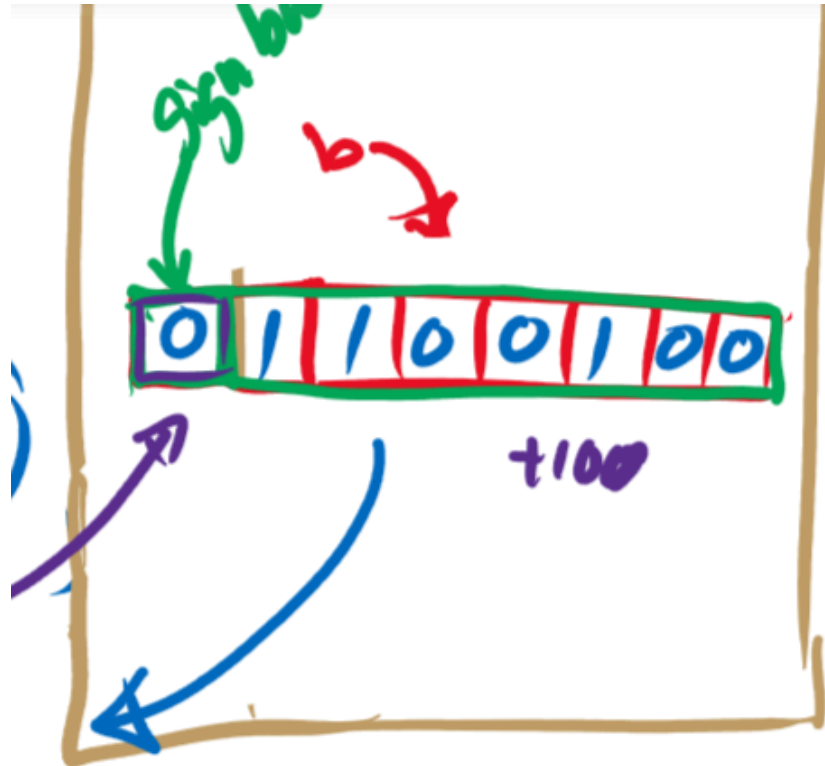
Q102



////////////////////////////////////

Q103 Negative Values

=====



main
working

100

4MB

(64

1100100

-100

10 → 2

111100100

⇒ ?

0 → 00000000

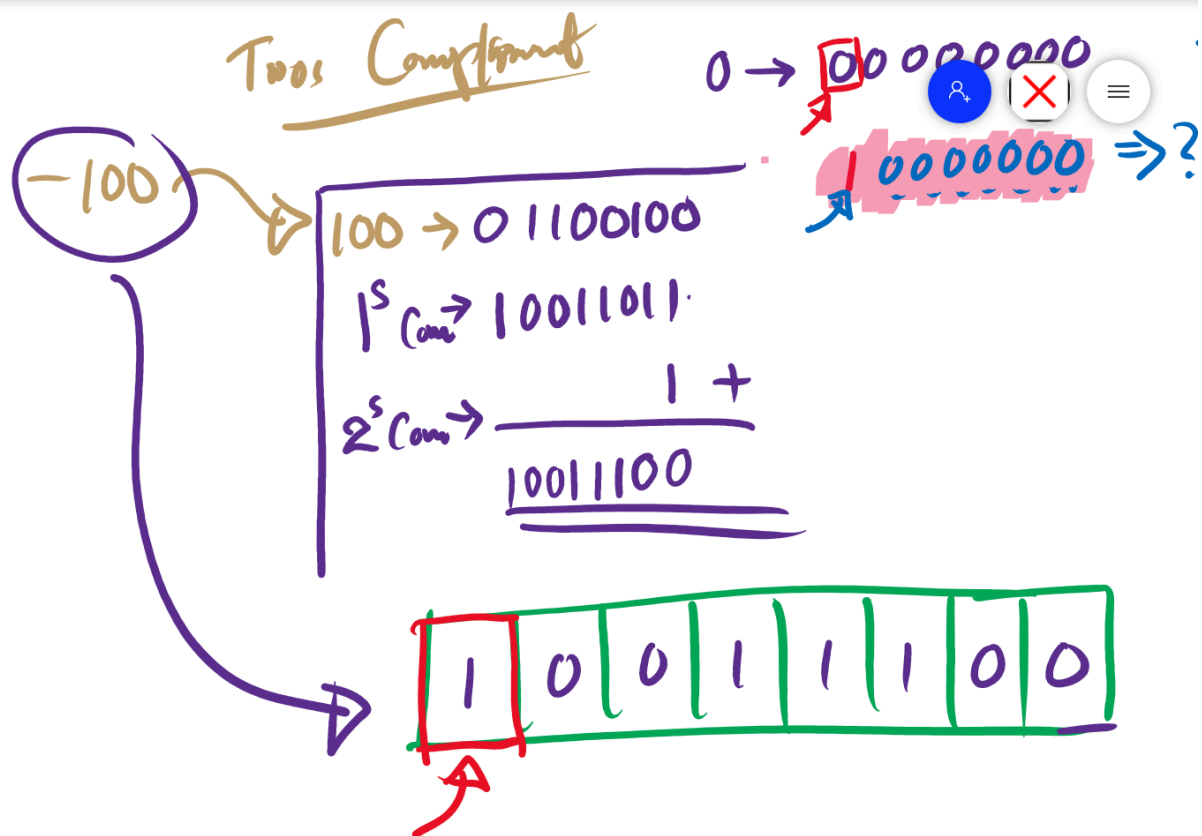
+0?

00000000

⇒ ? -0?

Q104 Negative values

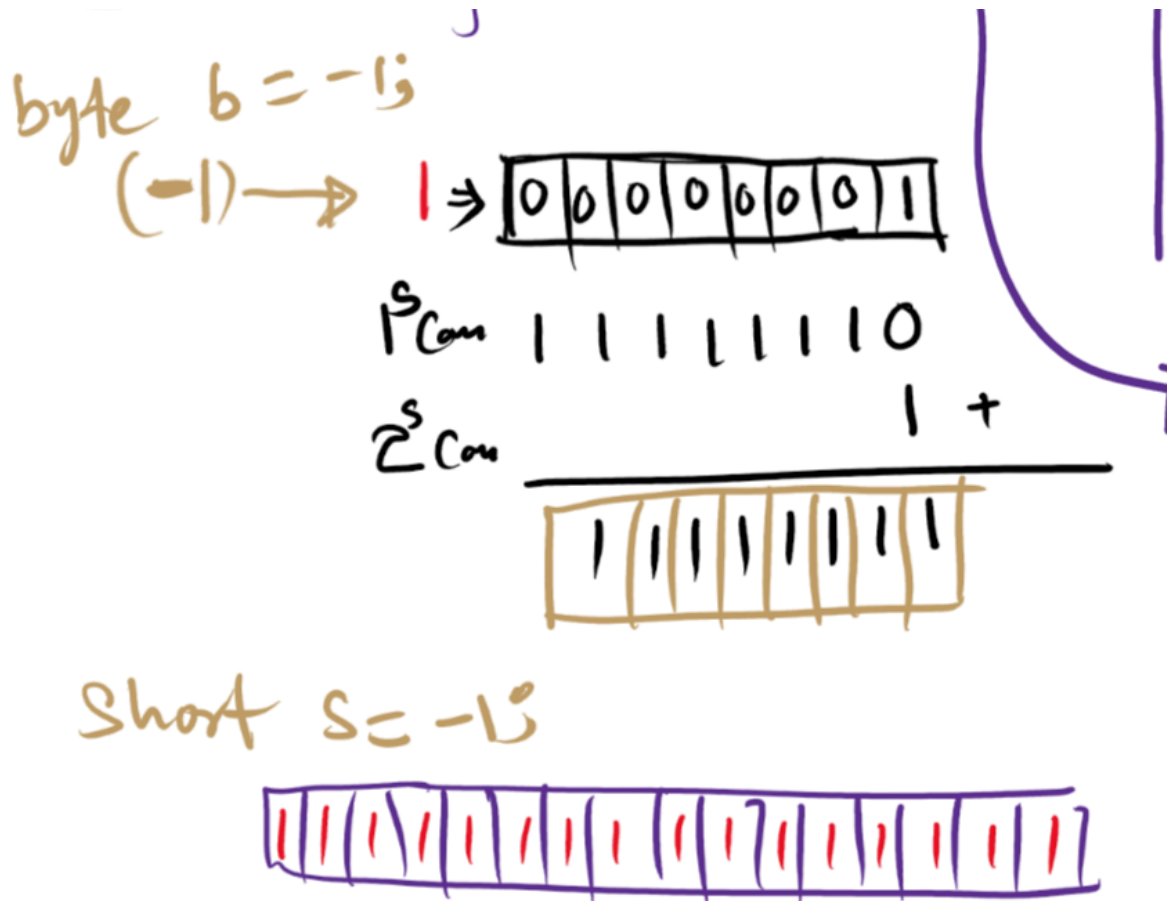
```
import java.util.*;
class Example{
    public static void main(String []args){
        byte b;
        b = -100;
        System.out.println(b);
    }
}
```



Q105 Max

		max	
byte	8	$2^7 - 1$	
Short	16	$2^{15} - 1$	
int	32	$2^{31} - 1$	
long	64	$2^{63} - 1$	
float	32		
double	64		
char	16		
boolean	1		

Q106



////////////////////////////////////

Q107 byte Min value

byte b = -128;

-128 → 128 10000000

1's comp 01111111 2's comp

2's comp

1 +

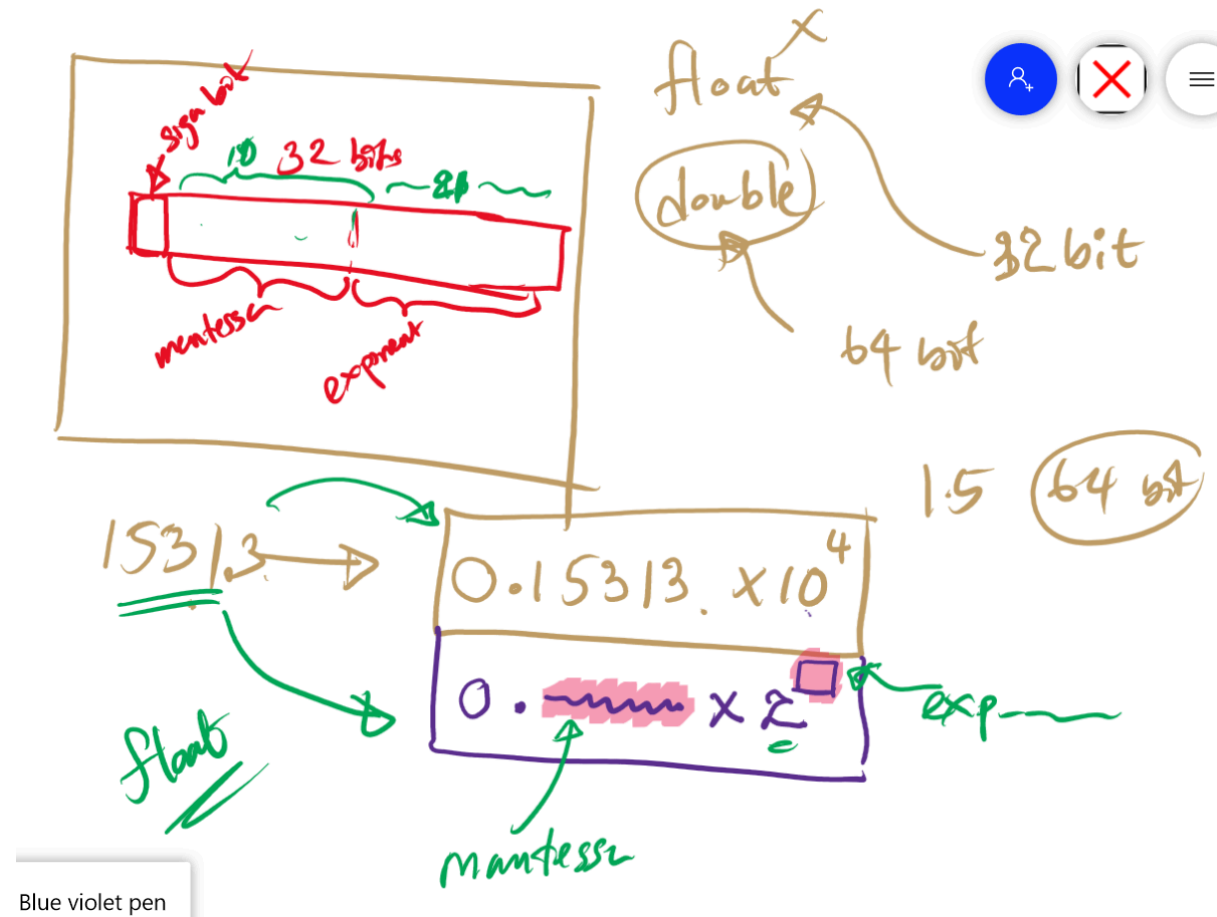


Sign Bit

Blue violet pen

		max	min
byte	8	$2^7 - 1$	-2^7
Short	16	$2^{15} - 1$	-2^{15}
int	32	$2^{31} - 1$	-2^{31}
long	64	$2^{63} - 1$	-2^{63}

Q108 Floating Point Numbers



Q109

```
import java.util.*;
class Example{
    public static void main(String []args){
        byte bMin = Byte.MIN_VALUE;
        byte bMax = Byte.MAX_VALUE;

        short sMin = Short.MIN_VALUE;
        short sMax = Short.MAX_VALUE;

        int xMin = Integer.MIN_VALUE;
        int xMax = Integer.MAX_VALUE;

        long lMin = Long.MIN_VALUE;
        long lMax = Long.MAX_VALUE;
```

```
}  
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q110

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String []args){
```

```
        // long - 64bits, float - 32bits
```

```
        float f = Long.MAX_VALUE; // ??
```

```
        System.out.println("long max value: " +  
Long.MAX_VALUE);
```

```
        System.out.println("f: " + f);
```

```
        long l = f; // Error
```

```
    }  
}
```

Q112 Character Representation

```
import java.util.*;
class Example{
    public static void main(String []args){
        char ch = 65;

        System.out.println(ch);
    }
}
```

Character 16 bits

'A' ASCII →

 → Integer
 value
 65



////////////////////////////////////
Q113

import java.util.*;

class Example{

public static void main(String []args){

char ch = 'h';

System.out.println(ch);

System.out.println(ch + 0); // ASCii Value of ch

: 'h'

System.out.println('A' + 0);

int x = ch;

System.out.println(x);

}

}

Q114 Boolean Representation

Boolean 1 bit
1 → true
0 → false

Conversion and Casting

Q115

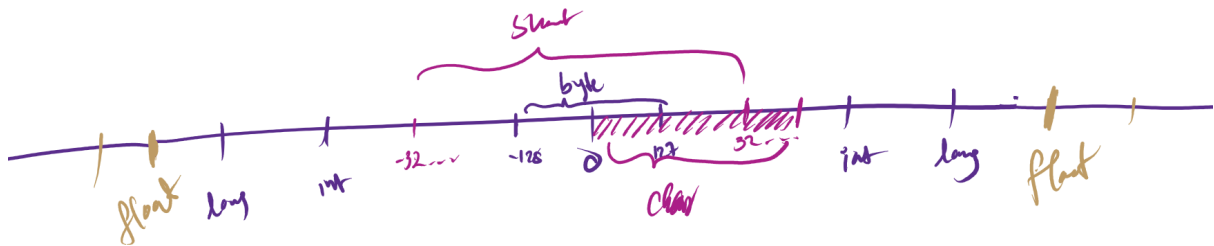
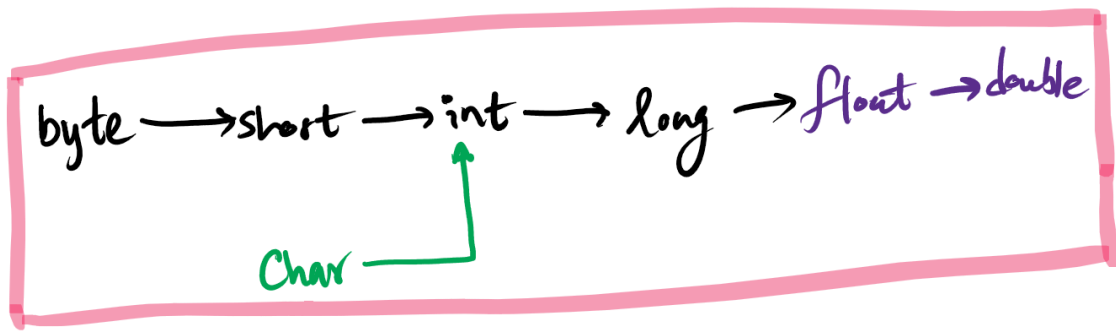
```
class Example{
    public static void main(String []args){
        byte b = 100;
        short s = b;
        int x = s;
        long l = x;

        float f = l;
        double d = f;

        char ch = 'A';

        byte b2 = ch; // error
        short s2 = ch; // error
        int y = ch; // legal
```

}
}



////////////////////////////////////

Q116

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String []args){
```

```
        int x = 100;
```

```
        byte b = (byte) x;
```

```
    }
```

```
}
```


2026-03-15

//

Q118 Exercise

=====

```
class Example{
    public static void main(String args[]){
        int x = 12;
        int y = 15;
        System.out.println(x + " : " + y); // 12 : 15

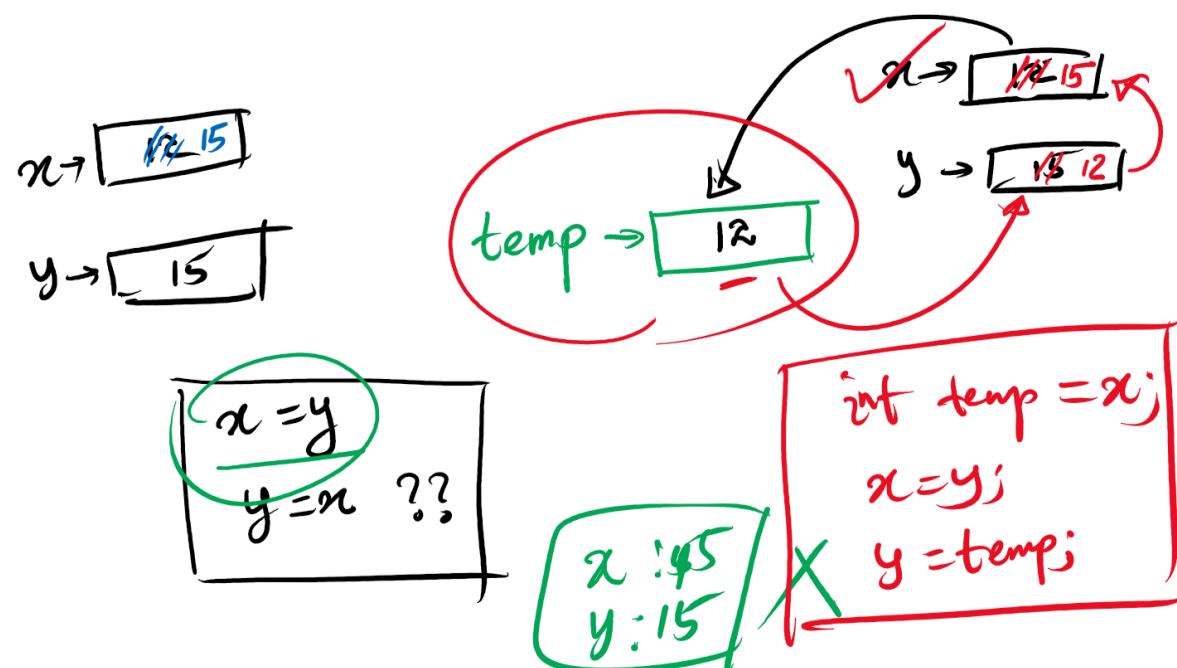
        //
        //
        //

        System.out.println(x + " : " + y); // 15 : 12
    }
}
```

//

Q119 From 118

=====



```

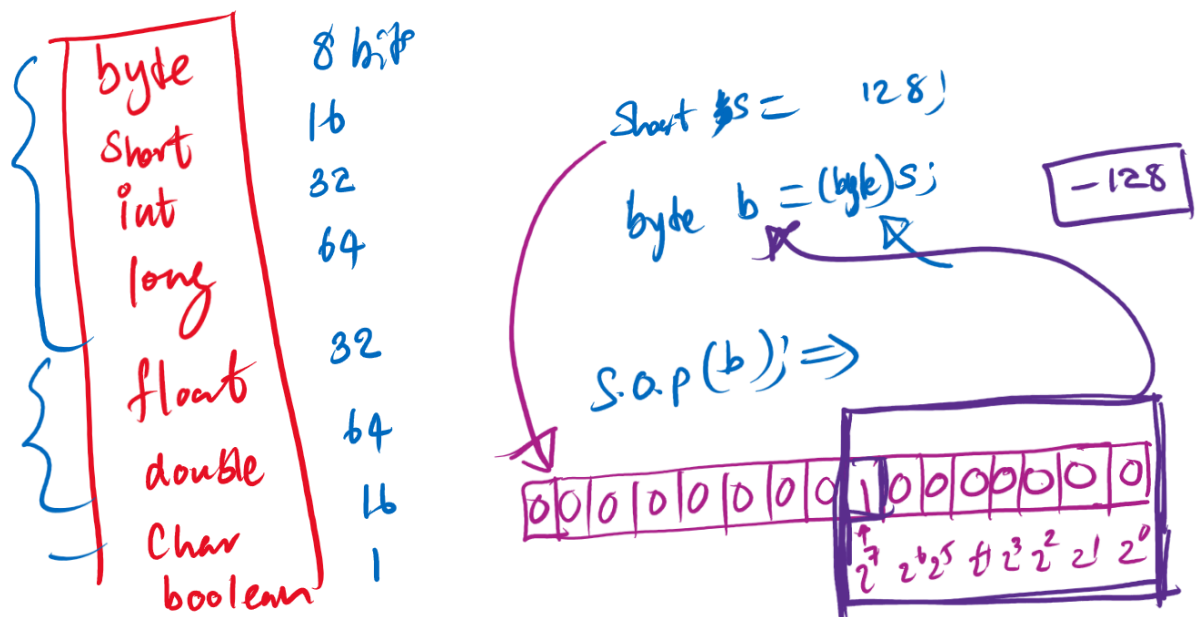
class Example{
    public static void main(String args[]){
        int x = 12;
        int y = 15;
        System.out.println(x + " : " + y); // 12 : 15

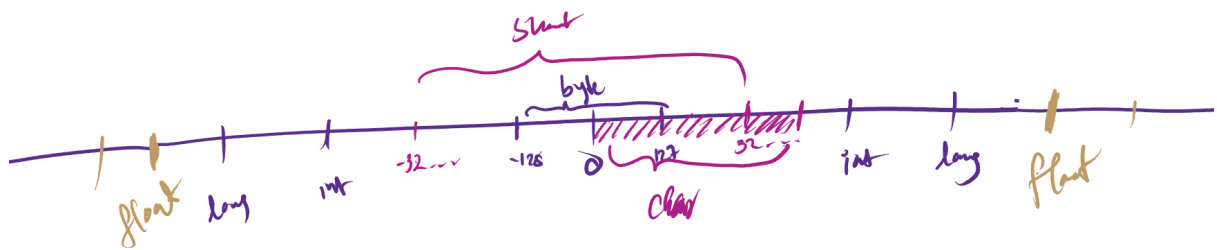
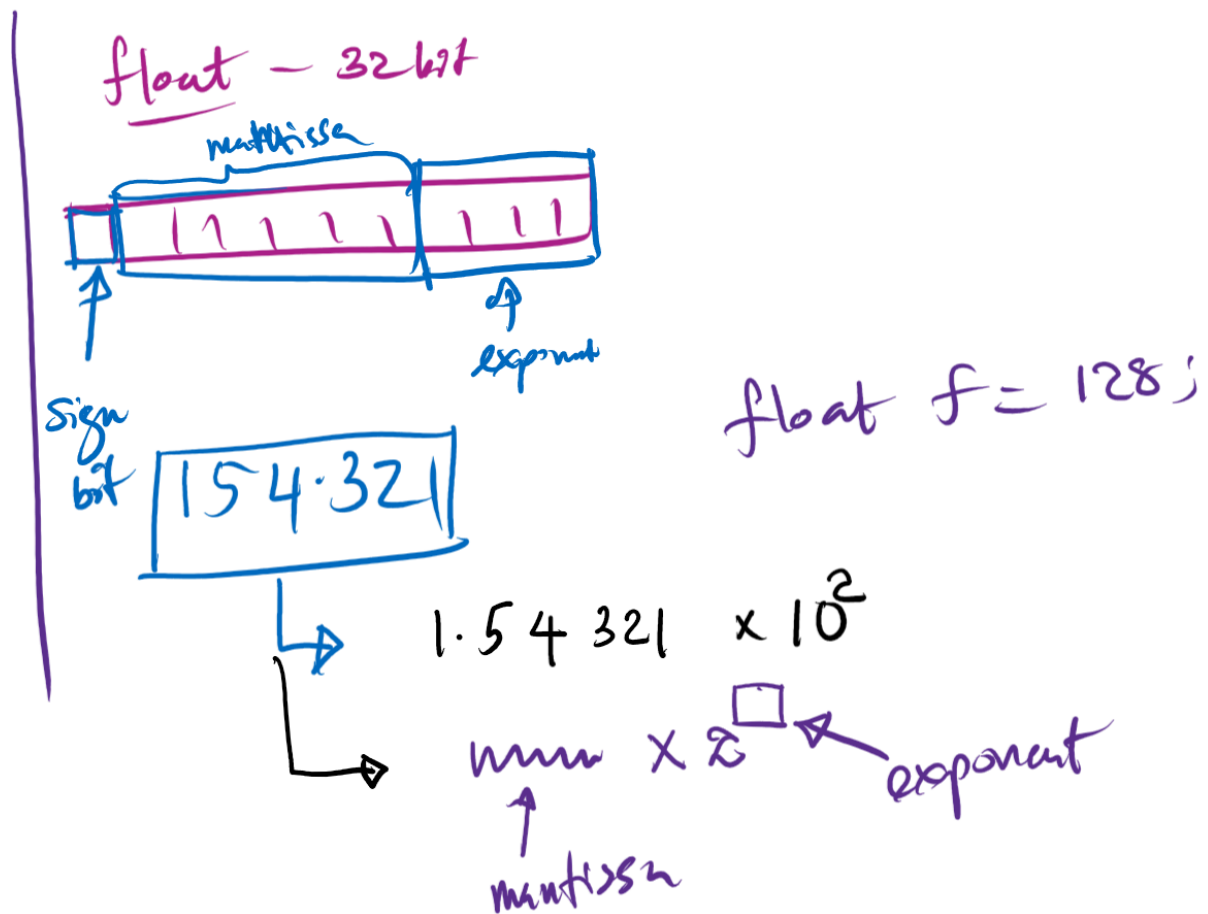
        int temp = x;
        x = y;
        y = temp;
        System.out.println(x + " : " + y); // 15 : 12
    }
}

```

////////////////////////////////////

Q120



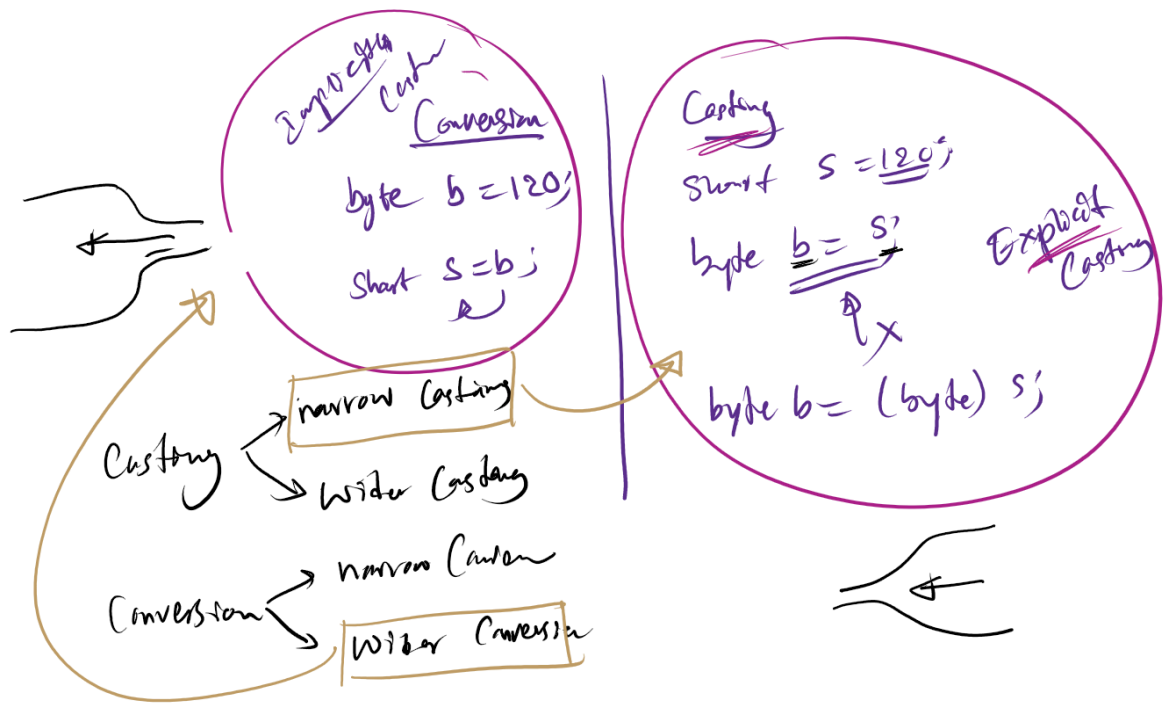


////////////////////////////////////

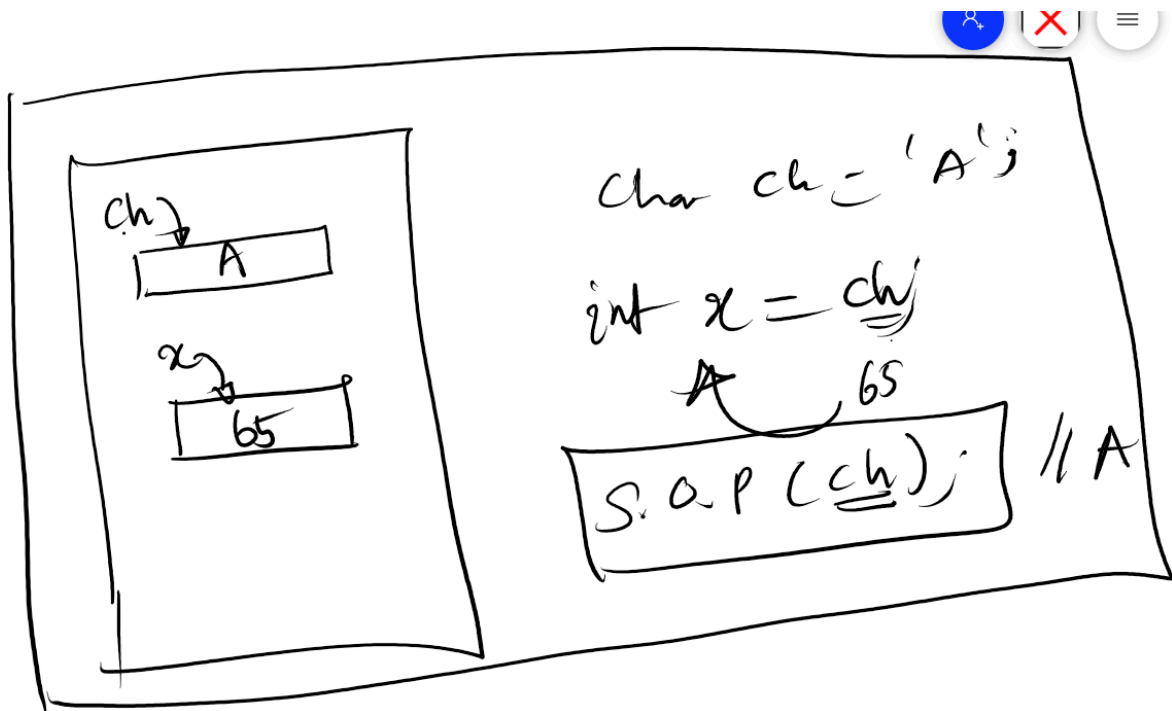
Q121

```
class Example{
    public static void main(String args[]){
        float f = 128; //
        int x = (int) f; //
        System.out.println(x); // 128
    }
}
```

Q122



Q123



```
class Example{
    public static void main(String args[]){
        char ch = 'A';
        int x = ch;

        System.out.println(ch); // A
    }
}
```

```
    }  
}
```

//

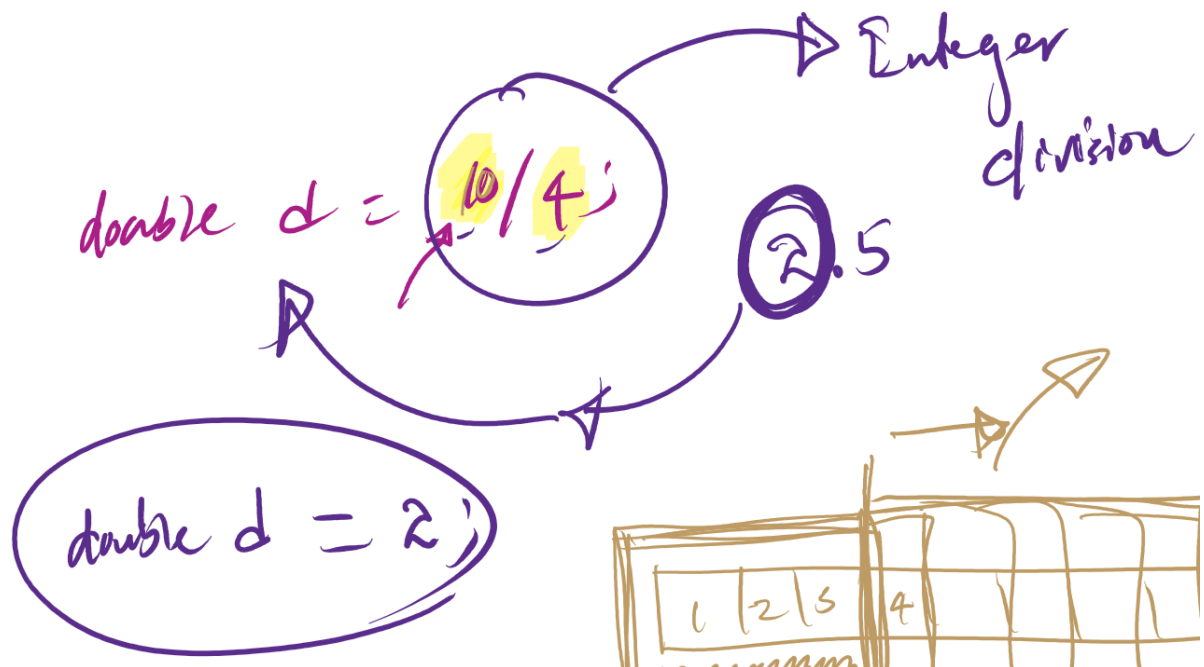
Q124

```
class Example{  
    public static void main(String args[]){  
        char ch = 'A';  
        int x = ch; // Wider Conversion  
  
        System.out.println(ch); // A  
        System.out.println(x); // 65  
    }  
}
```

//

Q125

```
class Example{  
    public static void main(String args[]){  
        double d = 10 / 4;  
        System.out.println(d); // 2.0  
    }  
}
```



Q126

```
class Example{
    public static void main(String args[]){
        double d1 = 10.0 / 4;
        double d2 = 10 / 4.0;

        double d3 = 10.0 / 4.0;

        System.out.println(d1); // 2.5
        System.out.println(d2); // 2.5
        System.out.println(d3); // 2.5
    }
}
```

Q127

```
class Example{
    public static void main(String args[]){
        double d = (double)(10 / 4);

        System.out.println(d); // 2.0
    }
}
```

//

Q128

```
class Example{
    public static void main(String args[]){
        double d1 = (double) 10 / 4;
        double d2 = 10 / (double) 4; // Wider Casting

        System.out.println(d1); // 2.5
        System.out.println(d2); // 2.5
    }
}
```

//

Q129

```
class Example{
    public static void main(String args[]){
        int total = 100;
        int n = 24;

        double avg = (double) total / n; // Wider Casting
        System.out.println(avg); // 4.0
    }
}
```

Arithmetic Operations



Arithmetic Operations

unary

+ plus sign
- minus sign

++

--

binary

+ addition

- subtraction

/ division

* multiplication

% modular / remainder

Q130 Unary Operations

```
class Example{
    public static void main(String args[]){
        int x = 100;
        System.out.println(+x);    //100
        System.out.println(+100); //100

        System.out.println(-x); //-100
        System.out.println(-100); //-100
    }
}
```

//

Q131

```
class Example{
    public static void main(String args[]){
        int x = 10;
        int y = 2;

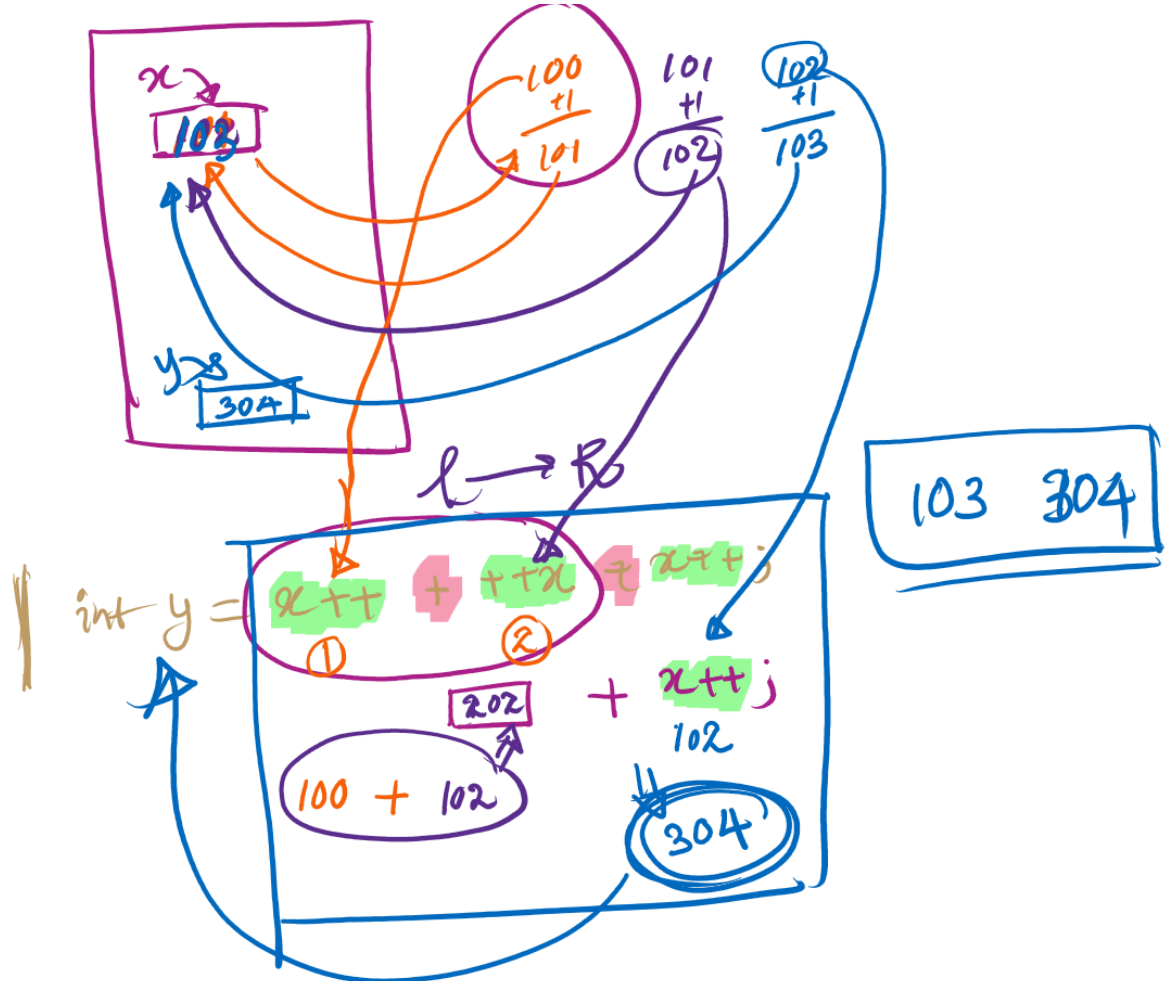
        System.out.println(x+y);//12
        System.out.println(x-y); //8
        System.out.println(x/y);//5
        System.out.println(x*y); //20

        System.out.println(x%y); // 0
    }
}
```

//

Q132 ++ and - -

```
class Example{
    public static void main(String args[]){
        int x = 100;
        x++;
        System.out.println(x); // 101
        x--;
        System.out.println(x); //
    }
}
```

////////////////////////////////////

Operator Precedence

() $l \rightarrow R$

++ --

* / %

+ , -

Q137

Input length: - 18857

mm = 1885751 =

1 Km 885m 75cm 1mm

1000530 mm

1 Km 0 m 53cm 0 mm

Operator

()

++ --

* / %

+ , -

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input length: ");
```

```
        int length = input.nextInt();
```

```
        int km, m, cm, mm;
```

```
        km = length / 1_000_000;
```

```
        length = length % 1_000_000;
```

```
        m = length / 1000;
```

```
        length = length % 1000;
```

Activate

```
cm = length / 10;
```

```
mm = length % 10;
```

```
System.out.println(km +"km, " + m + "m, " + cm  
+ "cm, " + mm + "mm ");
```

```
}
```

```
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q138

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input amount: ");
```

```
        int amount = input.nextInt();
```

```
        int r5000, r1000, r500, r100, r50, r20, r10, r5,  
r2, r1;
```

```
}
```

```
}
```

```
=====
```

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input amount: ");
```

```
        int amount = input.nextInt();
```

```
int r5000, r1000, r500, r100, r50, r20, r10, r5,  
r2, r1;
```

```
    r5000 = amount / 5000;  
    amount = amount % 5000;  
  
    }  
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q139 Operator Assignment

```
=====
```

```
import java.util.*;  
class Example{  
    public static void main(String args[]){  
        int x = 100;  
  
        x = x + 20;  
  
        x += 20; // Operator Assignment  
  
        x -= 10;  
  
        x /= 2;  
  
        x *= 3;  
  
        x %= 2;  
  
        System.out.println("x : " + x);  
    }  
}
```

////////////////////////////////////
Q140

import java.util.*;

class Example{

public static void main(String args[]){

int x = 100;

x = x + 20;

x += 20; // Operator Assignment

x -= 10;

x /= 2;

x *= 3;

x %= 2;

System.out.println("x : " + x);

}

}

Q141

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        short s = 100;
```

```
        byte b = 5;
```

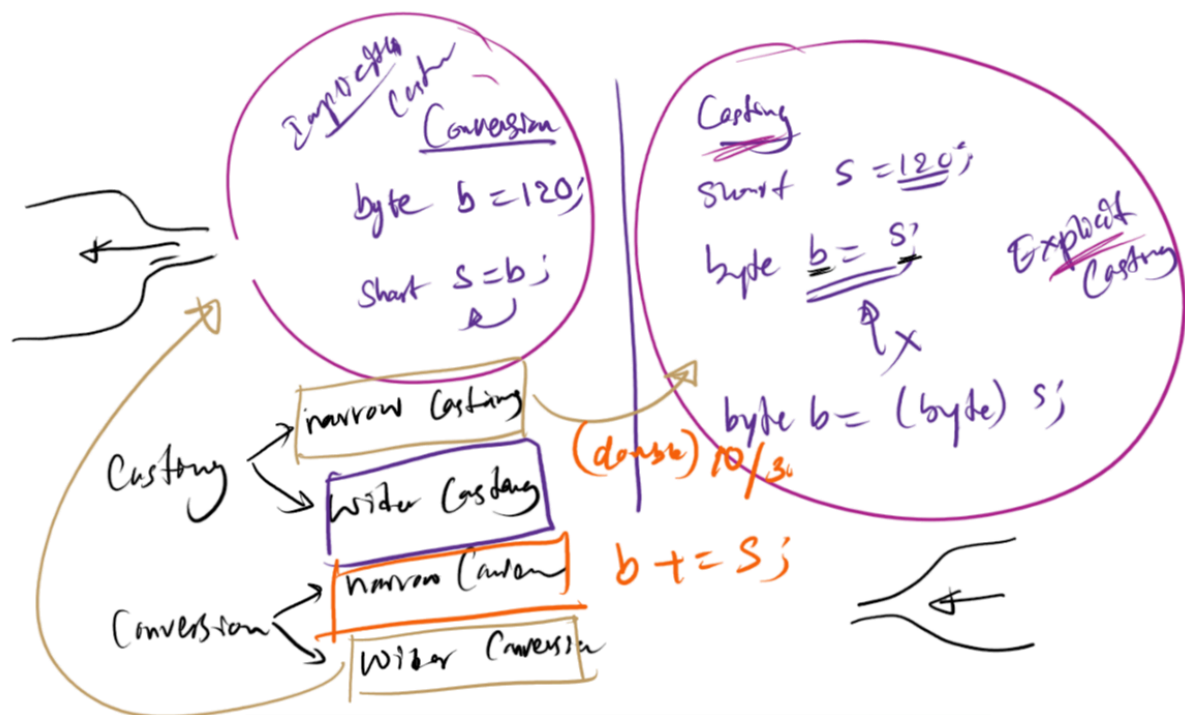
```
        // b = b + s; // Error
```

```
        b += s; // Narrow conversion
```

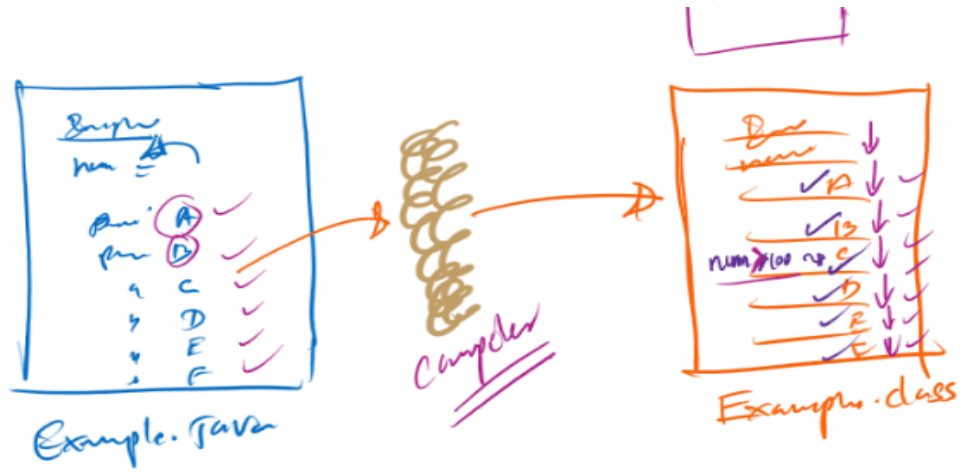
```
        System.out.println("b : " + b);
```

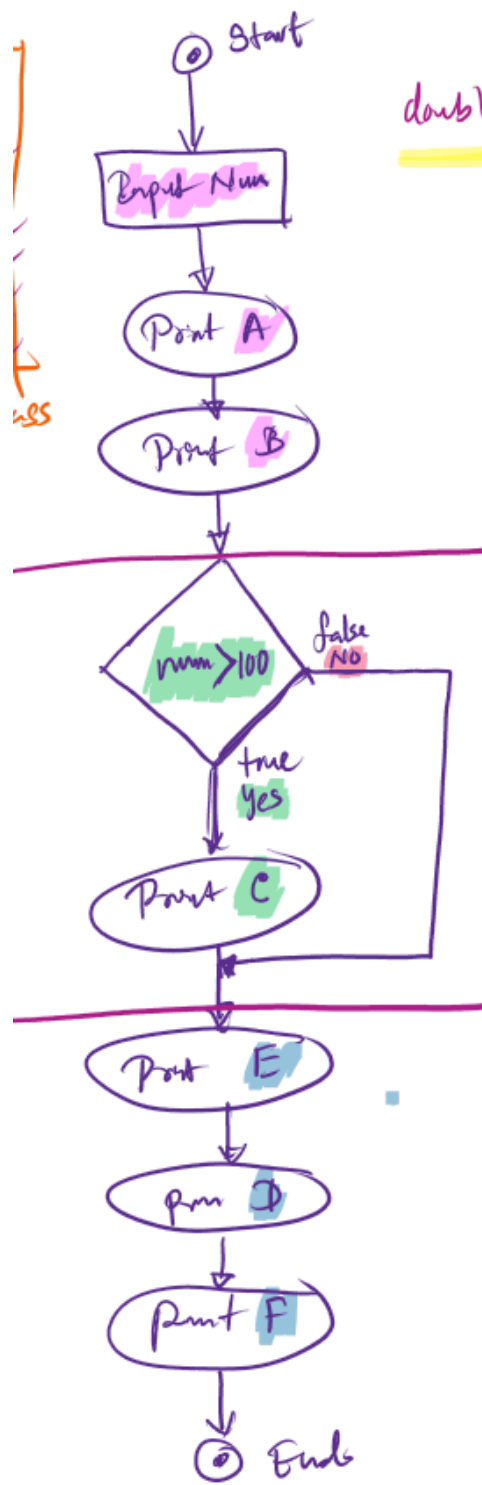
```
    }
```

```
}
```



Java Flow Control "if"





Q142

////////////////////////////////////

Q143

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Number: ");
        int num = input.nextInt();

        System.out.println("A");
        System.out.println("B");
        if(num > 100) System.out.println("C");
        System.out.println("D");
        System.out.println("E");
        System.out.println("F");
    }
}
```

```
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q144

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input Number: ");
```

```
        int num = input.nextInt();
```

```
        System.out.println("A");
```

```
        System.out.println("B");
```

```
        if(num > 100)
```

```
            System.out.println("C");
```

```
        System.out.println("D");
```

```
        System.out.println("E");
```

```
        System.out.println("F");
```

```
    }
```

```
}
```

//

Q145

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input Number: ");
```

```
        int num = input.nextInt(); // 99
```

```
        System.out.println("A");
```

```
        System.out.println("B");
```

```
        if(num > 100)
```

```
            System.out.println("C");
```

```
            System.out.println("D");
```

```
        System.out.println("E");
```

```
        System.out.println("F");
```

```
    }
```

```
}
```

//

Q146

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input Number: ");
```

```
        int num = input.nextInt(); // 99
```

```
        System.out.println("A");
```

```
        System.out.println("B");
```

```
        if(num > 100)
```

```
            {System.out.println("C");
```

```
            System.out.println("D");}
```

```
        System.out.println("E");
```

```
        System.out.println("F");
```

```

    }
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
Q147
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Number: ");
        int num = input.nextInt(); // 99

        System.out.println("A");
        System.out.println("B");
        if(num > 100){
            System.out.println("C");
            System.out.println("D");
        }
        System.out.println("E");
        System.out.println("F");
    }
}

```

Q148

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Number: ");
        int num = input.nextInt(); //

        System.out.println("A");
        System.out.println("B");
        if(num > 100) {
            System.out.println("C");
        }

        if(num < 100) {
            System.out.println("D");
            System.out.println("E");
        }
        System.out.println("F");
    }
}
```

//

Q149

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input Number: ");
```

```
        int num = input.nextInt(); // 99
```

```
        System.out.println("A");
```

```
        System.out.println("B");
```

```
        if(num > 100);
```

```
            System.out.println("C");
```

```
        System.out.println("D");
```

```
        System.out.println("E");
```

```
        System.out.println("F");
```

```
    }
```

```
}
```

//

Q150

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input Salary: ");
```

```
        int salary = input.nextInt();
```

```
        System.out.println("New Salary: " + ...);
```

```
    }
```

```
}
```

////////////////////////////////////

Q151

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Salary: ");
        int salary = input.nextInt();

        if(salary < 100_000) { // if block
            salary += 5000;
        }

        System.out.println("New Salary: " + salary);
    }
}
```

////////////////////////////////////

Q152

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Average Marks: ");
        double avg = input.nextDouble();

        if(avg > 50) System.out.println("Pass");

        if(avg < 50) System.out.println("Fail");
    }
}
```

//

Q153

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input Average Marks: ");
```

```
        double avg = input.nextDouble();
```

```
        if(avg > 50) {
```

```
            System.out.println("Pass");
```

```
        }
```

```
        if(avg < 50) {
```

```
            System.out.println("Fail");
```

```
        }
```

```
    }
```

```
}
```

Java Flow Control "if/else"

Q154

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Average Marks: ");
        double avg = input.nextDouble();

        if(avg > 50) {
            System.out.println("Pass");
        } else {
            System.out.println("Fail");
        }
    }
}
```

Q155 Exercise I

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Salary: ");
        double salary = input.nextDouble();

        //less than 100000 -> 10% increment
        //greater than 100000 -> 5% increment

        System.out.println("New Salary: " + salary);
    }
}
```

////////////////////////////////////

Q156 From Q155

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Salary: ");
        double salary = input.nextDouble();

        if(salary < 100_000) {
            // 10% increment
        } else {
            // 5% increment
        }

        if(salary > 100_000) {
            // 5% increment
        } else {
            // 10% increment
        }

        System.out.println("New Salary: " + salary);
    }
}
```

////////////////////////////////////
Q157 From Q156

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Salary: ");
        double salary = input.nextDouble();

        if(salary > 100_000) {
            // 5% increment
            salary = salary + (salary * 5/100);
        } else {
            // 10% increment
            salary = salary + (salary * 10/100);
        }

        System.out.println("New Salary: " + salary);
    }
}
```

////////////////////////////////////
Q158 From Q157

=====

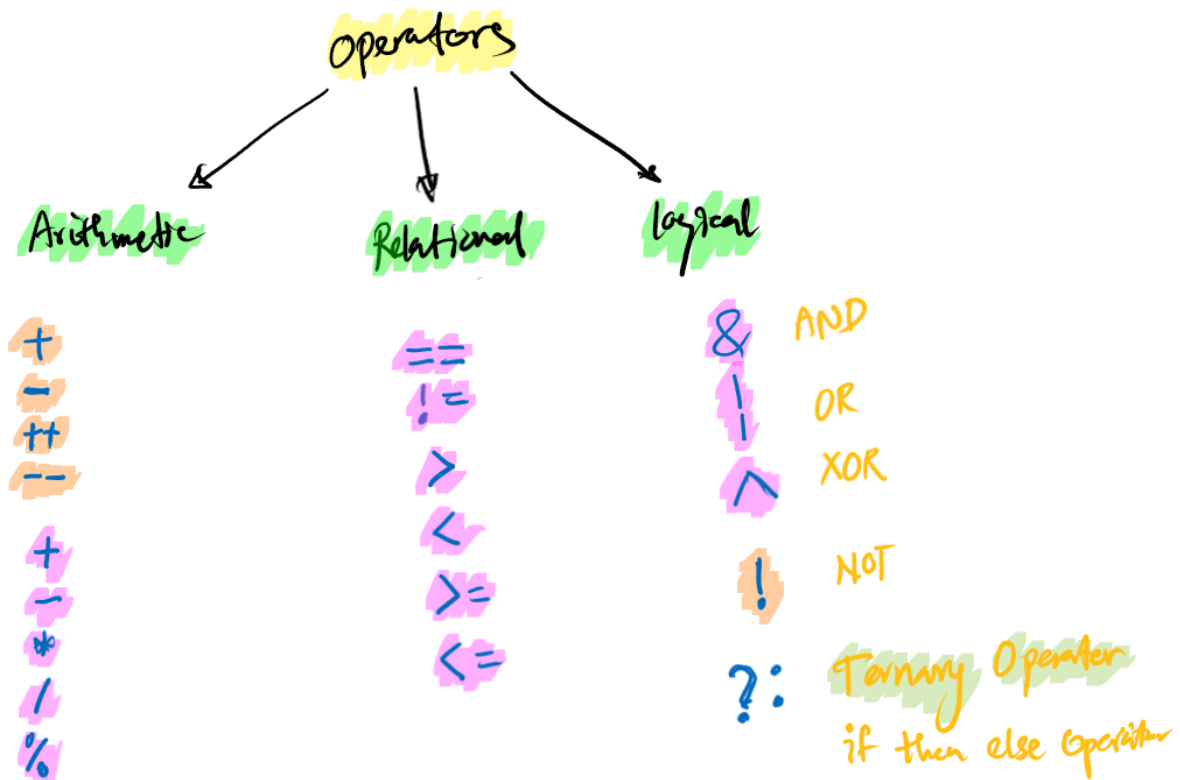
```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Salary: ");
        double salary = input.nextDouble();

        if(salary > 100_000) {
            // 5% increment
            // salary = salary + (salary * 5/100);
            // salary = salary * 105/100;
            salary *= 1.05;
        } else {
            // 10% increment
            // salary = salary + (salary * 10/100);
            salary *= 1.1;
        }

        System.out.println("New Salary: " + salary);
    }
}
```

Relational Operators

Q159



XOR

Input		Output
A	B	X
true	true	false
true	false	true
false	true	true
false	false	false

////////////////////////////////////

Q160

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        boolean b1 = true;
```

```
        boolean b2 = false;
```

```
        System.out.println(100 == 100); // true
```

```
        System.out.println(100 != 100); // false
```

```
        System.out.println(100 != 101); // true
```

```
        System.out.println(!b1); // false
```

```
        System.out.println(!b2); // true
```

```
        // System.out.println(!100); // Illegal
```

```
        System.out.println(b1 == b2); // false
```

```
        System.out.println(100 >= 101); // false
```

```
        // System.out.println(b1 > b2); // Illegal
```

```
        boolean b3 = 100 >= 101;
```

```
        System.out.println(!(100>=101)); // true
```

```
    }
```

```
}
```

Q161

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input a: ");
        int a = input.nextInt();
        System.out.print("Input b: ");
        int b = input.nextInt();
        System.out.print("Input c: ");
        int c = input.nextInt();
        System.out.print("Input d: ");
        int d = input.nextInt();
        System.out.print("Input e: ");
        int e = input.nextInt();

        int max;

        System.out.println("Maximum Value: " + max);
    }
}
```

////////////////////////////////////
Q162 From Q163

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input a: ");
        int a = input.nextInt();
        System.out.print("Input b: ");
        int b = input.nextInt();
        System.out.print("Input c: ");
        int c = input.nextInt();
        System.out.print("Input d: ");
        int d = input.nextInt();
        System.out.print("Input e: ");
        int e = input.nextInt();

        int max = a;

        if(max < b) {
            max = b;
        }
        // checked a, b

        if(max < c) {
            max = c;
        }

        if(max < d) {
            max = d;
        }

        if(max < e) {
```

```
max = e;
```

```
}
```

```
System.out.println("Maximum Value: " + max);
```

```
}
```

```
}
```

```
////////////////////////////////////
```

Q163

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        int x=99, y = 100, z=101;
```

```
        System.out.println(x == y); // 99 == 100 false
```

```
        System.out.println(x++ == y); // 99 == 100 false
```

```
        System.out.println(++x == y); // 101 == 100
```

false

```
        System.out.println(x == y--); // 101 == 100 false
```

```
        System.out.println(x == --y); // 101 == 98 false
```

```
        System.out.println(x++ == --y); // 101 == 97
```

false

```
}
```

```
}
```

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x=99, y = 100, z=101;

        System.out.println(x == y); // 99 == 100 false
        System.out.println(x++ == y); // 99 == 100 false
        System.out.println(++x == y); // 101 == 100 false
        System.out.println(x == y--); // 101 == 100 false
        System.out.println(x == --y); // 101 == 98 false
        System.out.println(x++ == --y); // 101 == 97 false
    }
}
```

x

99

100

101

y

100

$x == y$

$99 == 100 \rightarrow \text{false} \checkmark$

$x++ == y$

$99 == 100 \rightarrow \text{false}$

$++x == y$

$101 == 100 \rightarrow \text{false}$

////////////////////////////////////
Q164 "=" vs "=="

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x=99, y = 100, z=101;

        System.out.println(x=y); // 100
        System.out.println(x=100); // 100

        // System.out.println(x = y == z); // Illegal
        System.out.println((x=y) == z); // false

        // System.out.println((x=y) == z = 100); // Illegal
        System.out.println((x=y) == (z = 100)); // true
    }
}
```

////////////////////////////////////

Q165

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input marks: ");
        double avg = input.nextDouble();

        System.out.println("Grade A"); // 75 - 100
        System.out.println("Grade B"); // 65 - 75
        System.out.println("Grade C"); // 50 - 65
        System.out.println("Grade F"); // <50
    }
}
```

////////////////////////////////////
Q166 // Logical Issue

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input marks: ");
        double avg = input.nextDouble();

        if(avg >= 75) {
            System.out.println("Grade A"); // 75 - 100

        }
        if(avg >= 65){
            System.out.println("Grade B"); // 65 - 75
        }
        if(avg >=50){
            System.out.println("Grade C"); // 50 - 65
        } else {
            System.out.println("Grade F"); // <50
        }
    }
}
```

////////////////////////////////////
Q167 From Q166

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input marks: ");
        double avg = input.nextDouble();

        if(avg >= 75) {
            System.out.println("Grade A"); // 75 - 100

        } else
        if(avg >= 65){
            System.out.println("Grade B"); // 65 - 75
        } else
        if(avg >=50){
            System.out.println("Grade C"); // 50 - 65
        } else {
            System.out.println("Grade F"); // <50
        }
    }
}
```

if/else if

Q168 From Q169

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input marks: ");
        double avg = input.nextDouble();

        if(avg >= 75) {
            System.out.println("Grade A"); // 75 - 100
        } else if(avg >= 65){
            System.out.println("Grade B"); // 65 - 75
        } else if(avg >=50){
            System.out.println("Grade C"); // 50 - 65
        } else {
            System.out.println("Grade F"); // <50
        }
    }
}
```

Conditional Variable Initialization

Q169 Case I

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int y = 100;
        int x;

        if(y == 100) { // Illegal
            x = 200;
        }

        System.out.println(x);
    }
}
```

Q170 Case II

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int y = 100;
        int x;

        if(100 == 100) { // Legal
            x = 200;
        }

        System.out.println(x);
    }
}
```

```
////////////////////////////////////
```

Q171 Case III

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int y = 100;
        int x;

        if(y == 100) {
            x = 200;
        } else {
            x = 300;
        }

        System.out.println(x);
    }
}
```

```
////////////////////////////////////
```

Q172 Case IV

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int y = 100;
        int x;
        if(y == 100) {
            x = 200;
        } else {

        }

        System.out.println(x);
    }
}
```

//

Q173 Case V

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int y = 100;
        int x;
        if(y == 100) {
            x = 200;
        } else if(y > 150) {
            x = 300;
        }
        System.out.println(x);
    }
}
```

//

Q174 Case VI

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int y = 100;
        int x;
        if(y == 100) {
            x = 200;
        } else if(y > 150) {
            x = 300;
        } else {
            x = 400;
        }
        System.out.println(x);
    }
}
```

Logical Operators

logical

&

AND

|

OR

^

XOR

!

NOT

?:

Ternary Operator

if then else operator

Q175

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        boolean b1 = true;
```

```
        boolean b2 = false;
```

```
        System.out.println(b1 & b2); // false
```

```
        System.out.println(b1 & !b2); // true
```

```
        System.out.println(b1 | b2); // true
```

```
        System.out.println(b1 | !b2); // true
```

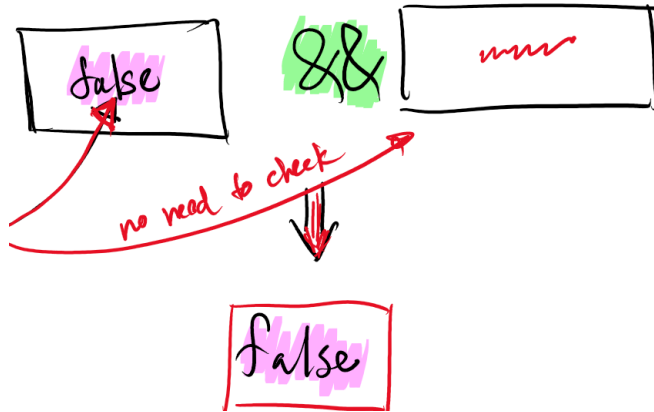
```
        System.out.println(b1 ^ b2); // true
```

```
        System.out.println(b1 ^ !b2); // false
```

```
    }
```

```
}
```

"&" vs "&&"



AND		Output
I ₁	I ₂	
true	true	True
false	true	false
true	false	false
false	false	false

Q176

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        int x = 100, y = 101;
```

```
        System.out.println(x++ == 100 & y-- == 100); //
```

```
false
```

```
        System.out.println("x: " + x + " | y: " + y); // 101
```

```
100
```

```
    }
```

```
}
```

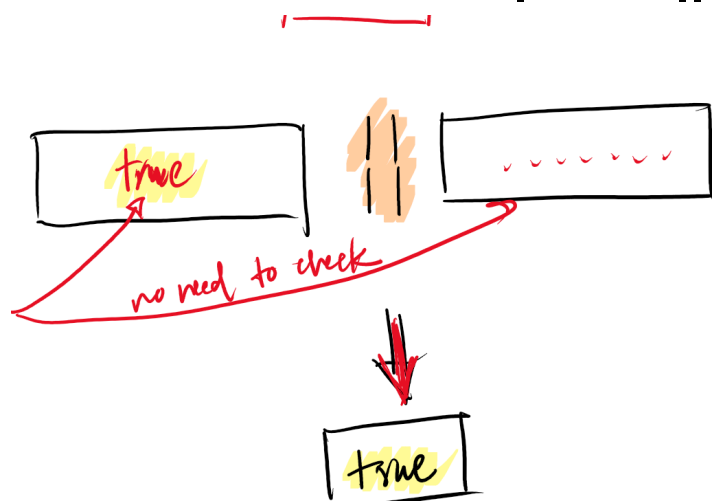
Q177 Short Circuit AND

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100, y = 101;

        //System.out.println(x++ & y--); // false
        System.out.println(x++ == 101 && y-- == 101);
        // false && ... => false

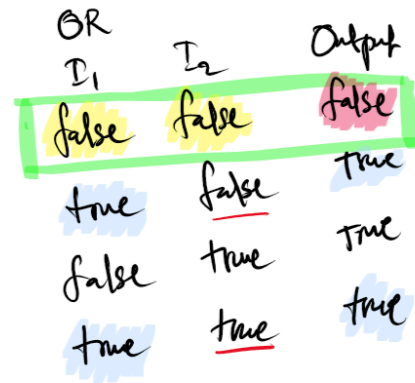
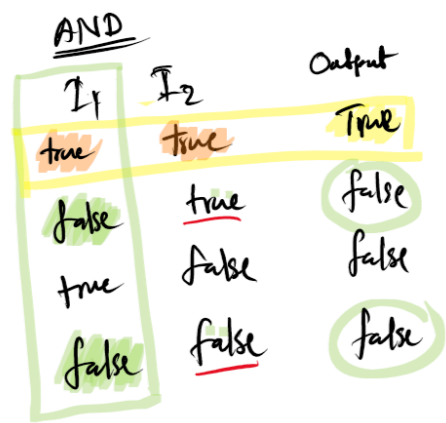
        System.out.println("x: " + x + " | y: " + y); // 101
    }
}
```

"|" vs "&&"



OR			Output
I_1	I_2		
false	false		false
true	false		true
false	true		true
true	true		true

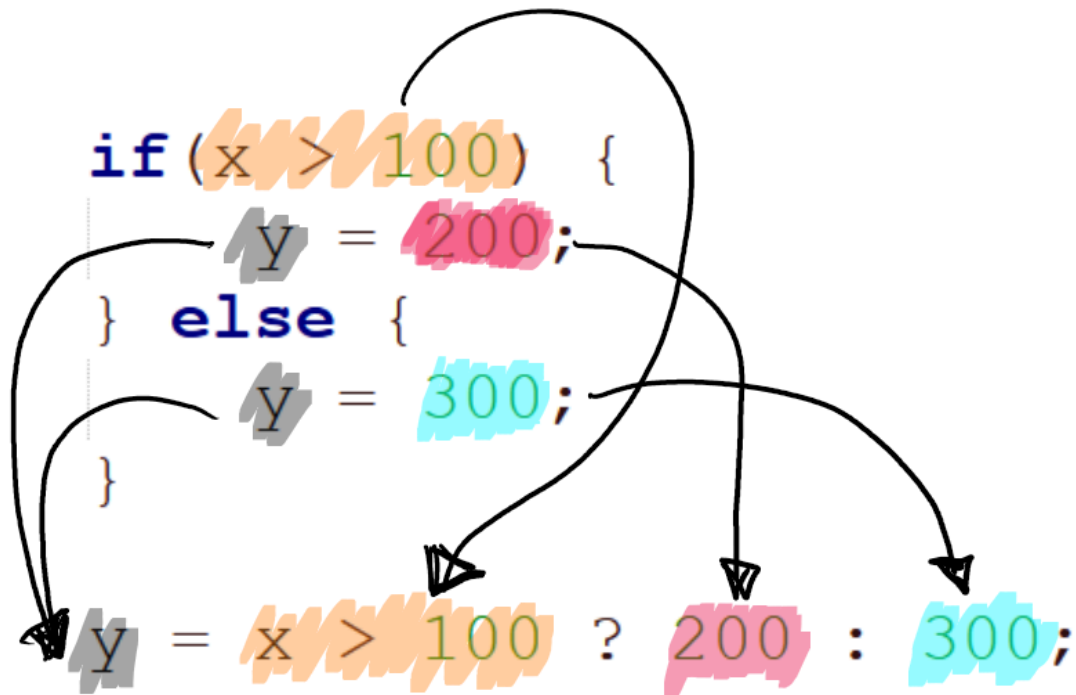
}



}

}

Q183



```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 99;
        int y;

        if(x > 100) {
            y = 200;
        } else {
            y = 300;
        }

        y = x > 100 ? 200 : 300;

        System.out.println("x: " + x + " | y: " + y); // x:  |
        y:
    }
}
```

//

Q184

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        int x = 99;
```

```
        int y;
```

```
        y = x > 100 ? 200 : 300;
```

```
        System.out.println("x: " + x + " | y: " + y); // x: |
```

```
y:
```

```
    }
```

```
}
```

//

Q185

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        int x = 99;
```

```
        int y = x > 100 ? 200 : 300;
```

```
        System.out.println("x: " + x + " | y: " + y); // x: |
```

```
y:
```

```
    }
```

```
}
```

//

Q186

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 99;
        int y, z;
        if(x > 100) { // Cannot use Ternary Operator
            y = 100;
        } else {
            z = 200;
        }
        System.out.println("x: " + x + " | y: " + y); // x: |
y:
    }
}
```

//

Q187 Exercise

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input Salary: ");
        double salary = input.nextDouble();
        if(salary > 100_000) {
            salary *= 1.05;
        } else {
            salary *= 1.1;
        }
        //
        System.out.println("New Salary: " + salary);
    }
}
```

Q188

```
class Example{
```

```
Scanner input = new Scanner(System.in);
```

```
double salary = input.nextDouble();
```

```
salary = salary * 1.05;
```

```
salary = salary * 1.1;
```

```
salary = salary > 100_000 ? salary * 1.05 :
```

```
System.out.println("New Salary: " + salary);
```

}

Q189

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Input marks: ");
```

```
        double avg = input.nextDouble();
```

```
        char ch;
```

```
        if(avg >= 75) {
```

```
            ch = 'A'; // 75 - 100
```

```
        } else if(avg >= 65){
```

```
            ch = 'B'; // 65 - 75
```

```
        } else if(avg >=50){
```

```
            ch = 'C'; // 50 - 65
```

```
        } else {
```

```
            ch = 'F'; // <50
```

```
        }
```

```
        System.out.println("Grade: " + ch);
```

```
    }
```

```
}
```

////////////////////////////////////

Q190 From Q189

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input marks: ");
        double avg = input.nextDouble();
        char ch;
        if(avg >= 75) {
            ch = 'A'; // 75 - 100
        } else
            if(avg >= 65){
                ch = 'B'; // 65 - 75
            } else
                if(avg >=50){
                    ch = 'C'; // 50 - 65
                } else {
                    ch = 'F'; // <50
                }

        ch = avg >= 75 ? 'A' : (avg>=65 ? 'B' :
        (avg>=50 ? 'C' : 'F'));

        System.out.println("Grade: " + ch);
    }
}
```



```
        System.out.println("Grade: " + ch);
    }
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q192

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100;

        if(x == 101 ? false : true) {

        }

        if(!(x == 101)){

        }

        if(x != 101){

        }
    }
}
```

Java Flow Control - Switch Statement

Q193

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input x: ");
        int x = input.nextInt();

        if(x == 100) {
            System.out.println("A");
        } else if(x == 200){
            System.out.println("B");
        } else if(x == 300){
            System.out.println("C");
        } else if(x == 400){
            System.out.println("D");
        } else {
            System.out.println("E");
        }
    }
}
```

Q194 From Q193

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input x: ");
        int x = input.nextInt();
        /*
        if(x == 100) {
            System.out.println("A");
        } else if(x == 200){
            System.out.println("B");
        } else if(x == 300){
            System.out.println("C");
        } else if(x == 400){
            System.out.println("D");
        } else {
            System.out.println("E");
        }
        */
        switch(x){
            case 100: System.out.println("A");
            case 200: System.out.println("B");
            case 300: System.out.println("C");
            case 400: System.out.println("D");
            default: System.out.println("E");
        }
    }
}
```

A
B
C
D
E

////////////////////////////////////

Q195

```
import java.util.*;
```

```
class Example{
```

```
public static void main(String args[]){
```

```
Scanner input = new Scanner(System.in);
```

```
System.out.print("Input x: ");
```

```
int x = input.nextInt();
```

```
switch(x){
```

case 100:

```
System.out.println("A");
```

case 200:

```
System.out.println("B");
```

case 300:

```
System.out.println("C");
```

case 400:

```
System.out.println("D");
```

default:

```
System.out.println("E");
```

}

}

}

////////////////////////////////////
Q196

import java.util.*;

class Example{

public static void main(String args[]){

Scanner input = new Scanner(System.in);

System.out.print("Input x: ");

int x = input.nextInt();

switch(x){

case 100:

System.out.println("A");

break;

case 200:

System.out.println("B");

break;

case 300:

System.out.println("C");

break;

case 400:

System.out.println("D");

break;

default:

System.out.println("E");

break;

}

}

}

Q197

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input x: ");
        int x = input.nextInt();

        switch(x){
            case 100:
                System.out.println("A");
                break;
            case 200:
                System.out.println("B");
                break;
            case 300:
                System.out.println("C");
                break;
            case 400:
                System.out.println("D");
                break;
            default:
                System.out.println("E");
                // break;
        }
    }
}
```

////////////////////////////////////

Q198 Exercise - Using "if" statement

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input month: ");
        int month = input.nextInt();

        System.out.println("31 days");
        System.out.println("30 days");
        System.out.println("28 days");
    }
}
```

////////////////////////////////////

Q199 From Q198

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input month: ");
        int month = input.nextInt();

        if(month == 1 || month == 3 || month == 5 ||
month == 7
                                || month == 8 || month == 10
|| month == 12) {
            System.out.println("31 days");
        } else if (month == 4 || month == 6 || month ==
9 || month == 11) {
            System.out.println("30 days");
        } else if (month == 2) {
            System.out.println("28 days");
        } else {
            System.out.println("Invalid Month");
        }
    }
}
```

////////////////////////////////////

Q200 From Q199 : Using "Switch/Case"

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input month: ");
        int month = input.nextInt();

        switch(month) {
            case 1:
            case 3:
            case 5:
            case 7:
            case 8:
            case 10:
            case 12:
                System.out.println("31 days");
                break;
            case 4:
            case 6:
            case 9:
            case 11:
                System.out.println("30 days");
                break;
            case 2:
                System.out.println("28 days");
                break;
            default:
                System.out.println("Invalid month");
        }
    }
}
```

////////////////////////////////////

Q201

```
import java.util.*;
```

```
class Example{
```

```
    public static void main(String args[]){
```

```
        char ch = 'A';
```

```
        switch(ch) {
```

```
            case 'A':
```

```
                System.out.println("---");
```

```
                break;
```

```
            case 'B':
```

```
                System.out.println(+++);
```

```
        }
```

```
    }
```

```
}
```

////////////////////////////////////

Q202 Case I

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){

        /* insert code */
        switch(x) {

        }

    }
}
```

//byte -

//short -

//int -

//long - Illegal

//float - Illegal

//double - Illegal

//char

//boolean - Illegal

//String

//enum

//

Q203 Case II

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100;

        switch(x) {

        }

    }
}
```

//

Q204 Case III

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100;

        switch(x) {
            case 100:
                System.out.println("---");
                break;
            case 200:
                System.out.println(+++);
                break;
        }
    }
}
```

////////////////////////////////////

Q205 Case IV

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100;

        switch(x) {
            case 100:
                System.out.println("---");
                break;
            default:
                System.out.println("default");
                break;
            case 200:
                System.out.println(+++);
                break;
        }
    }
}
```

////////////////////////////////////

Q206 Case V

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        int x = 100;

        switch(x) {
            case 100:
                System.out.println("---");
                break;
            case 200:
                System.out.println(+++);
                break;
            case 100: // Illegal
                System.out.println("--- 2");
                break;
            default:
                System.out.println("default");
                break;
        }
    }
}
```

////////////////////////////////////

Q207 Case VI

=====

```
class Example{
    public static void main(String args[]){
        int x = 100;

        switch(x) {
            case 100:
                System.out.println("---");
                break;
            case 'A':
                System.out.println(+++);
                break;
            default:
                System.out.println("default");
                break;
        }
    }
}
```

////////////////////////////////////

Q208 Case VII

=====

```
class Example{
    public static void main(String args[]){
        byte x = 100;

        switch(x) {
            case 100:
                System.out.println("---");
                break;
            case 200: // Illegal
                System.out.println(+++);
                break;
            default:
                System.out.println("default");
                break;
        }
    }
}
```

////////////////////////////////////
Q209 Case VIII

=====

```
class Example{
    public static void main(String args[]){
        int x = 100;

        int a = 100, b = 200, c = 300;
        switch(x) {
            case a:
                System.out.println("---a");
                break;
            case b:
                System.out.println(++b);
                break;
            case c:
                System.out.println(++c);
                break;
            default:
                System.out.println("default");
                break;
        }
    }
}
```

////////////////////////////////////
Q210 Final Variables - Constant

=====

```
class Example{
    public static void main(String args[]){
        final int x = 100;

        x = 200; // Illegal
    }
}
```

////////////////////////////////////

Q211 Final Variables - Constant

=====

```
class Example{
    public static void main(String args[]){
        final int x;
        x = 100; // Legal
        x = 200; // Illegal
    }
}
```

////////////////////////////////////

Q212 Case IX

=====

```
class Example{
    public static void main(String args[]){
        int x = 100;

        final int a = 100, b = 200, c = 300;
        switch(x) {
            case a:
                System.out.println("---a");
                break;
            case b:
                System.out.println(++b);
                break;
            case c:
                System.out.println(++c);
                break;
            default:
                System.out.println("default");
                break;
        }
    }
}
```

Q213

Variable Scope and Lifetime

Q214

```
class Example{
    public static void main(String args[]){
        int x = 100;

        int x = 200; // Illegal
    }
}
```

Q215

```
class Example{
    public static void main(String args[]){
        int x = 100;

        {
            int y = 200;

            System.out.println(y);
        }
        System.out.println(y); // Illegal
        //int x = 200;
    }
}
```

Q216

}

//

Q217

```
class Example{
    public static void main(String args[]){
        int x = 100;

        if(x > 100){
            int y = 200;
            System.out.println(y);
        }
        int y = 100;
        //int x = 200;
    }
}
```

//

Q218

```
class Example{
    public static void main(String args[]){
        int x = 100;
        int y = 100;
        {
            System.out.println(y);
            int y = 200; // Illegal
            System.out.println(y);
        }
    }
}
```

////////////////////////////////////
Q219

```
class Example{  
    public static void main(String args[]){  
        int x = 100;  
  
        {  
            int y = 200;  
            System.out.println(y);  
        }  
  
        {  
            int y = 100;  
            System.out.println(y);  
        }  
    }  
}
```

////////////////////////////////////

Variable Initialization

////////////////////////////////////

Q220

```
class Example{  
    public static void main(String args[]){  
        int x;  
  
        {  
            x = 200;  
        }  
  
        System.out.println(x);  
    }  
}
```

////////////////////////////////////
Q221

```
class Example{  
    public static void main(String args[]){  
        int x;  
  
        if(true){  
            x = 200;  
        }  
  
        System.out.println(x);  
    }  
}
```

////////////////////////////////////
Q222

```
class Example{  
    public static void main(String args[]){  
        int x;  
        boolean b = true;  
        if(b){  
            x = 200;  
        }  
  
        System.out.println(x);  
    }  
}
```

Q223

```
class Example{
    public static void main(String args[]){
        int x;
        final boolean b = true;
        if(b){
            x = 200;
        }

        System.out.println(x);
    }
}
```

Java Flow Control "for loop"

Q224

[illegible]

//

Q225

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<100; i++) System.out.println("A");
    }
}
```

//

Q226

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<100; i++)
            System.out.println("A");
        System.out.println("B");
    }
}
```

//

Q227

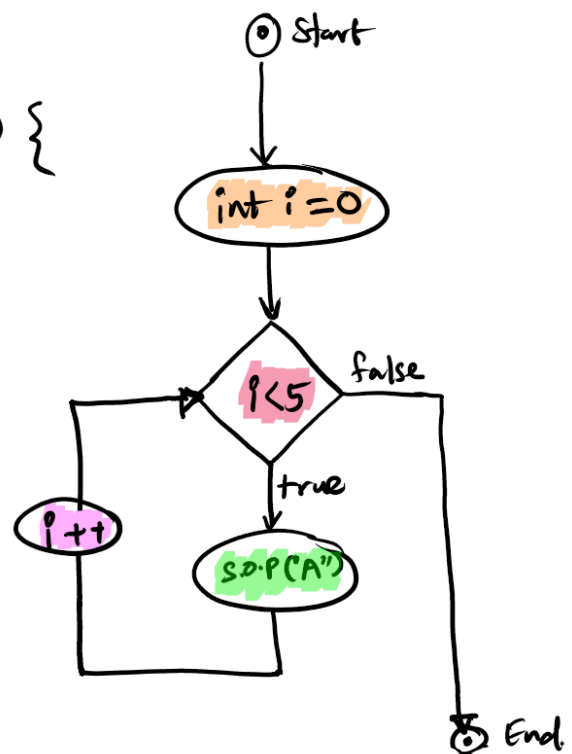
```
class Example{
    public static void main(String args[]){
        for(int i=0; i<100; i++)
            System.out.println("A");
        System.out.println("B");
    }
}
```

Q228

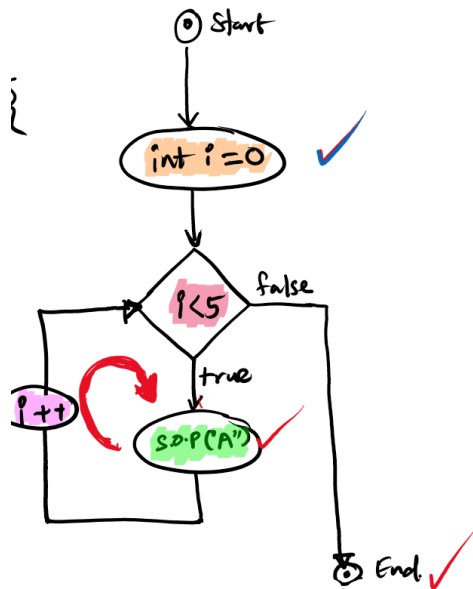
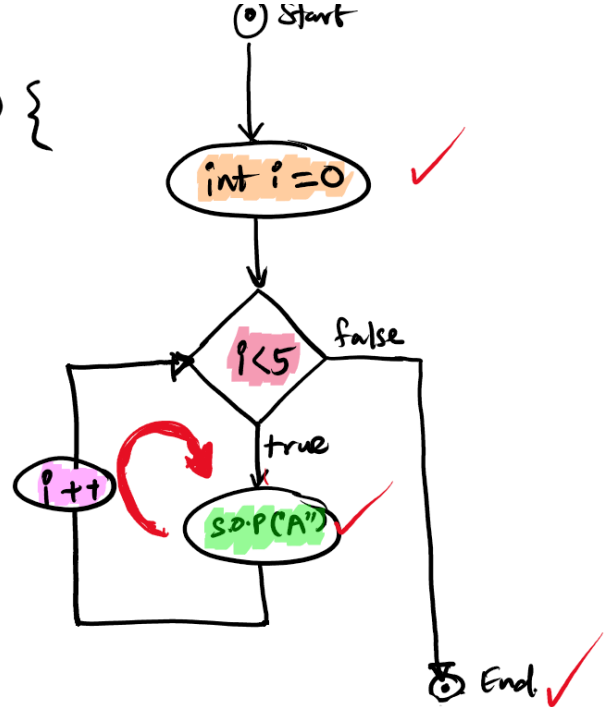
```
class Example{  
    public static void main(String args[]){  
        for(int i=0; i<100; i++){  
            System.out.println("A");  
            System.out.println("B");  
        }  
    }  
}
```

Q229

①
for (int i=0; i<5; i++) {
 S.O.P("A")
}
}



for (int i=0; i<5; i++) {
so.p("A")
 }



i → 0 1 2 3 4 5

⊗ ⊗ ⊗ ⊗ ⊗ ⊗

0 < 5 ✓
 1 < 5 ✓
 2 < 5 ✓
 3 < 5 ✓
 4 < 5 ✓
 5 < 5 ✗

A ✓
 A ✓
 A ✓
 A ✓
 A ✓

//

Q230 Case I

=====

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5; i++){
            System.out.println(i);
        }
    }
}
```

//

Q231 Case II

=====

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5; i++){
            System.out.println(i);

            System.out.println(i);
        }
    }
}
```

//

Q232 Case III

=====

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<=5; i++){
            System.out.println(i);
        }
    }
}
```

Q233 Case IV

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<=5; i+=2){
            System.out.println(i);
        }
    }
}
```

Q234 Case V

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<=5;){
            System.out.println(i);
        }
    }
}
```

Q235

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<=5;){
            System.out.println(i++);
        }
    }
}
```

//

Q236

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<=5;){
            System.out.println(++i);
        }
    }
}
```

//

Q237

```
class Example{
    public static void main(String args[]){
        int i = 0;
        for(;i<=5;){
            System.out.println(i);
        }
    }
}
```

//

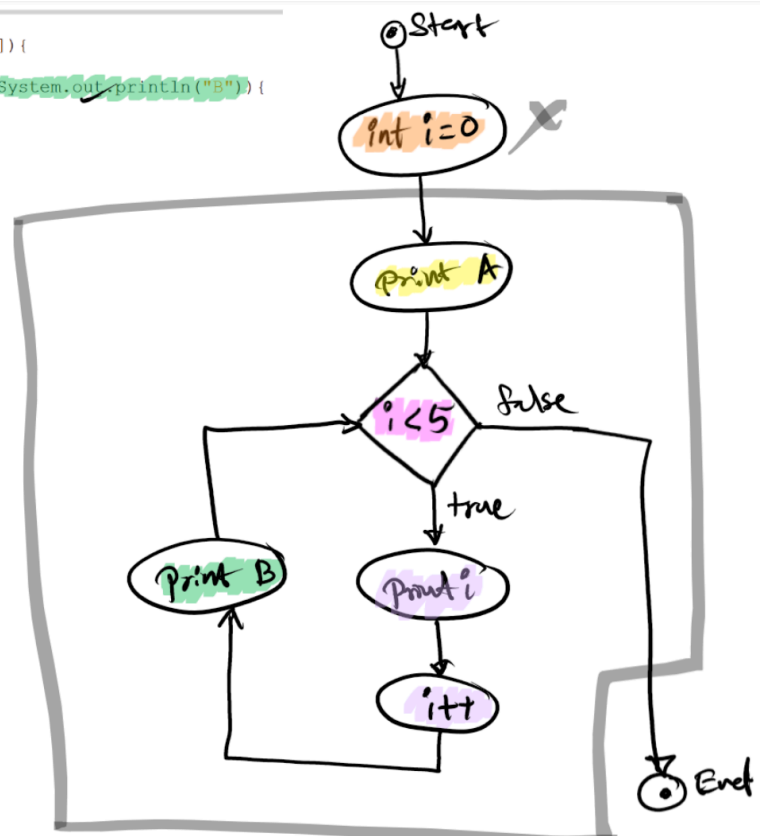
Q238

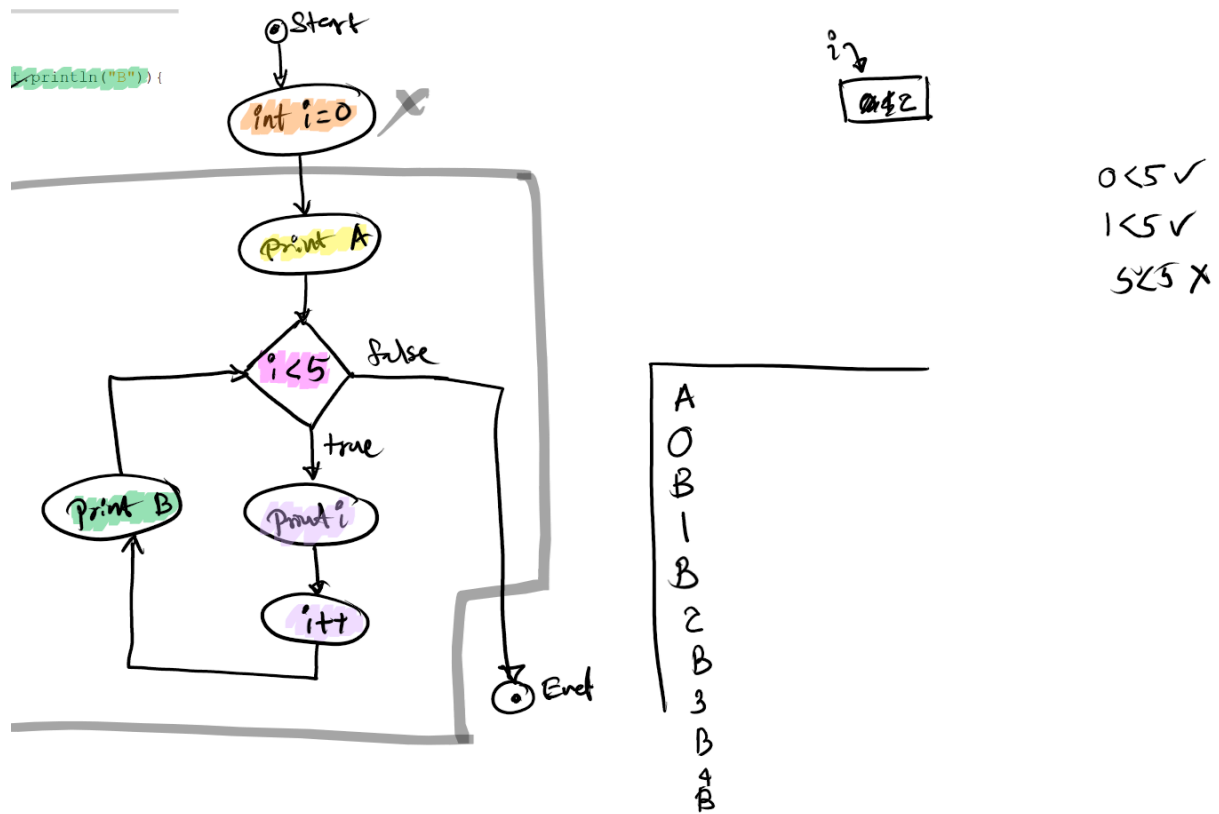
```
class Example{
    public static void main(String args[]){
        int i = 0;
        for(;i<5;){
            System.out.println(i);
            i++;
        }
    }
}
```

Q239

```
class Example{  
    public static void main(String args[]){  
        int i = 0;  
  
        for(System.out.println("A");i<5;System.out.println("B")){  
            System.out.println(i);  
            i++;  
        }  
    }  
}
```

```
class Example{  
    public static void main(String args[]){  
        int i = 0;  
        for(System.out.println("A");i<5;System.out.println("B")){  
            System.out.println(i);  
            i++;  
        }  
    }  
}
```





////////////////////////////////////

Q240

```

class Example{
    public static void main(String args[]){
        for(int i=2;i<5;i++){
            System.out.println(i);
        }
    }
}
  
```

////////////////////////////////////

Q241 Exercise I

=====

```

class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter no of subjects: ");
        int numofSubjects = input.nextInt();
        // Insert code here
    }
}
  
```

////////////////////////////////////

Q242 Exercise II

=====

```
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter num1: ");
        int num1 = input.nextInt();
        System.out.print("Enter num2: ");
        int num2 = input.nextInt();

        //[1, 2, 3, 4, 5]
    }
}
```

////////////////////////////////////

Q243 From 241

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter no of subjects: ");
        int numOfSubjects = input.nextInt();

        int total = 0;
        for(int i=0; i<numOfSubjects; i++){
            System.out.print("Enter marks for
subject" + (i+1) + ": ");
            int marks = input.nextInt();

            total += marks;
        }

        System.out.println("Total: " + total);
    }
}
```

```

        System.out.println("Avg: " +
(double)total/numOfSubjects);
    }
}

```

//

Q244 From 242

=====

```

import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter num1: ");
        int num1 = input.nextInt();
        System.out.print("Enter num2: ");
        int num2 = input.nextInt();

        System.out.print("[");

        if(num1 < num2){
            for(int i=num1; i<=num2; i++){
                System.out.print(i+", ");
            }
        } else {
            for(int i=num2; i<=num1; i++){
                System.out.print(i+", ");
            }
        }
        System.out.print("\b\b]");
    }
}

```

////////////////////////////////////
Q245

```
class Example{  
    public static void main(String args[]){  
        for(int i=0; i<10; i++){  
            if(i%2 == 0) {  
                System.out.println(i);  
            }  
        }  
    }  
}
```

////////////////////////////////////

Continue

////////////////////////////////////

Q246

```
class Example{  
    public static void main(String args[]){  
        for(int i=0; i<10; i++){  
  
            if(i%2 == 1){  
                continue;  
            }  
  
            System.out.println(i);  
        }  
    }  
}
```

```

graph TD
    Start((Start)) --> Init([init i = 0])
    Init --> Cond1{i < 10}
    Cond1 -- false --> End((End))
    Cond1 -- true --> Cond2{i % 2 == 1}
    Cond2 -- true --> Inc((i++))
    Cond2 -- false --> Print([Print i])
    Inc --> Cond1
    Print --> End
  
```

Q247

}

```
////////////////////////////////////
Q248
import java.util.*;
class Example{
    public static void main(String args[]){
        for(int i=0; i<10; i++){
            continue;
            System.out.println(i); // Illegal
        }
    }
}
////////////////////////////////////
```

Break

```
////////////////////////////////////
Q249
class Example{
    public static void main(String args[]){
        for(int i=0;;i++){
            if(i>=5){
                break;
            }
            System.out.println("A");
        }
    }
}
////////////////////////////////////
```

```
Q250
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Enter no of subjects: ");
        int numOfWorks = input.nextInt();
    }
}
```

```

        int total = 0;
        for(int i=0; i<numOfSubjects; i++){
            System.out.print("Enter marks for
subject"+(i+1)+" : ");
            int marks = input.nextInt();
            if(marks<50) {
                break;
            }
            total += marks;
        }

        System.out.println("Total: " + total);
        System.out.println("Avg: " +
(double)total/numOfSubjects);
    }
}

```

//

Q251

```

import java.util.*;
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;){
            System.out.println(i=i++);
        }
    }
}

```


////////////////////////////////////
Q253

```
import java.util.*;
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            if(i > 3) {
                return;
            }
            System.out.println(i);
        }

        System.out.println("End main()");
    }
}
```

////////////////////////////////////

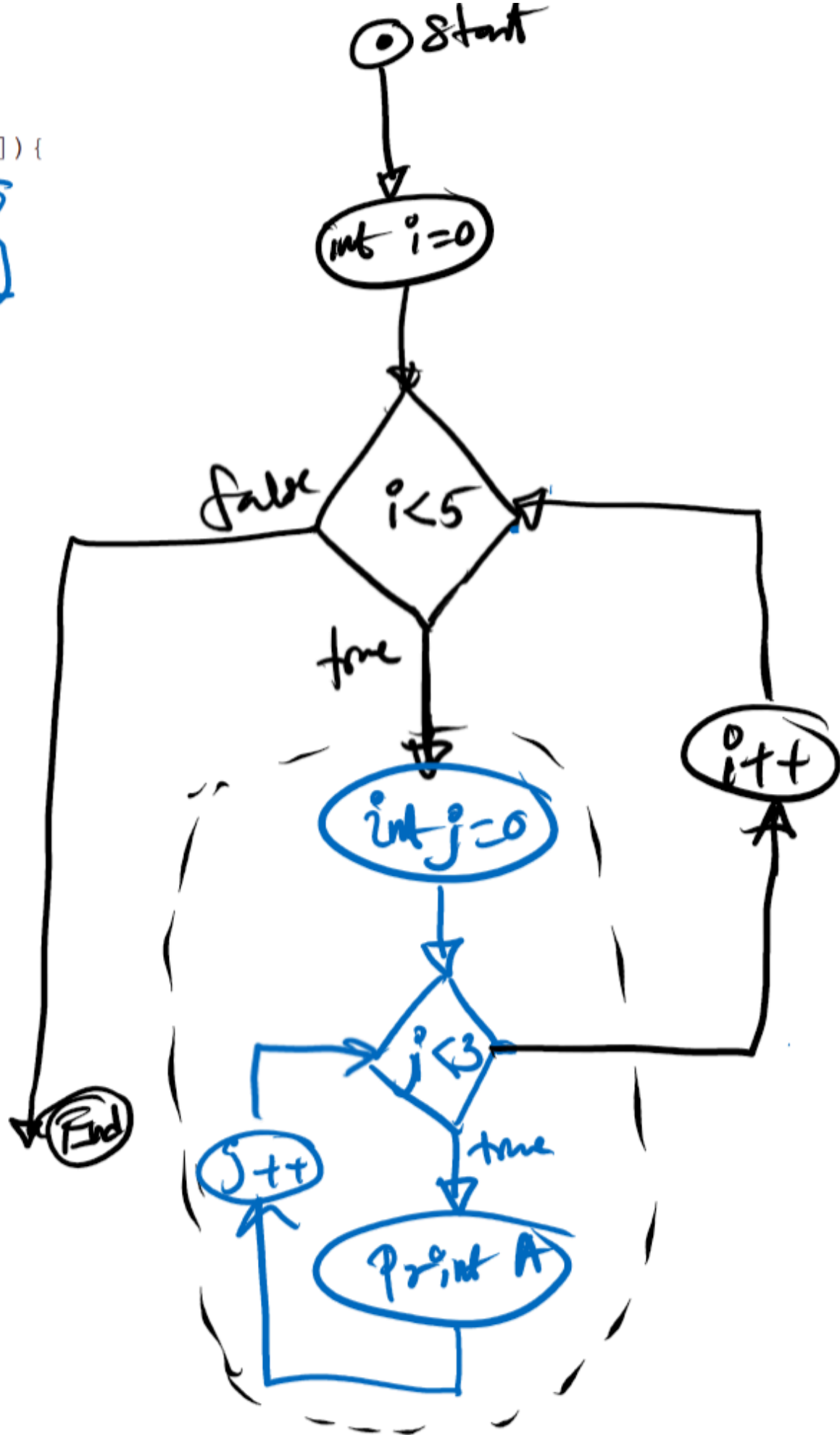
Nested Loops

////////////////////////////////////

Q254

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<3; j++) {
                System.out.println("A");
            }
        }
    }
}
```

[]) {
;
}





```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<3; j++){
                System.out.println(i + " : " + j);
            }
        }
    }
}
```



~~0:0~~
0:1
~~0:2~~

1:0
1:1
1:2

2:0
2:1
2:2

3:0
3:1
3:2

4:0
4:1
4:2

0<5 ✓

0<3 ✓

1<3 ✓

2<3 ✓

~~3<3~~ ✗

1<5 ✓

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<3; j++) {
                System.out.println(i + " : " + j);
            }
        }
    }
}
```

////////////////////////////////////

Q255

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<3; j++) {
```

```

        for(int k=0; k<2; k++) {
            }
        }
    }
}
////////////////////////////////////

```

Q256

```

class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=i; j<3; j++) {
                System.out.println(i + " : " + j);
            }
        }
    }
}

```



```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=i; j<3; j++) {
                System.out.println(i + " : " + j);
            }
        }
    }
}
```

0	1
0	2
1	1
1	2
2	2

Q257

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<i; j++) {
                System.out.println(i + " : " + j);
            }
        }
    }
}
```

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<i; j++) {
                System.out.println(i + " : " + j);
            }
        }
    }
}
```

1:0
2:0
2:1
3:0
3:1
3:2
4:0
4:1
4:2
4:3

?
✓ 0 → 0~~X~~0
✓ 1 → ~~0~~1
✓ 2 → j=0 j<2
✓ 3 → j=0 j<3
0,1,2
✓ 4 → j=0 j<4
0,1,2,3

//

Q258

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<i; j++) {

            }
            System.out.println(i + " : " + j);
        }
    }
}
```

//

Q259

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<i; j++) {

            }
            System.out.println(i + " : " + j); // j - compile
error
        }
    }
}
```

//

Q260

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<10;i++){
            for(int j=0; j<10; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }
    }
}
```

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<i; j++) {

                if(i>3){
                    break;
                }

                System.out.println("Inner Loop");
            }
            System.out.println("Outer Loop");
        }
    }
}
```

```
class Example{
    public static void main(String args[]){
        for(int i=0; i<5;i++){
            for(int j=0; j<i; j++) {

                if(i>3){
                    break;
                }

                System.out.println("Inner Loop");
            }
            System.out.println("Outer Loop");
        }
    }
}
```

0 $\rightarrow$ X Outer Loop
1 $\rightarrow$ $j=0$ $j < 1$ Inner Loop
Out Loop
2 $\rightarrow$ $j=0$ $j < 2$ In
In
Out
3 $\rightarrow$ $j=0$ $j < 3$ In
In
In
Out

4 $\rightarrow$ $j=0$ $j < 4$ Outer Loop

[illegible]

Q263

```
class Example{
    public static void main(String args[]){
        L1:for(int i=0; i<5;i++){
            L2:for(int j=0; j<i; j++) {

                if(i>3){
                    break L1;
                }

                System.out.println("Inner Loop");
            }
            System.out.println("Outer Loop");
        }

        System.out.println("End of main()");
    }
}
```

////////////////////////////////////
Q264

```
class Example{
    public static void main(String args[]){
        for(int i=5; i>0;i--){
            System.out.println(i);
        }
    }
}
```

////////////////////////////////////
Q265 Exercise

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);

        int total = 0;

        System.out.println("Total: " + total);
        System.out.println("Avg: " + );
    }
}
```

////////////////////////////////////
Q266

import java.util.*;

class Example{

public static void main(String args[]){

Scanner input = new Scanner(System.in);

int total = 0;

for(;;){

System.out.print("Enter number: ");

int num = input.nextInt();

if(num == -1) {

break;

}

total += num;

}

System.out.println("Total: " + total);

//System.out.println("Avg: " +);

}

}

```

////////////////////////////////////
Q267
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);

        int total = 0;
        int num = 0;
        for(;num != -1;){
            System.out.print("Enter number: ");
            num = input.nextInt();

            total += num;
        }

        System.out.println("Total: " + total);
        //System.out.println("Avg: " + );
    }
}

```

```

////////////////////////////////////

```

While Loop

```

////////////////////////////////////

```

```

Q268
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);

        int total = 1;
        int num = 0;
        while(num != -1){ // => for(;num!=-1;){
            System.out.print("Enter number: ");
            num = input.nextInt();

```

```

        total += num;
    }

    //total++;

    System.out.println("Total: " + total);
    //System.out.println("Avg: " + );
}
}
////////////////////////////////////

```

Q269

```

import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);

        int total = 1;
        int num;

        System.out.print("Enter number: ");
        num = input.nextInt();

        total += num;

        while(num != -1){ // => for(;num!=-1;){
            System.out.print("Enter number: ");
            num = input.nextInt();

            total += num;
        }
        System.out.println("Total: " + total);
    }
}

```

do/while loop

Q270

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        int total = 1;
        int num;
        do{
            System.out.print("Enter number: ");
            num = input.nextInt();

            total += num;
        }while(num != -1);
        System.out.println("Total: " + total);
    }
}
```

Q271

```
class Example{
    public static void main(String args[]){
        for(;;){

        } // legal
        while(){ // Illegal

        }
        do{

        }while(); // Illegal
    }
}
```

Q272 Exercise I

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.println("Enter 1 for DecimalToBinary");
        System.out.println("Enter 2 for DecimalToOctal");
        System.out.println("Enter 3 for DecimalToHexa");
        System.out.println("Enter 99 to Exit");

        System.out.print("Select the converter: ");

        System.out.println("DecimalToBinary Converter"); // 1
        System.out.println("Enter 0 to Back");
        System.out.println("Enter 99 to Exit");
        System.out.print("Select the action: ");

        System.out.println("DecimalToOctal Converter"); // 2
        System.out.println("Enter 0 to Back");
        System.out.println("Enter 99 to Exit");
        System.out.print("Select the action: ");

        System.out.println("DecimalToHexa Converter"); // 3
        System.out.println("Enter 0 to Back");
        System.out.println("Enter 99 to Exit");
        System.out.print("Select the action: ");

        System.out.println("Invalid Input...");

    }
}
```

////////////////////////////////////

Q273 From Q272

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);

        while(true) {
            System.out.println("Enter 1 for DecimalToBinary");
            System.out.println("Enter 2 for DecimalToOctal");
            System.out.println("Enter 3 for DecimalToHexa");
            System.out.println("Enter 99 to Exit");

            System.out.print("Select the converter: ");
            int option = input.nextInt();

            if(option == 1){
                System.out.println("DecimalToBinary Converter");
            } else if (option == 2){
                System.out.println("DecimalToOctal Converter");
            } else if (option == 3) {
                System.out.println("DecimalToHexa Converter");
            } else if (option == 99) {
                return;
                //System.exit(0);
            } else {
                System.out.println("Invalid Input...");
            }
        }
    }
}
```

////////////////////////////////////

Q274 From Q273

=====

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);

        L1:while(true) {
            System.out.println("Enter 1 for DecimalToBinary");
            System.out.println("Enter 2 for DecimalToOctal");
            System.out.println("Enter 3 for DecimalToHexa");
            System.out.println("Enter 99 to Exit");

            System.out.print("Select the converter: ");
            int option = input.nextInt();

            switch(option) {
                case 1:
                    while(true){

                        System.out.println("-----");
                        System.out.println("DecimalToBinary
                        Converter");

                        System.out.println("-----");
                        System.out.println("Enter 0 to Back");
                        System.out.println("Enter 99 to Exit");
                        System.out.print("Select the action:
                        ");

                        int action = input.nextInt();

                        if(action == 0) {
                            continue L1;
                        } else if (action == 99) {
                            return;
                        } else {
                            System.out.println("Invalid
                            Input...");
                        }
                    }
                case 2:
                    while(true){
```

```

System.out.println("-----");
                                System.out.println("DecimalToOctal
Converter");

System.out.println("-----");
                                System.out.println("Enter 0 to Back");
                                System.out.println("Enter 99 to Exit");
                                System.out.print("Select the action:

");

                                int action = input.nextInt();

                                if(action == 0) {
                                    continue L1;
                                } else if (action == 99) {
                                    return;
                                } else {
                                    System.out.println("Invalid
Input...");
                                }
                            }
                        case 3:
                            while(true){

System.out.println("-----");
                                System.out.println("DecimalToHexa
Converter");

System.out.println("-----");
                                System.out.println("Enter 0 to Back");
                                System.out.println("Enter 99 to Exit");
                                System.out.print("Select the action:

");

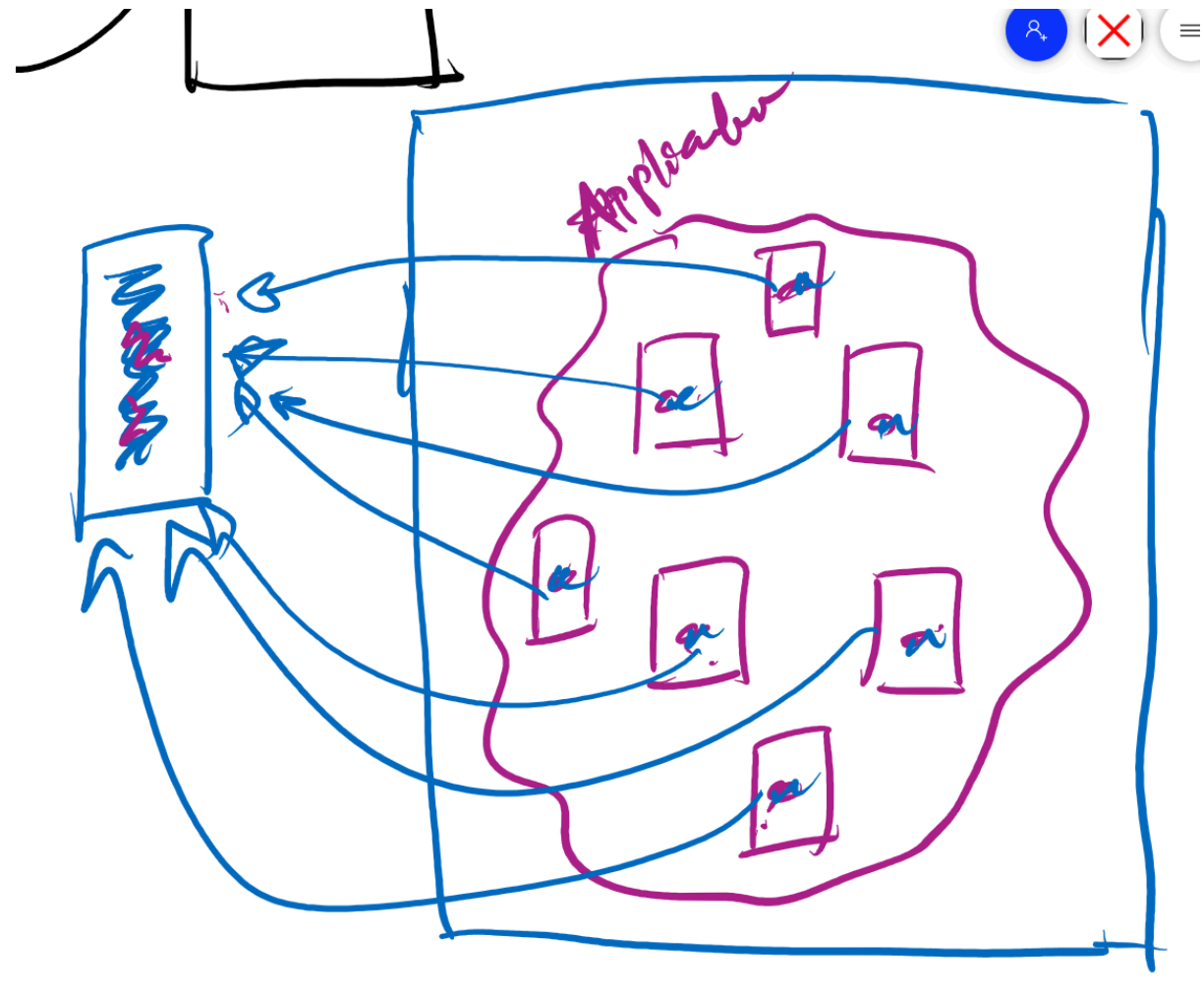
                                int action = input.nextInt();

                                if(action == 0) {
                                    continue L1;
                                } else if (action == 99) {
                                    return;
                                } else {
                                    System.out.println("Invalid
Input...");
                                }
                            }
                        }
                    }
                }
            }
        }
    }
}

```

```
case 99:  
    return;  
default:  
    System.out.println("Invalid Input...");  
}  
}  
}
```

////////////////////////////////////
Q275



Java Methods

Q276

```
import java.util.*;
class Example{
    public static void main(String args[]){

        System.out.println("Dinuka Kodithuwakku");
        System.out.println("No 23/5");
        System.out.println("Panadura Rd");
        System.out.println("Panadura");

        //
        //
        //

        System.out.println("Dinuka Kodithuwakku");
        System.out.println("No 23/5");
        System.out.println("Panadura Rd");
        System.out.println("Panadura");

        //
        //
        //
        //

        System.out.println("Dinuka Kodithuwakku");
        System.out.println("No 23/5");
        System.out.println("Panadura Rd");
        System.out.println("Panadura");
        //
        //
        //
        //
        //
        System.out.println("Dinuka Kodithuwakku");
        System.out.println("No 23/5");
        System.out.println("Panadura Rd");
        System.out.println("Panadura");

    }
}
```

}

`public static void printTotal() {`
// Method Body / Implementation
`}`

method name (pointing to `printTotal()`)

```
import java.util.*;
```

```
class Example{
```

```
    public static void printTotal(){
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Enter num1: ");
```

```
        int num1 = input.nextInt();
```

```
        System.out.print("Enter num2: ");
```

```
        int num2 = input.nextInt();
```

```
        int total = num1 + num2;
```

```
        System.out.println("Total of "+num1+" and "+num2+" => " +
total);
    }
```

```
    public static void main(String args[]){
```

```
        printTotal();
```

```
    }
```

```
}
```

}

Parameterized Methods

Q281

```
import java.util.*;
class Example{

    public static void printTotal(int num1, int num2){
        // ----- Process -----
        int total = num1 + num2;

        // ----- Output -----
        System.out.println("Total of "+num1+" and "+num2+" => " +
total);
    }

    public static void main(String args[]){
        // ----- Input -----
        Scanner input = new Scanner(System.in);
        System.out.print("Enter num1: ");
        int num1 = input.nextInt();
        System.out.print("Enter num2: ");
        int num2 = input.nextInt();

        printTotal(num1, num2);
    }
}
```

Parameters list

Public static void printTotal (int num1, int num2) {

// ----- Process -----

// ----- Output -----

System.out.println("Total of "+num1+" and "+num2+" => " +

total);

}

Public static void main (String args[]) {

// ----- Input -----

Scanner input = new Scanner (System.in);

System.out.print ("Enter num1: ");

int num1 = input.nextInt();

System.out.print ("Enter num2: ");

int num2 = input.nextInt();

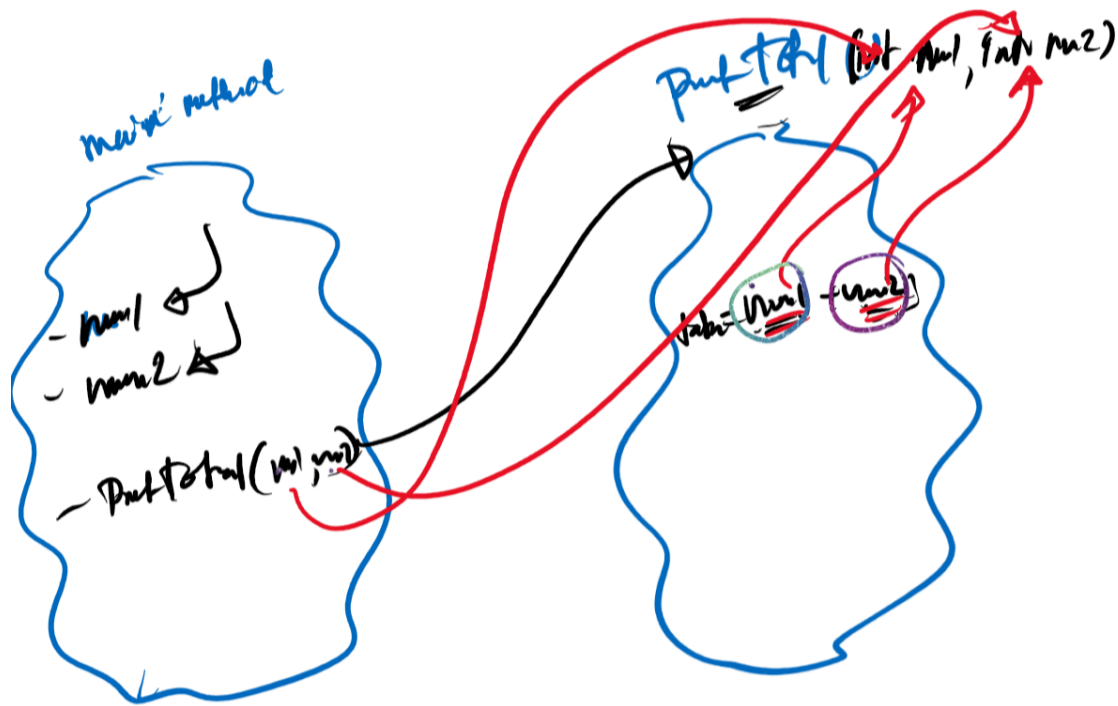
printTotal (num1, num2);

}

}

arguments

Q282



Q283

```
import java.util.*;
class Example{

    public static void printTotal(int a, int b){
        // ----- Process -----
        int total = num1 + num2;

        // ----- Output -----
        System.out.println("Total of "+num1+" and " +num2+" => " +
total);
    }

    public static void main(String args[]){
        // ----- Input -----
        Scanner input = new Scanner(System.in);
        System.out.print("Enter num1: ");
        int num1 = input.nextInt();
        System.out.print("Enter num2: ");
        int num2 = input.nextInt();

        printTotal(num1, num2);
    }
}
```

Q284

```
import java.util.*;
```

```
class Example{
```

```
    public static void printTotal(int a, int b){
```

```
        // ----- Process -----
```

```
        int total = a + b;
```

```
        // ----- Output -----
```

```
        System.out.println("Total of "+a+" and "+b+" => " + total);
```

```
    }
```

```
    public static void main(String args[]){
```

```
        // ----- Input -----
```

```
        Scanner input = new Scanner(System.in);
```

```
        System.out.print("Enter num1: ");
```

```
        int num1 = input.nextInt();
```

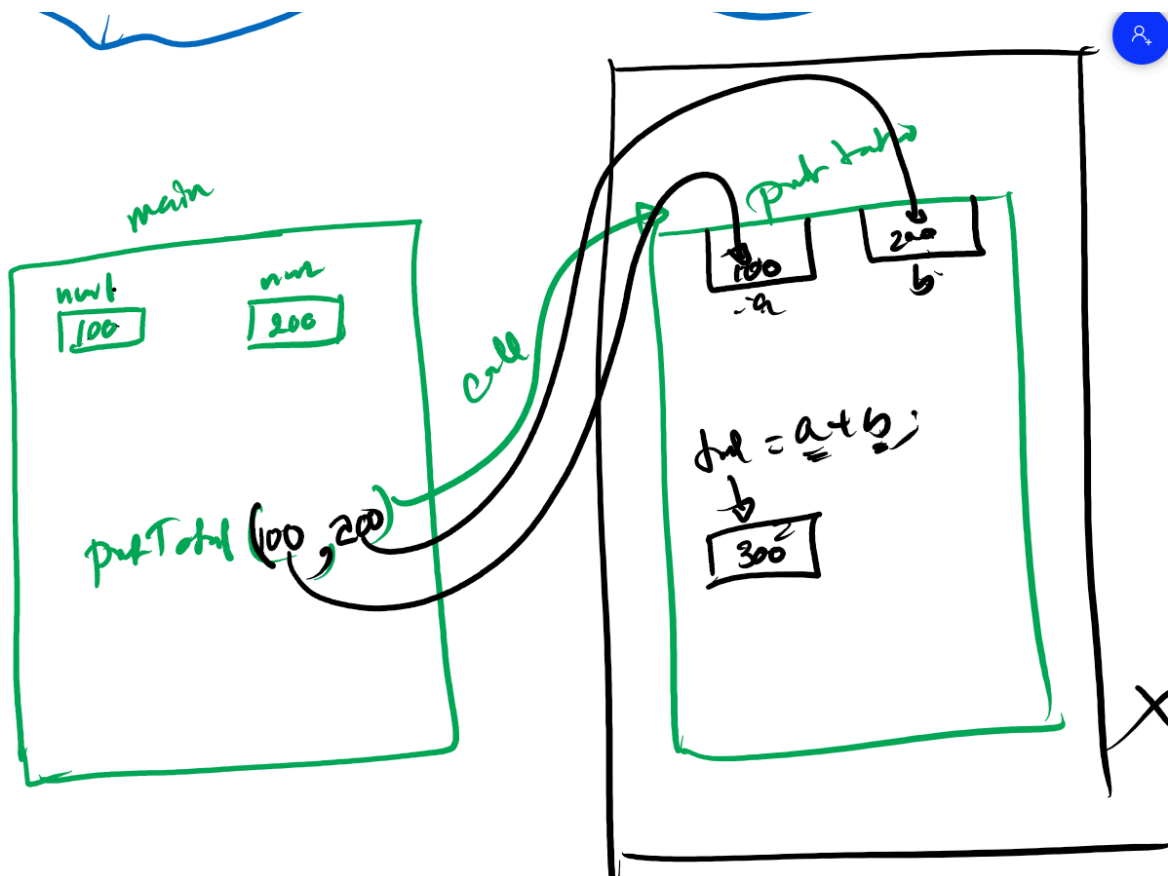
```
        System.out.print("Enter num2: ");
```

```
        int num2 = input.nextInt();
```

```
        printTotal(num1, num2);
```

```
    }
```

```
}
```



}

}

```
// 0! = 1;
```

```
Scanner input = new Scanner(System.in);
```

```
System.out.print("Enter a number: ");
int num = input.nextInt();
```

```
System.out.println("Reversed value of " + num + " => " +
reverse(num));
    }
}
```

```
////////////////////////////////////
```

Q293

```
import java.util.*;
```

```
class Example{
```

```
    public static boolean isPalindrom(int number) {
        if(number == reverse(number)) {
            return true;
        } else {
            return false;
        }
    }
}
```

```
    public static int reverse(int number) {
        int reversedNumber = 0;
        while(number != 0){
            reversedNumber = reversedNumber * 10 + (number %
10);
            number /= 10;
        }

        return reversedNumber;
    }
}
```

```
    public static void main(String args[]){
        // ----- Input -----
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = input.nextInt();

        System.out.println("Reversed value of " + num + " => " +
reverse(num));
    }
}
```

```
int reversedNumber = 0;
```

```

        while(number != 0){
            reversedNumber = reversedNumber * 10 + (number %
10);

            number /= 10;
        }

        return reversedNumber;
    }

    public static void main(String args[]){
        // ----- Input -----
        Scanner input = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = input.nextInt();
        System.out.println("Reversed value of " + num + " => " +
reverse(num));
        System.out.println("Is Palindrom: " + isPalindrom(num));
    }
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

```

Method Calling Stack

//

Q296

```

class Example{
    public static void methodA() {
        System.out.println("Start methodA()");
        methodB();
        System.out.println("End methodA()");
    }

    public static void methodB() {
        System.out.println("Start methodB()");

        System.out.println("End methodB()");
    }

    public static void main(String args[]){
        System.out.println("Start main()");

        methodA();

        System.out.println("End main()");
    }
}

```

Q297

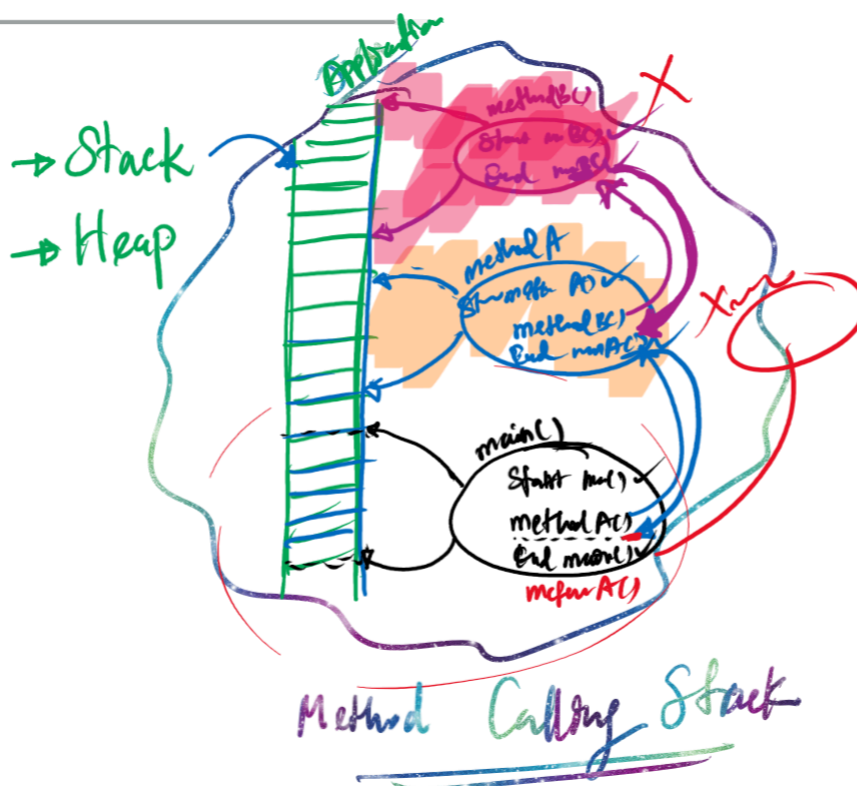
```

1 class Example{
2     public static void methodA() {
3         System.out.println("Start methodA()");
4         methodB();
5         System.out.println("End methodA()");
6     }
7
8     public static void methodB() {
9         System.out.println("Start methodB()");
10        System.out.println("End methodB()");
11    }
12
13    public static void main(String args[]){
14        System.out.println("Start main()");
15
16        methodA();
17
18        System.out.println("End main()");
19    }
20 }

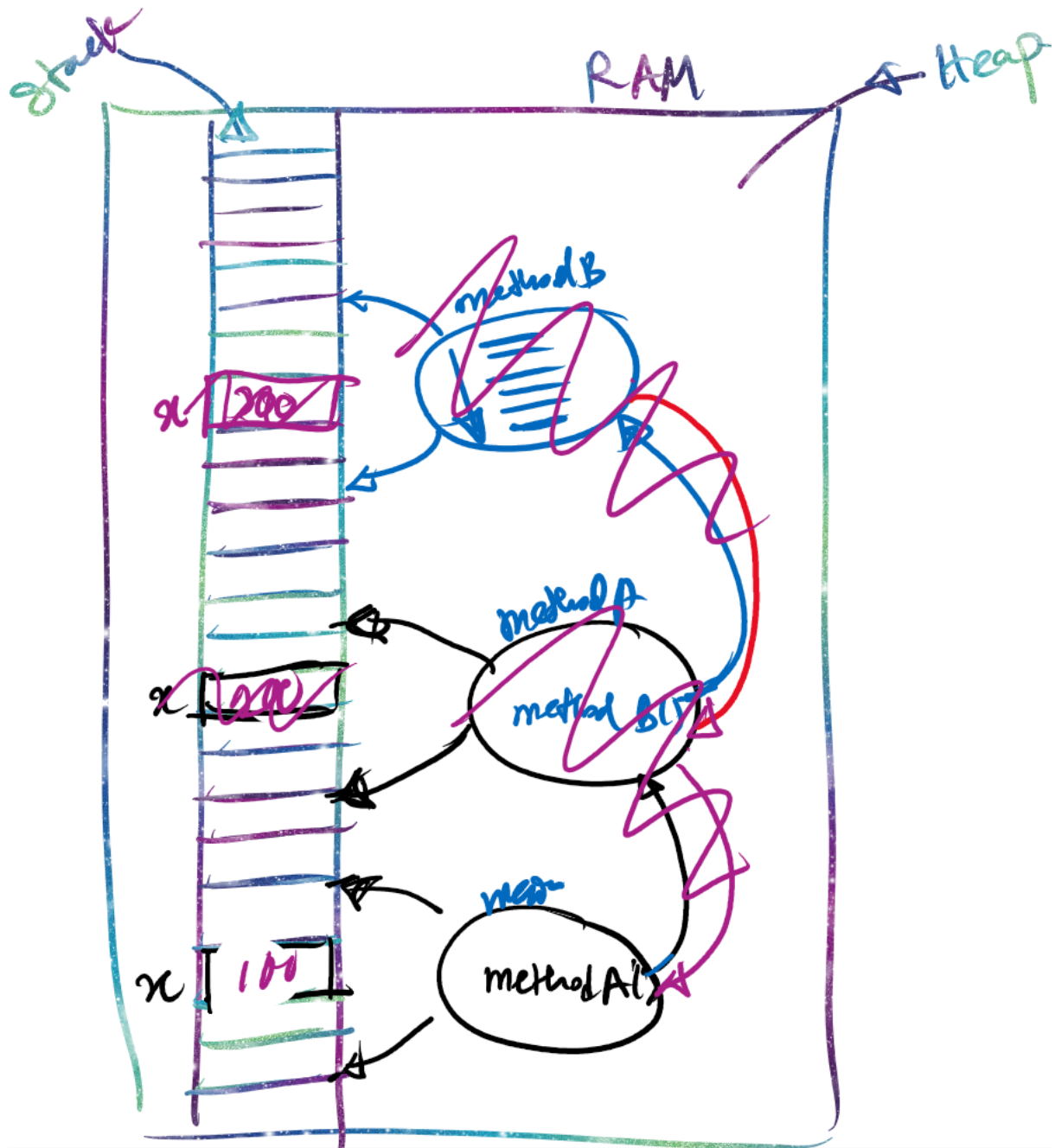
```

Start main();
 Start methodA();
 Start methodB();
 End methodB();
 End methodA();
 End main();

Q298



Start main()
 Start methodA()
 Start methodB()
 End methodB()
 End methodA()
 End main()



Method Definition Rules

Q300 Case I

=====

```
class Example{
    public static void methodA() {
        methodC();
    }

    public static void methodB() {

    }

    public static void main(String args[]){

    }

    public static void methodC() {

    }
}
```

Q301 Case II

=====

```
class Example{
    public static void methodA() {
        System.out.println("Start");

        System.out.println("End");
    }

    public static void main(String args[]){ // no output

    }
}
```

```
class Example{
    public static void methodA() {

    }

    public static void main(String args[]){
        methodA(100); // Illegal
    }
}
```

```
class Example{
    public static void methodA(int x) {

    }

    public static void main(String args[]){

        byte b = 100;
        short s = 100;
        int i = 100;
        long l = 100;
        float f = 100.0f;
    }
}
```

```
class Example{
    public static void methodA() {

    }

    public static void methodA() {

    }

    public static void main(String args[]){

    }

}
```

```

////////////////////////////////////
Q309 Case IX
=====
class A {
    public static void methodA() {

    }
}

```

```

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
Q310 Case X - Method Overloading
=====
class Example{
    public static void methodA() {

    }

    public static void methodA(int x) {

    }

    public static void methodA(int x, int y) {

    }

    public static void main(String args[]){

    }

}

```

```
class Example{
    public static int methodA() {
        byte b = 100;
        short s = 100;
        int i = 100;
        long l = 100;
        float f = 100.0f;
        double d = 100.0;
        char ch = 'A';
        boolean bool = false;

        return /* insert code here */;
    }

    public static void main(String args[]){
    }
```

```

}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
Q314 Case XIV
=====
class Example{
    public static int methodA() {
        byte b = 100;
        short s = 100;
        int i = 100;
        long l = 100;
        float f = 100.0f;
        double d = 100.0;
        char ch = 'A';
        boolean bool = false;

        return (int)l;
    }

    public static void main(String args[]){

    }
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

```

```

Q315 Case XV
=====
class Example{
    public static int methodA() {

    }

    public static boolean methodA() { // Illegal

    }

    public static void main(String args[]){

    }
}

```

```
class Example{
    public static int methodA(int x) {
        if( x > 100) {
            return 0;
        } else {
            return 1;
        }
    }

    public static void main(String args[]){

    }
}
```

```
class Example{
    public static int methodA(int x) {
        if( x > 100) {
            return 0;
        }
        return 1;
    }
}
```

```

        public static void main(String args[]){

            }
        }
////////////////////////////////////
Q322 Case XXII
=====
class Example{
    static int methodA() {
        return 0;
    }

    public static void main(String args[]){
        methodA();
    }
}

```

```

////////////////////////////////////
Q323 Case XXIII
=====
class Example{
    int methodA() {
        return 0;
    }

    public static void main(String args[]){
        methodA(); // Illegal
    }
}

```

```

////////////////////////////////////
Q324 Case XXIV
=====
class Example{
    int methodA() {
        return 0;
    }

    public static void main(String args[]){
        // methodA();
    }
}

```

```
class A {
    public static void methodB(){

    }
}

class Example{
    public static int methodA() {
        return 0;
    }

    public static void main(String args[]){
        methodA();

        A.methodB();
    }
}
```

```
class A {
    public void methodB(){

    }
}

class Example{
    public static int methodA() {
        A.methodB();
        return 0;
    }

    public static void main(String args[]){
        methodA();

        A.methodB();
    }
}
```

=====

```
class Example{
    methodA() { // Illegal

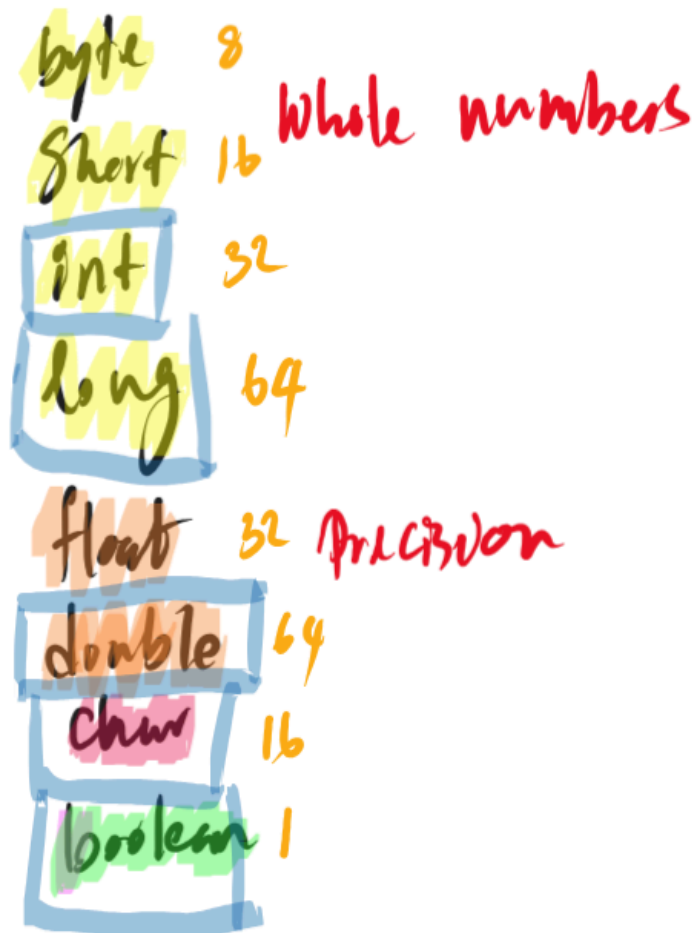
        return 0;
    }

    public static void main(String args[]){
        methodA();

        A.methodB();
    }
}
```

Revision

Primitive Data Types



Q331

```
class Example{
    public static void main(String args[]){
        int x = 100;
        String name = "John";
        double d = 10.25;
        double salary = 1000000;
        System.out.println(x);
        System.out.printf("The value of x: %10d | My name is %-10s
...\\n", x, name);
        System.out.printf("%2$-10s : %1$10d \\n", x, name);
        System.out.printf("%10.3f \\n", d);
        System.out.printf("%-10.1s END\\n", name);
        System.out.printf("%1$,015.2f \\n", salary);
        System.out.printf("%d \\n", x);
        System.out.println("-----");
        String formattedText = String.format("%1$,015.2f \\n", salary);
        System.out.println(formattedText);
    }
}
```

Print f

% [position #] [flag] [width] [precision] [type]

% *\$* *10* *.3* *d*
. *s*
 f
 b
 c

- alignment

0

(
+

Q332



byte⁸ → short¹⁶ → int³² → long⁶⁴ → float³² → double



Casting
by force
Explicitly

Conversion
auto
Implicitly

```
class Example{
    public static void main(String args[]){
        int x = 1000;
        byte b = (byte) x;

        System.out.println(b);
    }
}
```

Casting
by force
Explicitly

Conversion
auto
Implicitly

narrow Casting
wider Casting*

wider Conversion
narrow Conversion*

flower
don't
ch
bro

32
int → byte

Q333

⇒ int x = Condition? 100 : 200;

```
class Example{
    public static void main(String args[]){
        int x;
        int y = 100;
        if(condition) {
            x = 0;
        } else {
            x = y;
        }
        int x = condition ? 0 : y;
    }
}
```

Q334

```
class Example{
    public static void main(String args[]){
        int x;
        int y = 100;

        if(!condition) {
            x = 0;
        } else {
            // x += y;
            x = x + y;
        }
        int x = !condition ? 0 : x + y;

        if (condition) {
            x = x + y;
        } else {
            x = 0
        }

        int x = condition ? x + y : 0 ;

    }
}
```

////////////////////////////////////

Q335

```
class Example{
    public static void main(String args[]){
        System.out.println(10 & 5);
    }
}
```

Logical Operators

&

bit wise

&&

Short Circuit

true
both of operands are true

||

^

!

unary

10 $\Rightarrow$ (1010) &
5 $\Rightarrow$ (0101)
0000 $\Rightarrow$ 0

Q336

Arithmetic

Operation

+

-

/

*

%

++

--

unary

>>

<<

>>>

<<<

>>>

<<<

Relational Operators

==

!=

>

<

>=

<=

Logical Operators

&

bit wise

&&

Short Circuit

||

^

!

unary

Q337

50 < x < 150

50 < x && x < 150

```
class Example{
    public static void main(String args[]){
        int x = 100;

        if() {
            System.out.println("OK");
        } else {
            System.out.println("ERR");
        }
    }
}
```

```

    }
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
Q338
class Example{
    public static void main(String args[]){
        String country = "US"; // allowed values => "US" "CA"
        double salary = 75000;

        if(country == "CA" || salary > 100000) {
            System.out.println("OK");
        } else {
            System.out.println("ERR");
        }

        if(country == "US" && salary < 100000) {
            System.out.println("ERR");
        } else {
            System.out.println("OK");
        }

    }
}

```

```

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
Q339 Random Values
=====
import java.util.*;
class Example{
    public static void main(String args[]){
        Random r = new Random();

        int x = r.nextInt();

        System.out.println("x: " + x);
    }
}

```

```

////////////////////////////////////
Q340
import java.util.*;
class Example{
    public static void main(String args[]){
        Random r = new Random();

        int x = r.nextInt(10);

        System.out.println("x: " + x);
    }
}
////////////////////////////////////

```

```

Q341
class Example{
    public static void main(String args[]){
        System.out.println("\u001B[30m Hello!!! \u001B[0m");
        System.out.println("\u001B[31m Hello!!! \u001B[0m");
        System.out.println("\u001B[32m Hello!!! \u001B[0m");
        System.out.println("\u001B[33m Hello!!! \u001B[0m");
        System.out.println("\u001B[34m Hello!!! \u001B[0m");
        System.out.println("\u001B[35m Hello!!! \u001B[0m");
        System.out.println("\u001B[36m Hello!!! \u001B[0m");
    }
}
////////////////////////////////////

```

```

Q342
import java.util.*;

class Example{
    public static void main(String args[]){
        for(int i=0; i<7; i++) {
            System.out.println("\u001B[3"+i+"m Hello!!!
\u001B[0m");
        }
    }
}

```

```
class Example{
    public static void main(String args[]){
        Random r = new Random(); // Random numbers
        Scanner input = new Scanner(System.in); // User input

        int rNum = r.nextInt(100);
        int attempts = 0;
        while(true){
            System.out.print("Guess the number [0, 100): ");
            int guess = input.nextInt();

            if(rNum == guess) {
                break;
            }

            if(rNum > guess) {
                System.out.println("Higher");
            } else {
                System.out.println("Lower");
            }
            attempts++;
        }
        System.out.println("Congratulations! Correct guess with
        "+(attempts + 1)+" attempts.");
    }
}
```

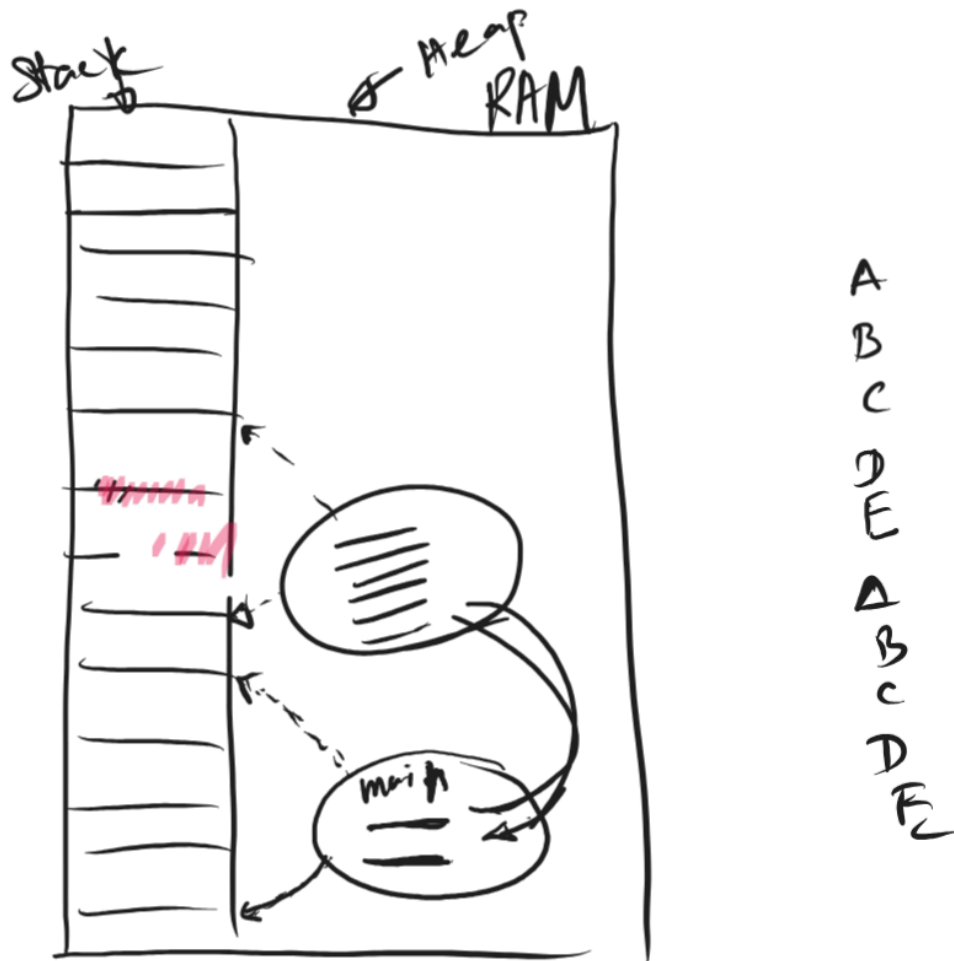
```
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in); // User input
        System.out.print("Input the number: ");
        int num = input.nextInt();
        int copyOfNum = num;

        int total = 0;
        while(num != 0){
            int digit = num % 10;
            int cubicValue = (int) Math.pow(digit, 3);
            total += cubicValue;
            num /= 10;
        }
        if(total == copyOfNum) {
            System.out.println("This number is Armstrong...");
        } else {
            System.out.println("This number is Not Armstrong...");
        }
    }
}
```

```
class Example{
    public static void main(String args[]){
        Random r = new Random();
        int num = r.nextInt(900) + 100;
        System.out.println("Vowel Count: " + vowelCount);
    }
}
```

Java Methods - Recap

Q347



```
class Example{

    public static void main(String args[]){
        System.out.println("Start main()");
        methodA();
        System.out.println("End main()");
    }

    static void methodA(){
        System.out.println("methodA()");
        methodB();
    }

    static void methodB(){
        System.out.println("methodB()");
    }

}
```

Q348

```
Example.java x
1 class Example{
2
3     public static void main(String args[]){
4         System.out.println("Start main()");
5         methodA();
6         System.out.println("End main()");
7     }
8
9     static void methodA(){
10        System.out.println("methodA()");
11        methodB();
12    }
13
14
15    static void methodB(){
16        System.out.println("methodB()");
17    }
18 }
```

Start main()
methodA();
methodB();
End main()



Q349

```
class Example{
```

```
    public static void main(String args[]){
        int x = getNumber();
```

```
        System.out.println(x);
    }
```

```
    static int getNumber(){
```

```
        return 100; // Return Statement
```

```
    }
```

```
}
```

Q350

```
class Example{
```

```
    public static void main(String args[]){
        byte x = (byte) getNumber();
```

```
        System.out.println(x);
    }
```

```
    static int getNumber(){
```

```
        return 100; // Return Statement
```

```
    }
```

```
}
```

```
////////////////////////////////////  
Q351  
class Example{  
    public static void main(String args[]){  
        long x = getNumber();  
        System.out.println(x);  
    }  
    static int getNumber(){ // Return Typed Method  
        return 100; // Return Statement  
    }  
}
```

```
////////////////////////////////////  
Q352  
class Example{  
  
    public static void main(String args[]){  
        int x = findTotal(100, 150); // Arguments  
  
        System.out.println(x);  
    }  
  
    static int findTotal(int a, int b){ // Parameterized method  
  
        return 100; // Return Statement  
    }  
}
```

```
////////////////////////////////////  
Q353  
class Example{  
  
    public static void main(String args[]){  
        int x = findTotal(100, 150); // Arguments  
  
        System.out.println(x);  
    }  
  
    static int findTotal(int a, int b){ // Parameterized method  
  
        return a + b; // Return Statement  
    }  
}
```

}

Q356

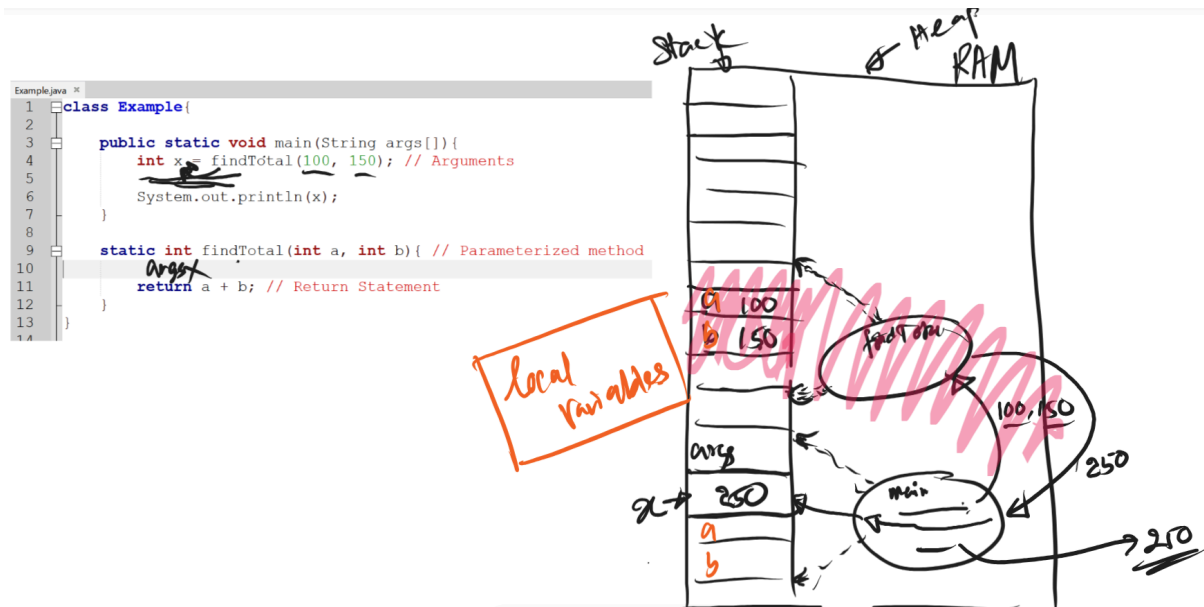
```
class Example{
    public static void main(String args[]){
        long l1 = 100;
        long l2 = 120;

        int x = findTotal((int)l1, (int)l2); // Arguments

        System.out.println(x);
    }
    static int findTotal(int a, int b){ // Parameterized method

        return a + b; // Return Statement
    }
}
```

Q357



Method Overloading / CompileTime Polymorphism

Q358

```
class Example{

    public static void main(String args[]){
        methodA();
    }

    static void methodA(){
        System.out.println("methodA()");
    }

    static void methodA(){ // Illegal
        System.out.println("methodA() second version");
    }
}
```

Q359

```
class Example{

    public static void main(String args[]){
        methodA();

        methodA(100);
    }

    static void methodA(){
        System.out.println("methodA()");
    }

    static void methodA(int a){
        System.out.println("methodA(int)");
    }
}
```

```
class Example{
    public static void main(String args[]){
        methodA();

        methodA(100.0);
    }

    static void methodA(){
        System.out.println("methodA()");
    }

    static void methodA(int a){
        System.out.println("methodA(int)");
    }
}
```

```
class Example{

    public static void main(String args[]){
        methodA();

        methodA(100);
    }

    static void methodA(){
        System.out.println("methodA()");
    }

    static int methodA(int a){
        System.out.println("methodA(int)");
        return 0;
    }

    static int methodA(int b){
        System.out.println("methodA(int)");
        return 0;
    }

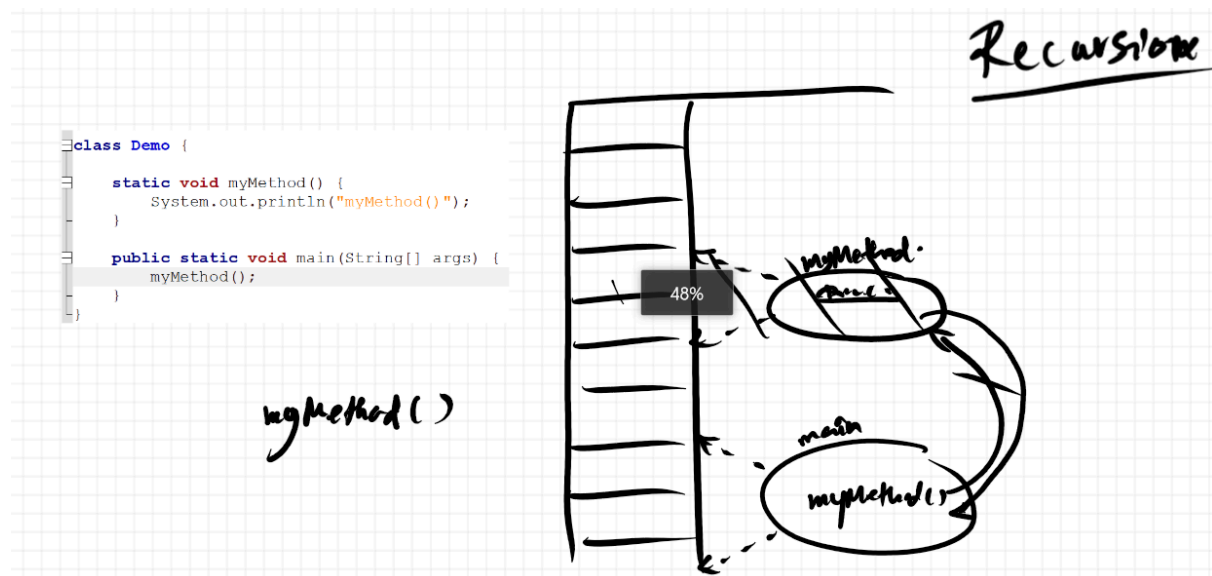
}
```

Recursion

Q362

```
import java.io.*;
import java.util.*;
```

```
class Demo {
    static void myMethod() {
        System.out.println("myMethod()");
    }
    public static void main(String[] args) {
        myMethod();
    }
}
```



Q363

```
import java.io.*;
import java.util.*;
```

```
class Demo {
    static void myMethod() {
        System.out.println("myMethod()");

        myMethod();
    }
    public static void main(String[] args) {
        myMethod();
    }
}
```

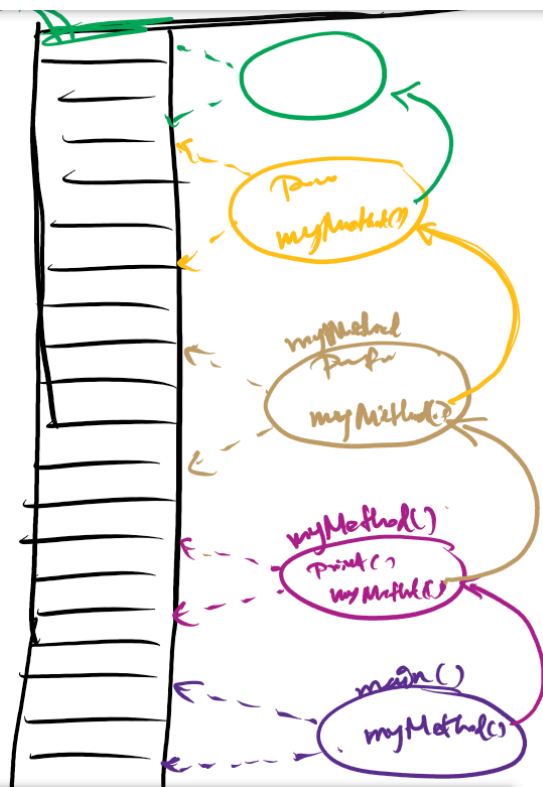
```

static void myMethod() {
    System.out.println("myMethod()");
    myMethod(); ✓
}

public static void main(String[] args) {
    myMethod();
}

```

myMethod()
myMethod()
myMethod()



////////////////////////////////////

Q364

```

import java.io.*;
import java.util.*;

```

```

class Demo {

```

```

    static int findTotal(int a, int b){
        return a + b;
    }

```

```

    static void myMethod() {
        if(condition){
            return;
        }
        System.out.println("myMethod()");
    }

```

```

    public static void main(String[] args) {
        myMethod();

```

```

        while(true){
            if(condition) {
                break;
            }

```

```

        }

```

```

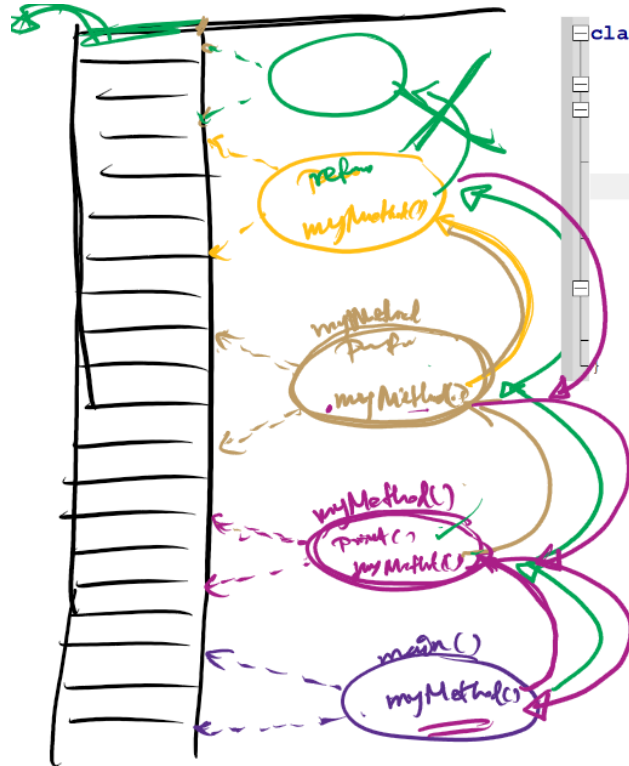
    }

```

```

}

```



```

class Demo {
    static void myMethod() {
        if(condition) {
            return;
        }
        System.out.println("myMethod()");
        myMethod();
    }

    public static void main(String[] args) {
        myMethod();
    }
}

```

////////////////////////////////////

Q365

class Demo {

```

    static void myMethod(int n) {
        System.out.println("myMethod() - n: " + n);
        myMethod(n - 1);
    }

```

```

    public static void main(String[] args) {
        myMethod(10);
    }
}

```

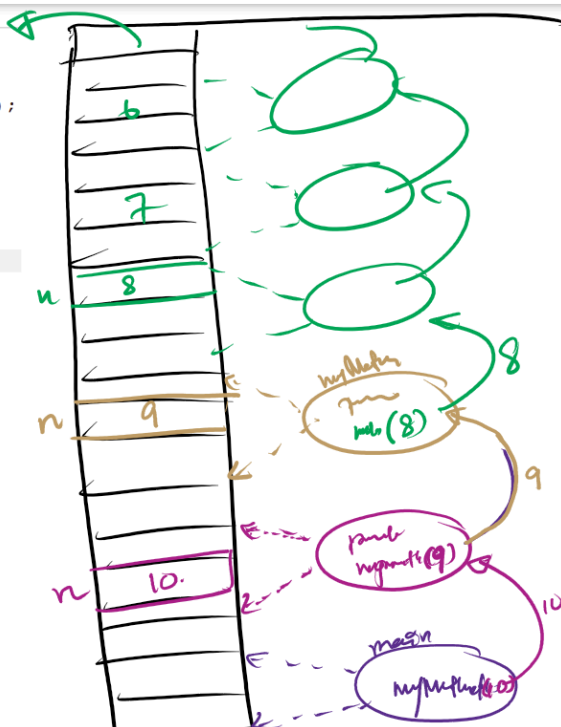
```

class Demo {
    static void myMethod(int n) {
        System.out.println("myMethod() - n: " + n);
        myMethod(n - 1);
    }

    public static void main(String[] args) {
        myMethod(10);
    }
}

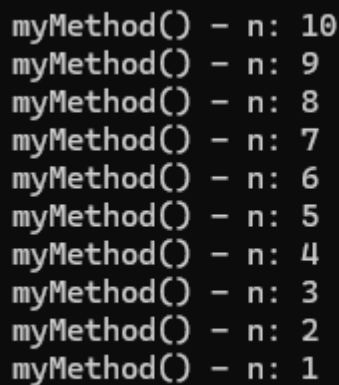
```

~~~~~ : 10  
 ~~~~~ : 9  
 ~~~~~ : 8  
 ↓



Q366

```
class Demo {  
  
    static void myMethod(int n) {  
        if(n == 0) {  
            return;  
        }  
  
        System.out.println("myMethod() - n: " + n);  
        myMethod(n - 1);  
    }  
  
    public static void main(String[] args) {  
        myMethod(10);  
    }  
}
```



```
myMethod() - n: 10  
myMethod() - n: 9  
myMethod() - n: 8  
myMethod() - n: 7  
myMethod() - n: 6  
myMethod() - n: 5  
myMethod() - n: 4  
myMethod() - n: 3  
myMethod() - n: 2  
myMethod() - n: 1
```

Q367

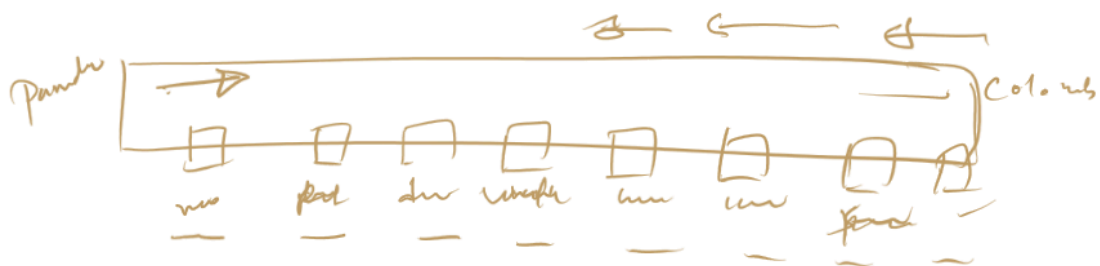
```
class Demo {  
  
    static void myMethod(int n) {  
        if(n == 0) { // No change 0  
            return;  
        }  
  
        System.out.println("myMethod() - n: " + n);  
        myMethod(n - 1); // No change n-1  
    }  
  
    public static void main(String[] args) {  
        myMethod(10); // No change 10  
    }  
}
```

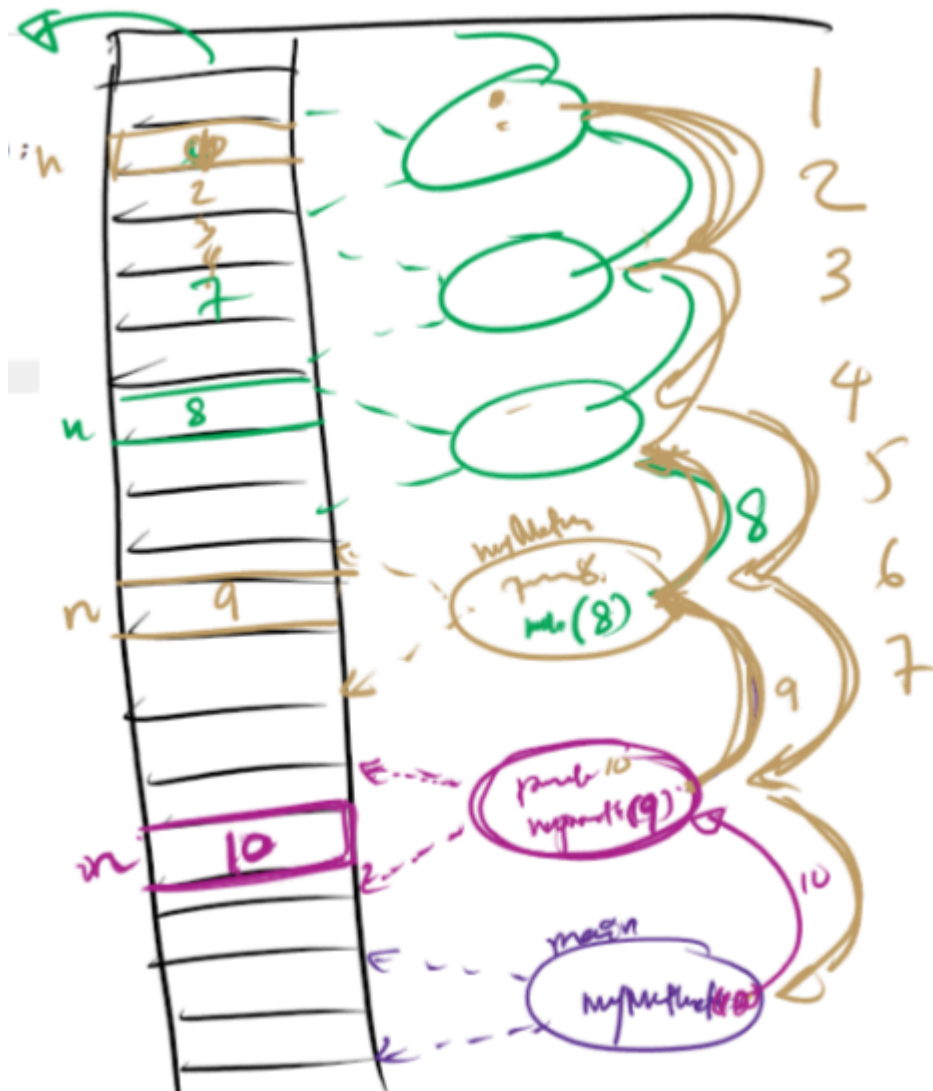
Q368

```
class Demo {
```

```
    static void myMethod(int n) {  
        if(n == 0) { // No change 0  
            return;  
        }  
  
        myMethod(n - 1); // No change n-1  
        System.out.println("myMethod() - n: " + n);  
    }
```

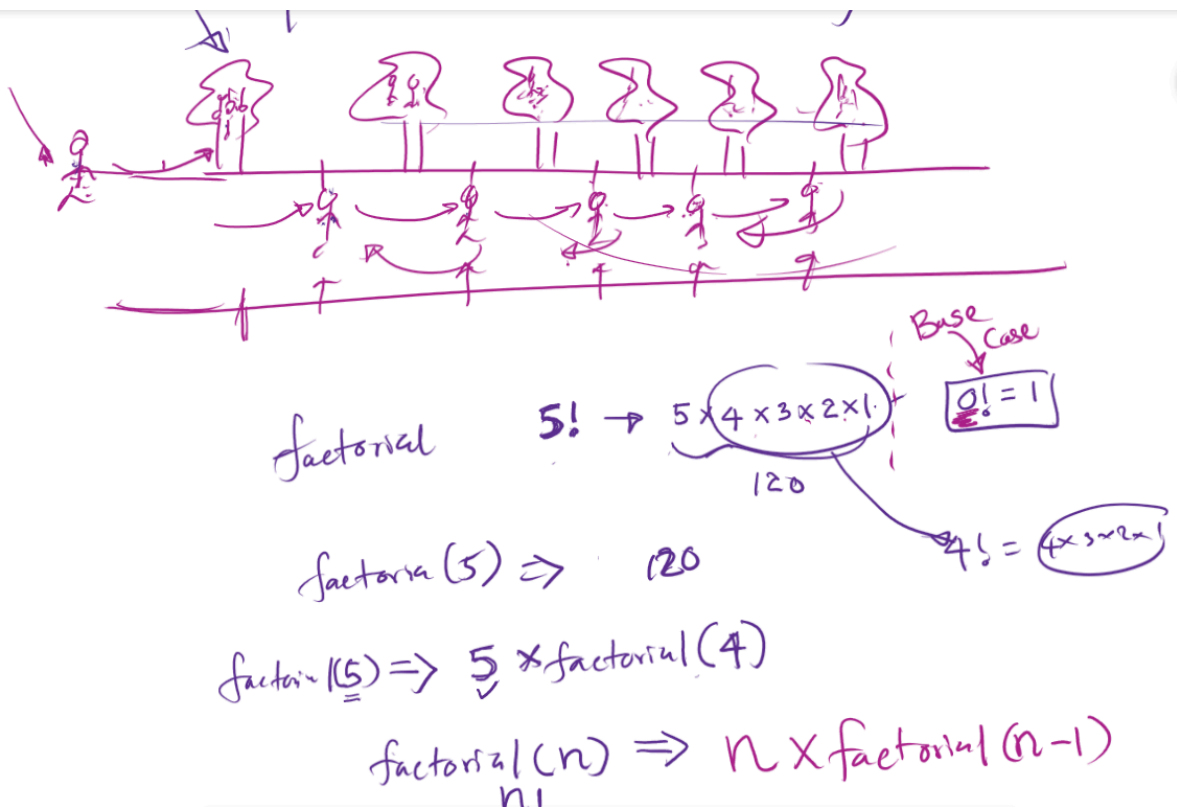
```
    public static void main(String[] args) {  
        myMethod(10); // No change 10  
    }  
}
```





////////////////////////////////////

Q369



```
static int factorial(int n) {  
  
}  
  
public static void main(String[] args) {  
    int fact = factorial(10);  
  
    System.out.println(fact);  
}  
}
```

=====

```
class Demo {  
  
    static int factorial(int n) {  
  
        if(n == 0) {  
            return 1;  
        }  
  
        return n * factorial(n-1);  
    }  
  
    public static void main(String[] args) {  
        int fact = factorial(5);  
  
        System.out.println(fact);  
    }  
}
```

```
class Demo {
    static int fib(int n) {

    }

    public static void main(String[] args) {
        int fibonacci = fib(5);
        System.out.println(fibonacci);
    }
}
```



# Arrays

Q373

```
import java.util.*;
class Demo {

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        int a0, a1, a2, a3, a4, a5, a6, a7, a8, a9;

        // ----- Read Numbers -----

        System.out.print("Enter number 1: ");
        a0 = input.nextInt();

        System.out.print("Enter number 2: ");
        a1 = input.nextInt();

        System.out.print("Enter number 3: ");
        a2 = input.nextInt();

        System.out.print("Enter number 4: ");
        a3 = input.nextInt();

        System.out.print("Enter number 5: ");
        a4 = input.nextInt();

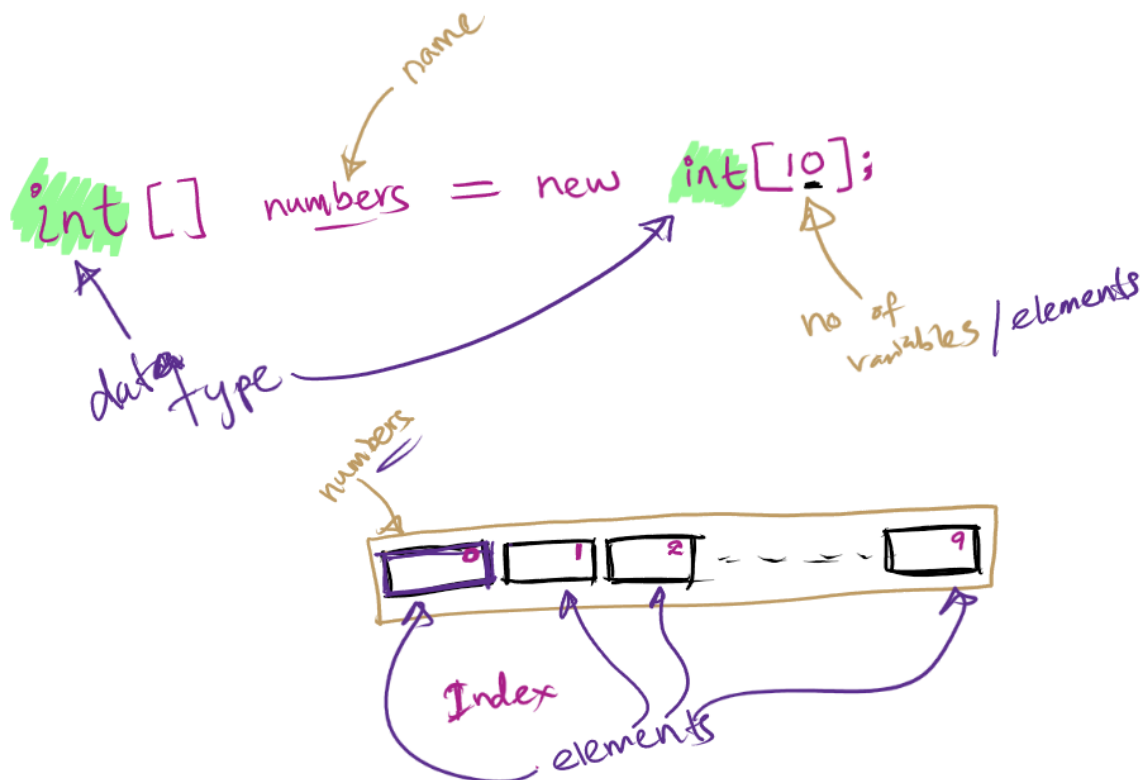
        System.out.print("Enter number 6: ");
        a5 = input.nextInt();

        System.out.print("Enter number 7: ");
        a6 = input.nextInt();

        System.out.print("Enter number 8: ");
        a7 = input.nextInt();

        System.out.print("Enter number 9: ");
        a8 = input.nextInt();

        System.out.print("Enter number 10: ");
        a9 = input.nextInt();
```





```
a6 = input.nextInt();
```

```
System.out.print("Enter number 8: ");  
a7 = input.nextInt();
```

```
System.out.print("Enter number 9: ");  
a8 = input.nextInt();
```

```
System.out.print("Enter number 10: ");  
a9 = input.nextInt();
```

```
// ----- Find Total -----
```

```
int total = 0;
```

```
total += a0;  
total += a1;  
total += a2;  
total += a3;  
total += a4;  
total += a5;  
total += a6;  
total += a7;  
total += a8;  
total += a9;
```

```
System.out.println("Total: " + total);
```

```
    }  
}
```



```
int total = 0;

total += ar[0];
total += ar[1];
total += ar[2];
total += ar[3];
total += ar[4];
total += ar[5];
total += ar[6];
total += ar[7];
total += ar[8];
total += ar[9];

System.out.println("Total: " + total);
}
}
```

```
import java.util.*;
class Demo {

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        int[] ar = new int[10];

        // ----- Read Numbers -----
        for(int i=0; i<10; i++) {
            System.out.print("Enter number 1: ")
            ar[i] = input.nextInt();
        }

        // ----- Find Total -----

        int total = 0;

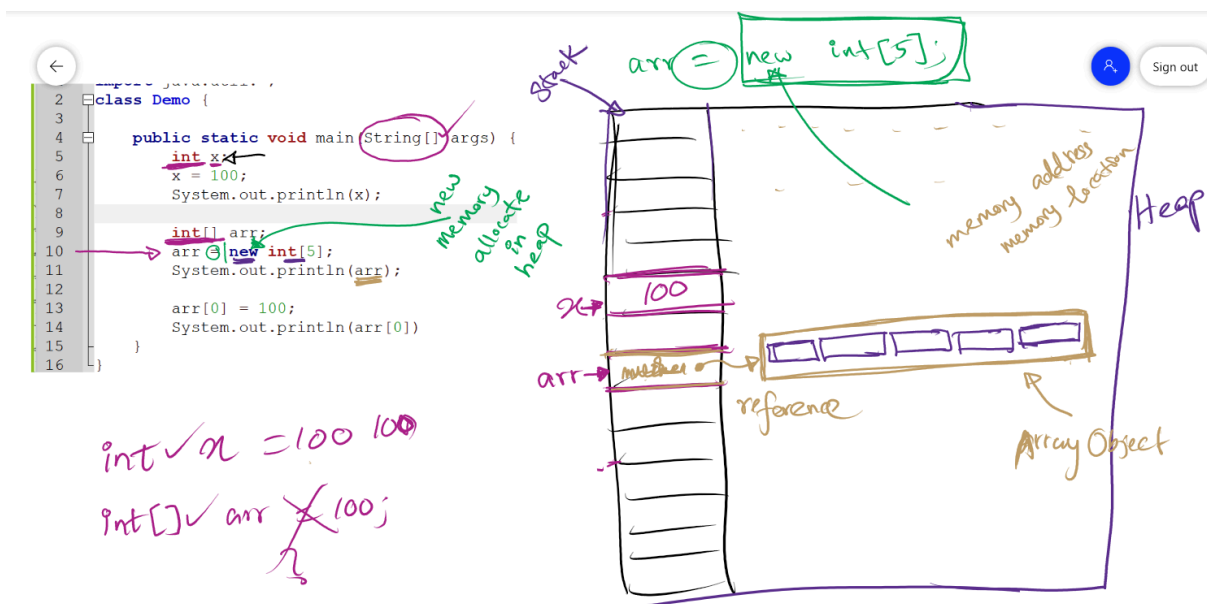
        for(int i=0; i<10; i++) {
            total += ar[i];
        }
        System.out.println("Total: " + total);
    }
}
```

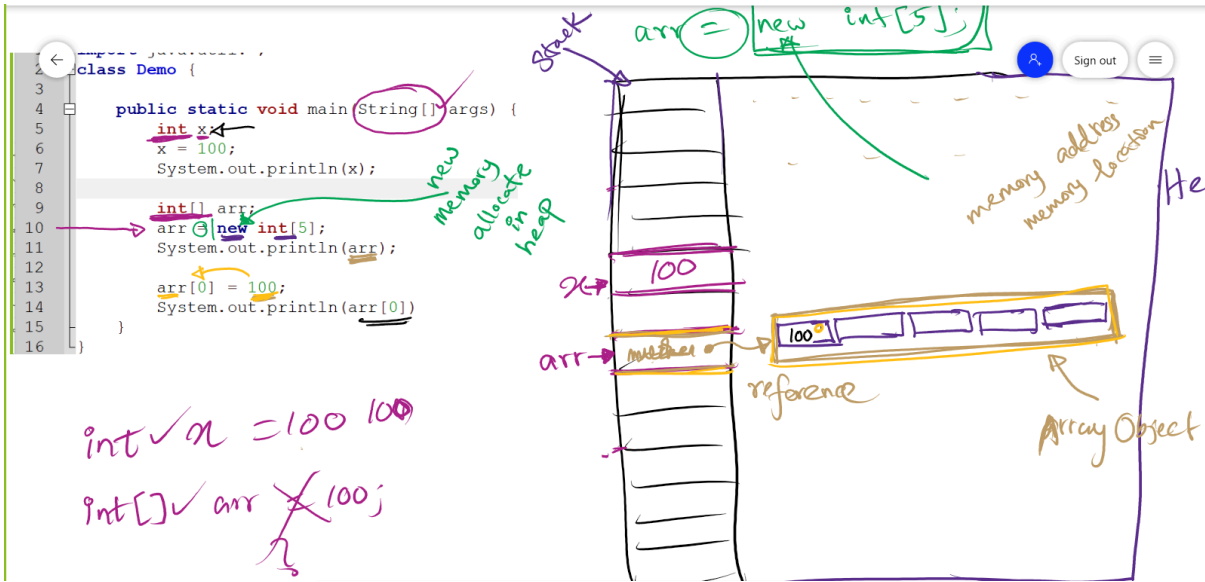
## Q378 Case I

=====

```
class Demo {  
  
    public static void main(String[] args) {  
        // int x = 100;  
  
        int x;  
        x = 100;  
        System.out.println(x);  
  
        // int[] arr = new int[5];  
  
        int[] arr;  
        arr = new int[5];  
        System.out.println(arr);  
  
        arr[0] = 100;  
        System.out.println(arr[0])  
    }  
}
```

```
100  
[I@73a28541
```





Q379

```

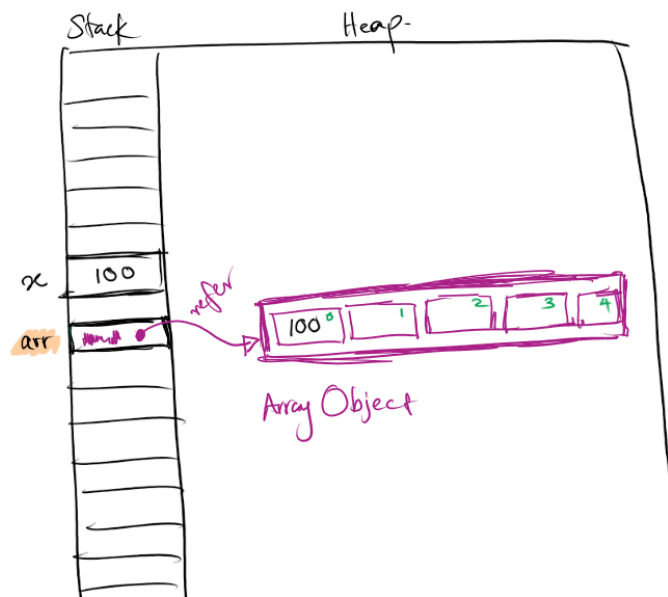
class Demo {
    public static void main(String[] args) {
        int x;
        x = 100;
        System.out.println(x);

        int[] arr;
        arr = new int[5];
        System.out.println(arr);

        arr[0] = 100;
        System.out.println(arr[0]);
    }
}

```

100  
100  
100



Q380 Case II

=====

import java.util.\*;

class Demo {

public static void main(String[] args) {

int[] xrr = new int[5];

int[] arr;

arr = new int[5];

}

}

```

////////////////////////////////////
Q381 Case III
=====
class Demo {

    public static void main(String[] args) {
        int[] xrr = new int[3];

        xrr[0] = 100;
        xrr[1] = 200;
        xrr[2] = 300;

        System.out.println(xrr[0]); // 100
        System.out.println(xrr[1]); // 200
        System.out.println(xrr[2]); // 300
    }
}

```

```

////////////////////////////////////
Q382 Case IV - Default Value
=====
class Demo {
    public static void main(String[] args) {
        int x;

        // System.out.println(x); // Illegal
        int[] xrr = new int[3];

        System.out.println(xrr[0]); // Default Value 0
        System.out.println(xrr[1]); // Default Value 0
        System.out.println(xrr[2]); // Default Value 0
    }
}

```

```

////////////////////////////////////
Q383 Case IV - From Q382
=====
class Demo {
    public static void main(String[] args) {
        double[] xrr = new double[3];

        System.out.println(xrr[0]); // Default Value 0.0
        System.out.println(xrr[1]); // Default Value 0.0
        System.out.println(xrr[2]); // Default Value 0.0
    }
}

```

```

class Demo {
    public static void main(String[] args) {
        char[] xrr = new char[3];

        System.out.println((int)xrr[0]);
        System.out.println(xrr[1]);
        System.out.println(xrr[2]);
    }
}

```

## Default Values

|         |       |                                                                                                 |      |
|---------|-------|-------------------------------------------------------------------------------------------------|------|
| byte    | 0     | <div>String</div> <div>{</div> <div>—</div> <div>—</div> <div>—</div> <div>—</div> <div>}</div> | null |
| short   | 0     |                                                                                                 |      |
| int     | 0     |                                                                                                 |      |
| long    | 0     |                                                                                                 |      |
| float   | 0.0   |                                                                                                 |      |
| double  | 0.0   |                                                                                                 |      |
| char    | —     |                                                                                                 |      |
| boolean | false |                                                                                                 |      |

```

class Demo {
    public static void main(String[] args) {
        String[] xrr = new String[3];

        System.out.println(xrr[0]);
        System.out.println(xrr[1]);
        System.out.println(xrr[2]);
    }
}

```

```
////////////////////////////////////
Q384 Case V
=====
class Demo {
    public static void main(String[] args) {
        byte b = 100;

        int x = b;

        int[] xrr = new byte[3]; // Illegal

        System.out.println(xrr[0]);
        System.out.println(xrr[1]);
        System.out.println(xrr[2]);
    }
}
////////////////////////////////////
```

```
Q385 Case VI
=====
class Demo {
    public static void main(String[] args) {
        byte[] xrr = new byte[3];

        xrr[0] = 100;

        int[] arr = new int[3];

        arr[0] = xrr[0];

        System.out.println(arr[0]);
    }
}
```

## Q386 Case VII

=====

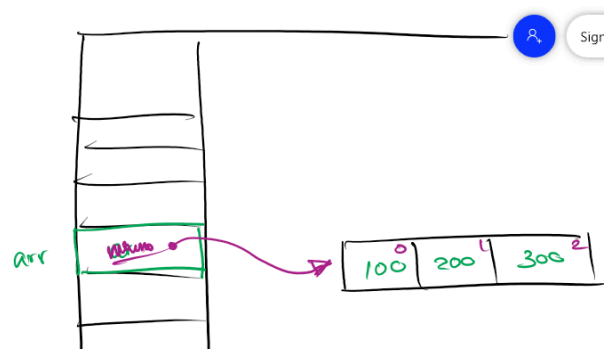
```
class Demo {  
    public static void main(String[] args) {  
        int[] arr1; // Preferred  
  
        int []arr2;  
  
        int [] arr3;  
  
        int arr4[];  
  
        int arr5 [];  
  
        int[5] arr6; // Illegal  
    }  
}
```

## Q387 Case VIII

=====

```
class Demo {  
    public static void main(String[] args) {  
        int[] arr = {100, 200, 300};  
  
        System.out.println(arr[0]);  
        System.out.println(arr[1]);  
        System.out.println(arr[2]);  
    }  
}
```

```
1 class Demo {  
2     public static void main(String[] args) {  
3         int[] arr = {100, 200, 300};  
4  
5         System.out.println(arr[0]);  
6         System.out.println(arr[1]);  
7         System.out.println(arr[2]);  
8     }  
9 }
```



```
class Demo {  
    public static void main(String[] args) {  
        int[] arr;  
        arr = new int[]{100, 200, 300};  
  
        System.out.println("No of Elements/Size/Length: " +  
arr.length);  
  
        System.out.println(arr[0]);  
        System.out.println(arr[1]);  
        System.out.println(arr[2]);  
    }  
}
```

```

////////////////////////////////////
Q391
class Demo {
    public static void main(String[] args) {
        int[] arr;
        arr = new int[]{100, 200, 300};

        String name = "Dinuka";

        System.out.println("No of Elements/Size/Length: " +
arr.length);
        System.out.println("Length of String: " + arr.length());
    }
}
////////////////////////////////////

```

#### Q392 Case XII

=====

```

class Demo {
    public static void main(String[] args) {
        int[] arr;
        arr = new int[]{100, 200, 300};

        System.out.println("Length of String: " + arr.length);
        System.out.println(arr[0]);
        System.out.println(arr[1]);
        System.out.println(arr[2]);

        System.out.println(arr[3]); //
ArrayIndexOutOfBoundsException
        System.out.println(arr[-1]); //
ArrayIndexOutOfBoundsException
    }
}

```

#### Q393 Case XIII

=====

```

class Demo {
    public static void main(String[] args) {
        int[] arr = {100, 200, 300, 400, 500, 102.2};
    }
}

```

```

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
Q394 Exercise
=====
class Demo {
    public static void main(String[] args) {
        int[] arr = {100, 200, 300, 400, 500};

        // ----- Print Array -----
    }
}

```

```
Q395 From Q394
=====
class Demo {
    public static void main(String[] args) {
        int[] arr = {100, 200, 300, 400, 500};

        // ----- Print Array -----
        for(int i=0; i<arr.length; i++) {
            System.out.println(arr[i]);
        }
    }
}
```

```
Q396 From Q395
=====
class Demo {
    public static void main(String[] args) {
        int[] arr = {100, 200, 300, 400, 500};

        // ----- Print Array -----
        System.out.print("[");
        for(int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + " , ");
        }
        System.out.println("\b\b]");
    }
}
```

# Passing array as Argument

Q397

```
class Demo {
```

```
    static void printArray(int[] arr){
        System.out.print("[");
        for(int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + ", ");
        }
        System.out.println("\b\b]");
    }
}
```

```
public static void main(String[] args) {
    int[] arr = {100, 200, 300, 400, 500};
```

```
    // ----- Print Array -----
    printArray(arr);
```

```
    int[] xr = {1,5,7,8,6,95,2,1};
```

```
    printArray(xr);
```

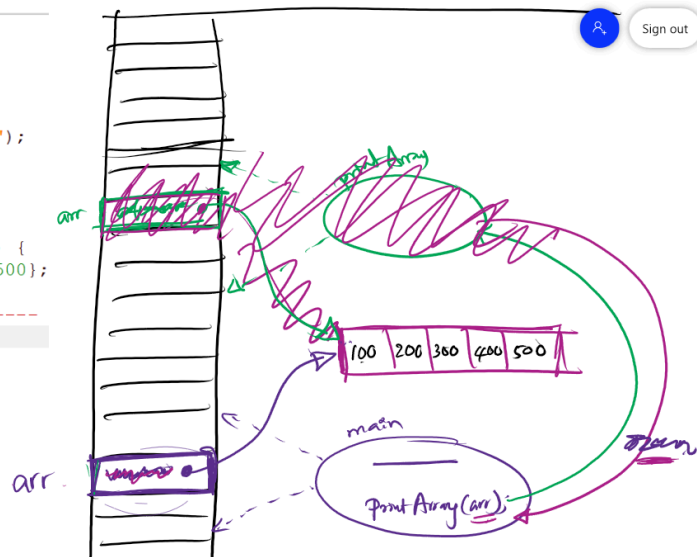
```
    }
}
```

```
< class Demo {
    static void printArray(int[] arr){
        System.out.print("[");
        for(int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + ", ");
        }
        System.out.println("\b\b]");
    }

    public static void main(String[] args) {
        int[] arr = {100, 200, 300, 400, 500};

        // ----- Print Array -----
        printArray(arr);
    }
}
```

[I@mmm



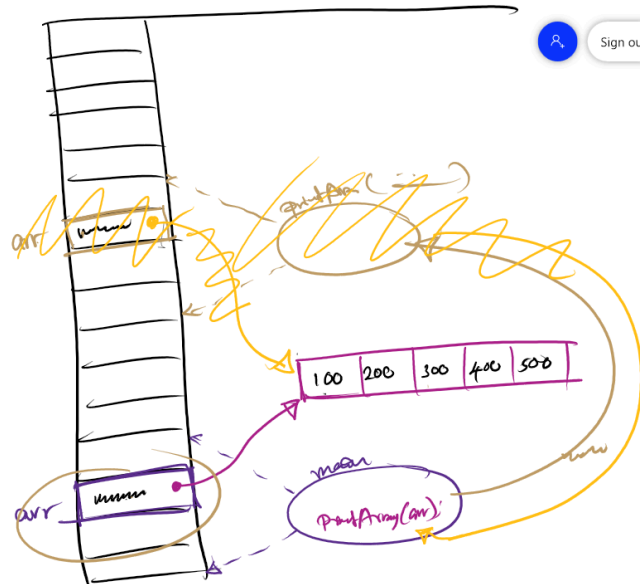
## Q398

```

class Demo {
    static void printArray(int[] arr){
        System.out.print("[");
        for(int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + ", ");
        }
        System.out.println("\b\b");
    }

    public static void main(String[] args) {
        int[] arr = {100, 200, 300, 400, 500};
        // ----- Print Array -----
        printArray(arr);
    }
}

```



## Q399 Exercise

=====

```

class Demo {

```

```

    static void myMethod(int[] xr){
        xr[2] = 125;
    }

```

```

    static void printArray(int[] arr){
        System.out.print("[");
        for(int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + ", ");
        }
        System.out.println("\b\b");
    }

```

```

    public static void main(String[] args) {
        int[] arr = {100, 200, 300, 400, 500};

```

```

        printArray(arr);

```

```

        myMethod(arr);

```

```

        printArray(arr);

```

```

    }

```

```

}

```

## For-Each | Enhanced For Loop

Q400

```
class Demo {

    static void printArray(int[] arr){
        System.out.print("[");
        //for(int i=0; i<arr.length; i++) {
        //    System.out.print(arr[i] + ", ");
        //}

        for(int element: arr) { // For Each | Enhanced For Loop
            System.out.print(element + ", ");
        }

        System.out.println("\b\b");
    }

    public static void main(String[] args) {
        int[] arr = {100, 200, 300, 400, 500};

        printArray(arr);
    }
}
```

Q401

```
class Demo {
    public static void main(String[] args) {
        int[] arr = {100, 200, 300, 400, 500};
        for(int element: arr) { // For Each | Enhanced For Loop
            System.out.print(element);
        }
    }
}
```

Q402

```
class Demo {
    public static void main(String[] args) {
        int[] arr = {100, 200, 300, 400, 500};

        int element;
        for(element: arr) { // Illegal
            System.out.print(element);
        }
    }
}
```

```

    }
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
Q403
class Demo {
    public static void main(String[] args) {
        int[] arr = {100, 200, 300, 400, 500};

        for(long element: arr) {
            System.out.print(element);
        }
    }
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

```

```

Q404
class Demo {
    public static void main(String[] args) {
        String[] fruits = new String[]{"Apple", "Orange", "Pineapple"};

        for(String fruit: fruits) {
            //
        }
    }
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

```

Q405 Exercise  
=====

```

class Demo {
    static void printArray(int[] arr){
        System.out.print("[");
        for(int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + ", ");
        }
        System.out.println("\b\b]");
    }
    public static void main(String[] args) {
        int[] arr = {100, 200, 300, 400, 500};

        for(int element: arr) {
            element++;
        }

        printArray(arr);
    }
}

```

[illegible]

=====

```
// Find index of 51
```

[illegible]

=====

```
public static void main(String[] args) {
    int[] arr = {1,5,78,6,51,2,8,39,54};
```

} }

```
class Demo {

    static int max(int[] arr) {
        return 0;
    }

    static int indexOf(int[] arr, int value){
        for(int i=0; i<arr.length;i++){
            if(arr[i] == value) {
                return i;
            }
        }
        return -1; // default case: no value found
    }

    public static void main(String[] args) {
        int[] arr = {1,5,78,6,51,2,8,39,54};

        System.out.println("Index of 51: " + indexOf(arr, 51));
        System.out.println("Max : " + max(arr));
    }
}
```

Q409

```
class Demo {
```

```
    static int indexOf(int[] arr, int value){
        for(int i=0; i<arr.length;i++){
            if(arr[i] == value) {
                return i;
            }
        }
        return -1; // default case: no value found
    }
}
```

```
    static int max(int[] arr) {

        int max = arr[0];

        for(int i=1; i<arr.length; i++){
            if(arr[i] > max) {
                max = arr[i];
            }
        }

        return max;
    }
}
```

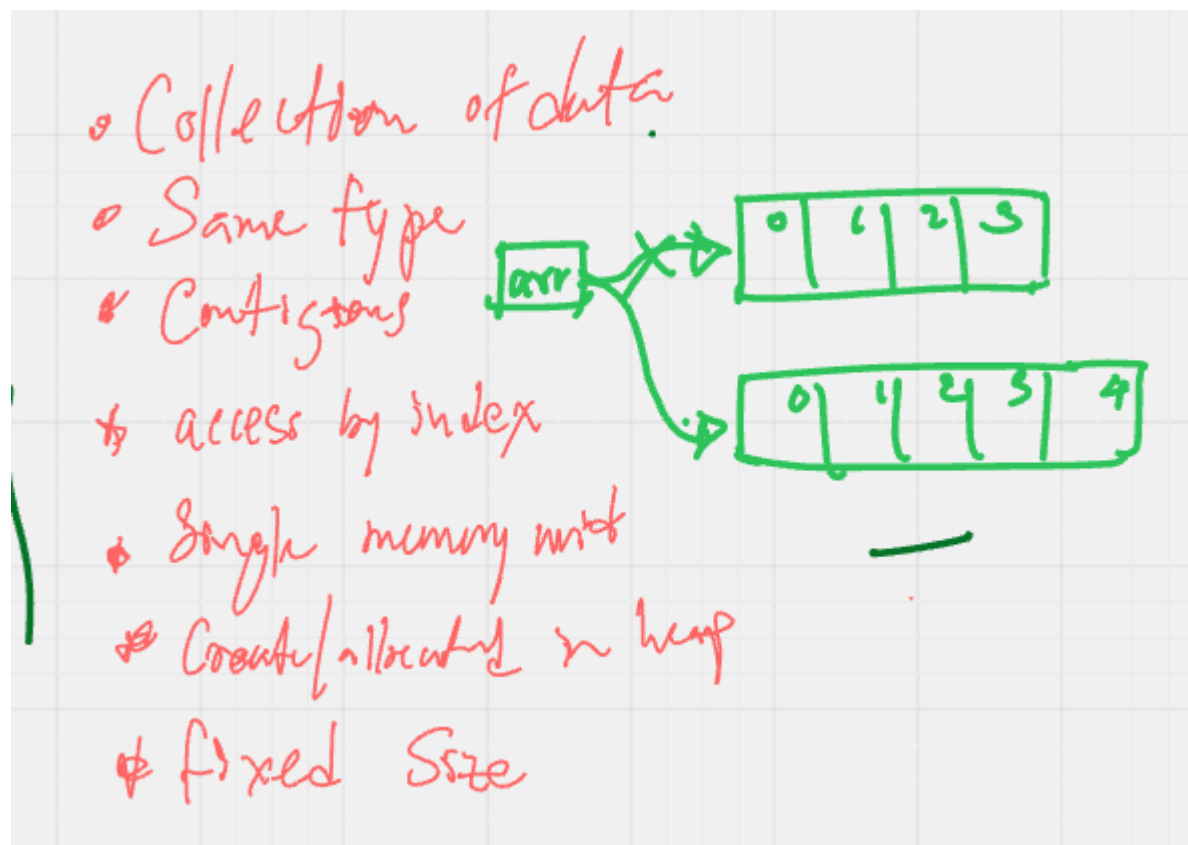
```
    public static void main(String[] args) {
        int[] arr = {1,5,78,6,51,2,8,39,54};

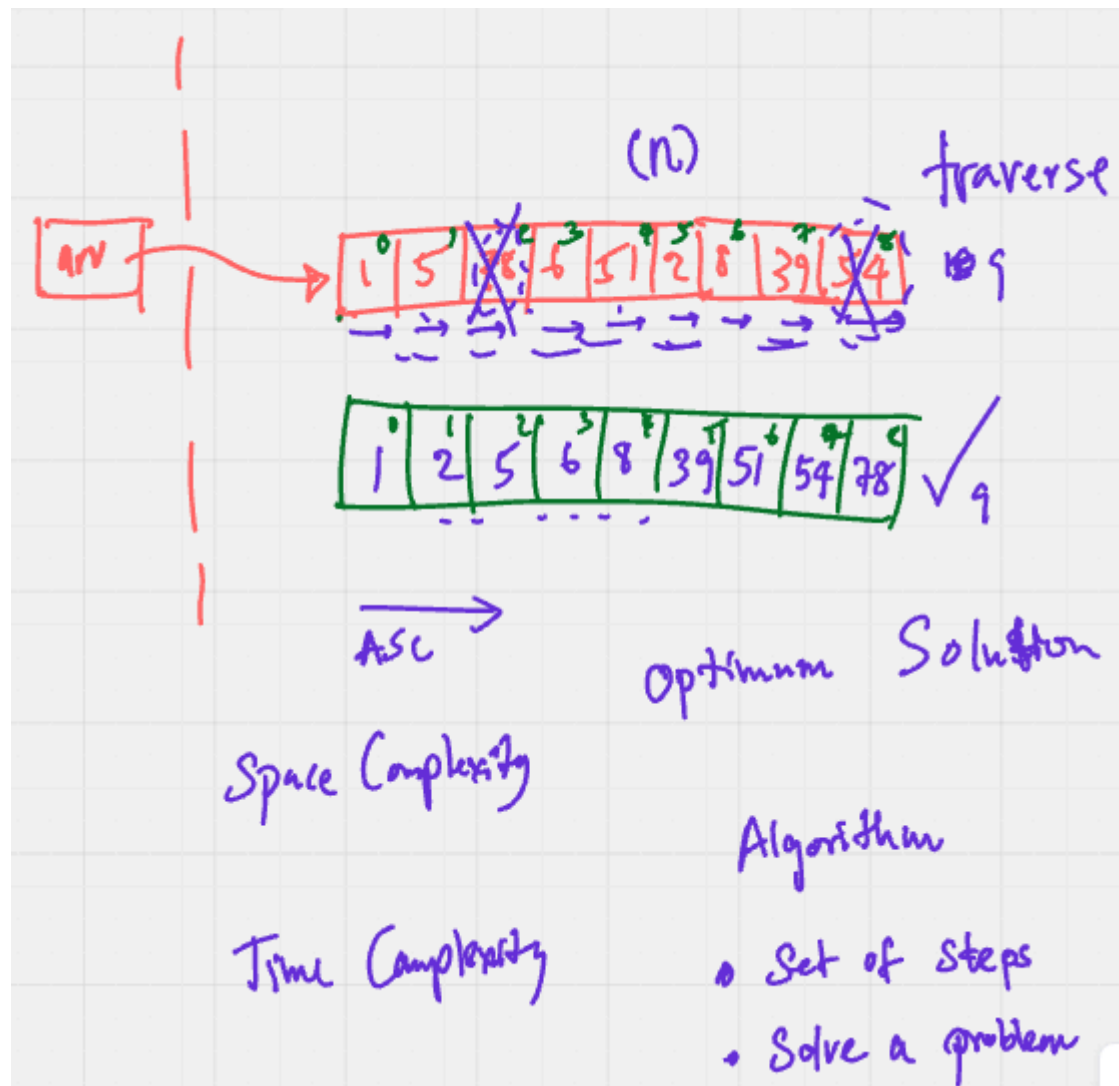
        System.out.println("Index of 51: " + indexOf(arr, 51));
        System.out.println("Max : " + max(arr));
    }
}
```

## Q410 Array Sorting

=====

```
class Demo {  
    static void printArray(int[] arr){  
        System.out.print("[");  
        for(int i=0; i<arr.length; i++) {  
            System.out.print(arr[i] + ", ");  
        }  
        System.out.println("\b\b]");  
    }  
  
    public static void main(String[] args) {  
        int[] arr = {1,5,78,6,51,2,8,39,54};  
  
        printArray(arr);  
  
        // Sort the array  
  
        printArray(arr);  
    }  
}
```





////////////////////////////////////  
Q412 Exercise

=====

```
class Demo {
    static void printArray(int[] arr){
        System.out.print("[");
        for(int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + ", ");
        }
        System.out.println("\b\b]");
    }

    public static void main(String[] args) {
        int[] arr = {1,5,78,6,51,2,8,39,54};

        printArray(arr);

        // Sort the array
        int max = 0;
        int indexOfMax = -1;
        for(int i=0; i<arr.length; i++){
            if(max < arr[i]){
                max = arr[i];
                indexOfMax = i;
            }
        }

        // Swap

        System.out.println("Max: " + max);
        System.out.println("indexOfMax: " + indexOfMax);

        printArray(arr);
    }
}
```

////////////////////////////////////

Q413 From Q412

=====

```
class Demo {
    static void printArray(int[] arr){
        System.out.print("[");
        for(int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + ", ");
        }
        System.out.println("\b\b]");
    }

    public static void main(String[] args) {
        int[] arr = {1,5,78,6,51,2,8,39,54};

        printArray(arr);

        // Sort the array
        int max = 0;
        int indexOfMax = -1;
        for(int i=0; i<arr.length; i++){
            if(max < arr[i]){
                max = arr[i];
                indexOfMax = i;
            }
        }

        int temp = arr[arr.length-1];
        arr[arr.length-1] = arr[indexOfMax];
        arr[indexOfMax] = temp;

        //System.out.println("Max: " + max);
        //System.out.println("indexOfMax: " + indexOfMax);

        printArray(arr);
    }
}
```

////////////////////////////////////

=====

```

class Demo {
    static void printArray(int[] arr){
        System.out.print("[");
        for(int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + ", ");
        }
        System.out.println("\b\b");
    }

    public static void main(String[] args) {
        int[] arr = {1,5,78,6,51,2,8,39,54};

        printArray(arr);

        // Sort the array
        int max = 0;
        int indexOfMax = -1;

        // First Traverse
        for(int i=0; i<arr.length; i++){
            if(max < arr[i]){
                max = arr[i];
                indexOfMax = i;
            }
        }

        int temp = arr[arr.length-1];
        arr[arr.length-1] = arr[indexOfMax];
        arr[indexOfMax] = temp;
        printArray(arr);

        // Second Traverse
        for(int i=0; i<arr.length - 1; i++){
            if(max < arr[i]){
                max = arr[i];
                indexOfMax = i;
            }
        }

        temp = arr[arr.length - 1 - 1];
        arr[arr.length-1 - 1] = arr[indexOfMax];
    }
}

```

```

        arr[indexOfMax] = temp;
        printArray(arr);

    }
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
Q415 - Selection Sorting
=====
class Demo {
    static void printArray(int[] arr){
        System.out.print("[");
        for(int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + ", ");
        }
        System.out.println("\b\b]");
    }

    public static void main(String[] args) {
        int[] arr = {1,5,78,6,51,2,8,39,54};

        printArray(arr);

        // Sort the array

        for(int i=0; i<arr.length; i++){
            int max = 0;
            int indexOfMax = -1;

            for(int j=0; j<arr.length - i; j++){
                if(max < arr[j]){
                    max = arr[j];
                    indexOfMax = j;
                }
            }

            int temp = arr[arr.length - 1 - i];
            arr[arr.length-1-i] = arr[indexOfMax];
            arr[indexOfMax] = temp;
            printArray(arr);
        }
    }
}

```

## Q416 Homework

- ////////////////////////////////////

```
class Demo {
    static void printArray(int[] arr){
        System.out.print("[");
        for(int i=0; i<arr.length; i++) {
            System.out.print(arr[i] + ", ");
        }
        System.out.println("\b\b]");
    }

    static void changeArray(int[] xrr){
        xrr[0] = 100;
    }

    public static void main(String[] args) {
        int[] arr = {1,5,78,6,51,2,8,39,54};

        printArray(arr); // [1,5,78,6,51,2,8,39,54]

        changeArray(arr);

        printArray(arr); // [100,5,78,6,51,2,8,39,54]
    }
}
```

```
System.out.print("\033[H\033[2J");
System.out.flush();
```

////////////////////////////////////

Q420 java.util.Arrays

=====

import java.util.\*;

class Demo {

public static void main(String[] args) {

int[] arr = {1,5,78,6,51,2,8,39,54};

System.out.println(Arrays.toString(arr));

// Sort the array

Arrays.sort(arr);

System.out.println(Arrays.toString(arr));

//Arrays.reverse(arr);

//System.out.println(Arrays.toString(arr));

}

}

} }

```
printBoard(board);
```

}



```

        System.out.println(x2drr);
        System.out.println(x2drr[0]);
        System.out.println(x2drr[1]);
        System.out.println(x2drr[2]);
    }
}
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

```

#### Q426 Char array Print

```

class Demo {
    public static void main(String[] args) {
        int[] xr = {1, 2, 3};
        System.out.println(xr); // [I@092340

        char[] ch = {'A', 'B', 'C'};
        System.out.println(ch); //

        char[] ch2 = new char[3];
        System.out.println(ch2); //

    }
}

```

```

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////

```

#### Q427

```

import java.util.*;

class Demo {

    static void printBoard(int[][] board){
        System.out.println("=====");
        System.out.println(Arrays.toString(board[0]));
        System.out.println(Arrays.toString(board[1]));
        System.out.println(Arrays.toString(board[2]));
        System.out.println("=====");
    }

    public static void main(String[] args) {
        int[][] x2drr = {{1,2,3}, {4,5,6}, {7,8,9}};

        printBoard(x2drr);

    }
}

```

Q428

```
import java.util.*;
```

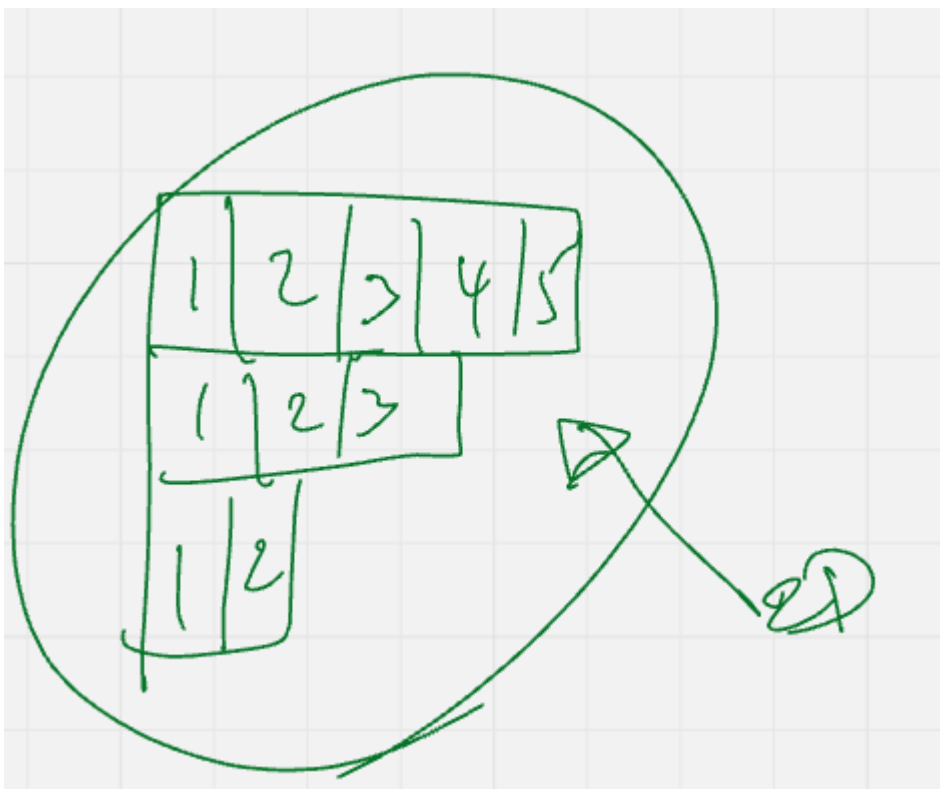
```
class Demo {
```

```
    static void printBoard(int[][] board){  
        System.out.println("=====");  
        System.out.println(Arrays.toString(board[0]));  
        System.out.println(Arrays.toString(board[1]));  
        System.out.println(Arrays.toString(board[2]));  
        System.out.println("=====");  
    }
```

```
    public static void main(String[] args) {  
        int[][] x2drr = {{1,2,3,4,5}, {4,5,6}, {7,8}};
```

```
        printBoard(x2drr);
```

```
    }  
}
```





} }

```
static void printBoard(char[][] board){
```

```

        System.out.println("=====");
        System.out.println(Arrays.toString(board[0]));
        System.out.println(Arrays.toString(board[1]));
        System.out.println(Arrays.toString(board[2]));
        System.out.println("=====");
    }

```

```

static int checkWinner(char[][] board){

```

```

}

```

```

public static void main(String[] args) {
    Scanner input = new Scanner(System.in);

```

```

    char[][] board = new char[3][3];

```

```

    printBoard(board);

```

```

    for(int i=0; i<4; i++){
        // User 1 Input
        System.out.print("User 1(X): ");
        int x1 = input.nextInt();

```

```

        board[x1/10][x1%10] = 'X';

```

```

        printBoard(board);

```

```

        int res = checkWinner(board);
        if(res != 0) {

```

```

            break;
        }

```

```

        // User 2 Input
        System.out.print("User 2(O): ");
        int o1 = input.nextInt();

```

```

        board[o1/10][o1%10] = 'O';

```

```

        printBoard(board);

```

```

        res = checkWinner(board);
        if(res != 0) {

```

} }

////////////////////////////////////

The diagram illustrates a 3x3 board state with rows and columns indexed 0 to 2. The board contains the following values:

|   |   |   |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 0 | 1 |
| 2 | 0 | 1 |

Red lines connect the values in the board to their corresponding indices in the code below. A green oval highlights the top row (row 0).

The code defines the board state and its indices:

```
board[0][0] == 'x' && board[0][1] == 'x' && board[0][2] == 'x'
board[1][0] == 'x' && board[1][1] == 'x' && board[1][2] == 'x'
```

```
static int checkWinner(char[][] board){
    if(
```

```

        (board[0][0] == 'X' && board[0][1] == 'X' && board[0][2] ==
'X') ||
        (board[1][0] == 'X' && board[1][1] == 'X' && board[1][2] ==
'X') ||
        (board[2][0] == 'X' && board[2][1] == 'X' && board[2][2] ==
'X') ||
        (board[0][0] == 'X' && board[1][0] == 'X' && board[2][0] ==
'X') ||
        (board[0][1] == 'X' && board[1][1] == 'X' && board[2][1] ==
'X') ||
        (board[0][2] == 'X' && board[1][2] == 'X' && board[2][2] ==
'X') ||
        (board[0][0] == 'X' && board[1][1] == 'X' && board[2][2] ==
'X') ||
        (board[0][2] == 'X' && board[1][1] == 'X' && board[2][0] ==
'X')){
            return 1;
        }

        if(
'O') ||
        (board[1][0] == 'O' && board[1][1] == 'O' && board[1][2] ==
'O') ||
        (board[2][0] == 'O' && board[2][1] == 'O' && board[2][2] ==
'O') ||
        (board[0][0] == 'O' && board[1][0] == 'O' && board[2][0] ==
'O') ||
        (board[0][1] == 'O' && board[1][1] == 'O' && board[2][1] ==
'O') ||
        (board[0][2] == 'O' && board[1][2] == 'O' && board[2][2] ==
'O') ||
        (board[0][0] == 'O' && board[1][1] == 'O' && board[2][2] ==
'O') ||
        (board[0][2] == 'O' && board[1][1] == 'O' && board[2][0] ==
'O')){
            return 2;
        }

        return 0;
    }

    public static void main(String[] args) {

```

```
Scanner input = new Scanner(System.in);
```

```
char[][] board = new char[3][3];
```

```
printBoard(board);
```

```
for(int i=0; i<4; i++){
```

```
    // User 1 Input
```

```
    System.out.print("User 1(X): ");
```

```
    int x1 = input.nextInt();
```

```
    board[x1/10][x1%10] = 'X';
```

```
    printBoard(board);
```

```
    int res = checkWinner(board);
```

```
    if(res != 0) {
```

```
        break;
```

```
    }
```

```
    // User 2 Input
```

```
    System.out.print("User 2(O): ");
```

```
    int o1 = input.nextInt();
```

```
    board[o1/10][o1%10] = 'O';
```

```
    printBoard(board);
```

```
    res = checkWinner(board);
```

```
    if(res != 0) {
```

```
        break;
```

```
    }
```

```
}
```

```
}
```

```
}
```

Q435

```
import java.util.*;
```

```
class Demo {
```

```
    static void printBoard(char[][] board){
        System.out.println("=====");
        System.out.println(Arrays.toString(board[0]));
        System.out.println(Arrays.toString(board[1]));
        System.out.println(Arrays.toString(board[2]));
        System.out.println("=====");
    }
```

```
    static int checkWinner(char[][] board){
        if(
            (board[0][0] == 'X' && board[0][1] == 'X' && board[0][2] ==
            'X') ||
            (board[1][0] == 'X' && board[1][1] == 'X' && board[1][2] ==
            'X') ||
            (board[2][0] == 'X' && board[2][1] == 'X' && board[2][2] ==
            'X') ||
            (board[0][0] == 'X' && board[1][0] == 'X' && board[2][0] ==
            'X') ||
            (board[0][1] == 'X' && board[1][1] == 'X' && board[2][1] ==
            'X') ||
            (board[0][2] == 'X' && board[1][2] == 'X' && board[2][2] ==
            'X') ||
            (board[0][0] == 'X' && board[1][1] == 'X' && board[2][2] ==
            'X') ||
            (board[0][2] == 'X' && board[1][1] == 'X' && board[2][0] ==
            'X')){
            return 1;
        }
```

```
        if(
            (board[0][0] == 'O' && board[0][1] == 'O' && board[0][2] ==
            'O') ||
            (board[1][0] == 'O' && board[1][1] == 'O' && board[1][2] ==
            'O') ||
            (board[2][0] == 'O' && board[2][1] == 'O' && board[2][2] ==
            'O') ||
            (board[0][0] == 'O' && board[1][0] == 'O' && board[2][0] ==
            'O') ||
```

```

        (board[0][1] == 'O' && board[1][1] == 'O' && board[2][1] ==
'O') ||
        (board[0][2] == 'O' && board[1][2] == 'O' && board[2][2] ==
'O') ||
        (board[0][0] == 'O' && board[1][1] == 'O' && board[2][2] ==
'O') ||
        (board[0][2] == 'O' && board[1][1] == 'O' && board[2][0] ==
'O')){
            return 2;
        }

        return 0;
    }
}

```

```

public static void main(String[] args) {
    Scanner input = new Scanner(System.in);

    char[][] board = new char[3][3];

    Arrays.fill(board[0], ' ');
    Arrays.fill(board[1], ' ');
    Arrays.fill(board[2], ' ');

    printBoard(board);

    int res = 0;

    for(int i=0; i<4; i++){
        // User 1 Input
        System.out.print("User 1(X): ");
        int x1 = input.nextInt();

        board[x1/10][x1%10] = 'X';

        printBoard(board);

        res = checkWinner(board);
        if(res != 0) {
            break;
        }

        // User 2 Input
        System.out.print("User 2(O): ");
    }
}

```



```

        (board[1][0] == 'X' && board[1][1] == 'X' && board[1][2] ==
'X') ||
        (board[2][0] == 'X' && board[2][1] == 'X' && board[2][2] ==
'X') ||
        (board[0][0] == 'X' && board[1][0] == 'X' && board[2][0] ==
'X') ||
        (board[0][1] == 'X' && board[1][1] == 'X' && board[2][1] ==
'X') ||
        (board[0][2] == 'X' && board[1][2] == 'X' && board[2][2] ==
'X') ||
        (board[0][0] == 'X' && board[1][1] == 'X' && board[2][2] ==
'X') ||
        (board[0][2] == 'X' && board[1][1] == 'X' && board[2][0] ==
'X')){
            return 1;
        }

        if(
'O') ||
        (board[1][0] == 'O' && board[1][1] == 'O' && board[1][2] ==
'O') ||
        (board[2][0] == 'O' && board[2][1] == 'O' && board[2][2] ==
'O') ||
        (board[0][0] == 'O' && board[1][0] == 'O' && board[2][0] ==
'O') ||
        (board[0][1] == 'O' && board[1][1] == 'O' && board[2][1] ==
'O') ||
        (board[0][2] == 'O' && board[1][2] == 'O' && board[2][2] ==
'O') ||
        (board[0][0] == 'O' && board[1][1] == 'O' && board[2][2] ==
'O') ||
        (board[0][2] == 'O' && board[1][1] == 'O' && board[2][0] ==
'O')){
            return 2;
        }

        return 0;
    }

```

```

public static void main(String[] args) {
    Scanner input = new Scanner(System.in);

```

```
char[][] board = new char[3][3];
```

```
Arrays.fill(board[0], ' ');
```

```
Arrays.fill(board[1], ' ');
```

```
Arrays.fill(board[2], ' ');
```

```
printBoard(board);
```

```
int res = 0;
```

```
for(int i=0; i<4; i++){
```

```
    // User 1 Input
```

```
    System.out.print("User 1(X): ");
```

```
    int x1 = input.nextInt();
```

```
    board[x1/10][x1%10] = 'X';
```

```
    printBoard(board);
```

```
    res = checkWinner(board);
```

```
    if(res != 0) {
```

```
        break;
```

```
    }
```

```
    // User 2 Input
```

```
    System.out.print("User 2(O): ");
```

```
    int o1 = input.nextInt();
```

```
    board[o1/10][o1%10] = 'O';
```

```
    printBoard(board);
```

```
    res = checkWinner(board);
```

```
    if(res != 0) {
```

```
        break;
```

```
    }
```

```
}
```

```
if(res == 1) {
```

```
    System.out.println("The Winner is User 1(X)");
```

```
} else if(res == 2) {
```

```
    System.out.println("The Winner is User 2(O)");
```

```
} else {
```

```
    System.out.println("Draw!!!");
```

}

## Q437 Homework

## How to check the winner with loops?

Q438

Q439

```
static void increment(){
    X++; // Compile Error
}
```

```
static void add10(){
    x += 10; // Compile Error
}
```

```
System.out.println(x);
increment();
System.out.println(x);
```

} }

Q440

```
static int x = 100;
```

```
static void increment(){
    x++;
}
```



## Variable Arguments / var-args

Q442

```
import java.util.*;
```

```
class Demo {
```

```
    static int findTotal(int a, int b){
```

```
    }
```

```
    static int findTotal(int a, int b, int c){
```

```
    }
```

```
    static int findTotal(int a, int b, int c, int d){
```

```
    }
```

```
    static int findTotal(int a, int b, int c, int d, int e){
```

```
    }
```

```
    ...
```

```
    public static void main(String[] args) {
```

```
        findTotal(10,20);
```

```
        findTotal(10,20,30)
```

```
        findTotal(10,20,30,40,50,60);
```

```
        findTotal(10,20,30,40,50,60,70,80,90,100);
```

```
        findTotal(10,20,30,40,50,60,70,80,90,100,10,20,30,40,50,60,70,80,90,100);
```

```
    }
```

```
}
```

////////////////////////////////////

Q443

class Demo {

```
    static int findTotal(int ... a){  
        return 0;  
    }
```

```
    public static void main(String[] args) {  
        findTotal(10,20);  
        findTotal(10,20,30);  
        findTotal(10,20,30,40,50,60);  
        findTotal(10,20,30,40,50,60,70,80,90,100);  
        findTotal(10,20,30,40,50,60,70,80,90,100, 101);
```

```
findTotal(10,20,30,40,50,60,70,80,90,100,10,20,30,40,50,60,70,80,90,10  
0);  
    }
```

```
}
```

////////////////////////////////////

Q444 Case I

=====

class Demo {

```
    static int findTotal(int ... a){  
        System.out.println(a);  
        return 0;  
    }
```

```
    public static void main(String[] args) {  
        findTotal(10,20);  
        findTotal(10,20,30);  
        findTotal(10,20,30,40,50,60);  
        findTotal(10,20,30,40,50,60,70,80,90,100);  
        findTotal(10,20,30,40,50,60,70,80,90,100, 101);
```

```
findTotal(10,20,30,40,50,60,70,80,90,100,10,20,30,40,50,60,70,80,90,10  
0);  
    }
```

```
}
```

## Q445 Case II

```
class Demo {
```

```
static int findTotal(int ... a){
    System.out.println(a);
}
```

```
for(int x: a){
    System.out.print(x + ", ");
}
```

```
}
System.out.println("\b\b");
return 0;
```

}

```
public static void main(String[] args) {
```

```
findTotal(10,20);
```

```
findTotal(10,20,30);
```

```
findTotal(10,20,30,40,50,60);
```

```
findTotal(10,20,30,40,50,60,70,80,90,100);
```

```
findTotal(10,20,30,40,50,60,70,80,90,100, 101);
```

```
findTotal(10,20,30,40,50,60,70,80,90,100,10,20,30,40,50,60,70,80,90,100);
```

}

}

### Q446 Case III

```
class Demo {
```

```
static int findTotal(int ... a){
    System.out.println(a);
}
```

```
int total = 0;
```

```
for(int x: a){
    total += x;
}
```

}

```
return total;
```

}

```
public static void main(String[] args) {
```

```
findTotal(10,20);
findTotal(10,20,30);
findTotal(10,20,30,40,50,60);
findTotal(10,20,30,40,50,60,70,80,90,100);
findTotal(10,20,30,40,50,60,70,80,90,100, 101);
```

```
findTotal(10,20,30,40,50,60,70,80,90,100,10,20,30,40,50,60,70,80,90,10
0);
    }
}
```

```
////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
```

Q447 Case IV

=====

```
class Demo {

    static int findTotal(int ... a){
        System.out.println(a.length);

        int total = 0;
        for(int x: a){
            total += x;
        }

        return total;
    }

    public static void main(String[] args) {
        findTotal();
    }
}
```

}

Q457

////////////////////////////////////  
Q458

////////////////////////////////////  
Q459

////////////////////////////////////  
Q410